# Helfi.Space

Manual for users, agents and administrators

Helfi.Space is a lean ticket system for IT departments and institutes. It runs as a single Docker container, signs users in via Active Directory or single sign-on and turns emails into tickets.

**Users**
Report problems, follow requests, reply – chapter 2

**Agents**
Work on tickets, templates, reports, keyboard shortcuts – chapter 3

**IT service management**
Incidents, SLA, problems, changes, knowledge, service catalog, devices, satisfaction – chapter 4

**Administrators**
Installation, AD, SSO, 2FA, email, updates, backup – chapters 5 and 6

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All names, addresses and computers in this manual are examples.

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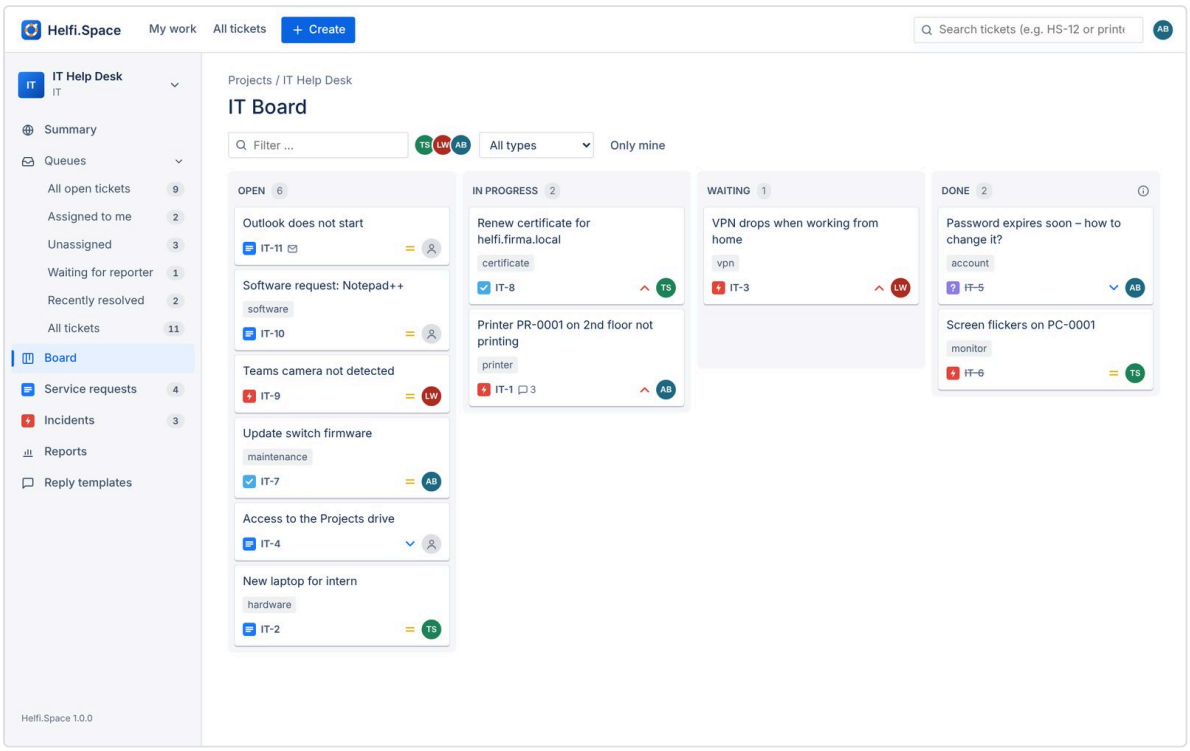
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# 1 Overview

In Helfi.Space every concern becomes a **ticket** – whether someone reports it in the browser or sends an email to the help desk mailbox. Agents see all tickets of their projects, assign them, reply and finally set them to “Done”. Reporters receive every reply by email and can answer it directly.




Board of a project with the four status columns

## 1.1 Roles

| Role | What it may do |
|---|---|
| **User** | Report and follow their own tickets, see tickets they participate in, reply and attach files. Internal notes are never shown to users. |
| **Agent** | See and work on the tickets of their projects – all projects by default: status, assignee, priority, type, labels, internal notes, reply templates, bulk actions, reports. |
| **Administrator** | Everything agents may do, plus *Administration*: projects, users, AD, SSO, security, email, appearance, import, system. |

Agents can be limited to single projects, so that e.g. facilities do not see IT tickets (Agents per project). With Active Directory the role is derived from the AD groups at every sign-in (chapter 5.4). The initial setup also creates a **local administrator** that keeps working when AD is unreachable.

## 1.2 Statuses, types and priorities

| Status | Meaning |
|---|---|
| Open | New or not started yet. |
| In progress | Someone is working on it. |
| Waiting | Waiting for the reporter. If they reply, the ticket reopens. |
| Done | Finished. The reporter gets an email. If they reply to it, the ticket reopens (unless it was rejected or closed as a duplicate). |

**Types:** Incident (something is broken), Request (something is needed), Task (internal work), Question; for agents also Problem (common cause of several incidents) and Change (planned change), see chapter 4. **Priorities:** Highest, High, Medium, Low.

Every ticket has a **key** made of the project key and a running number, e.g. `IT-42`. It also appears in the subject of all emails, e.g. `[IT-42]`.

## 1.3 The interface

- **Header:** logo, *My work*, *All tickets*, the **Create** button, search (ticket key or keyword) and, on the right, the user menu with profile, language (Deutsch/English), theme and sign-out.
- **Sidebar (agents):** current project, summary, queues with live counts, board, service requests, incidents, problems, changes, my approvals, change calendar, service catalog, devices & services, knowledge, reports, reply templates and, for admins, settings.
- **Sidebar (users):** *My requests*, depending on the setup *Request a service* and *My approvals*, and *Help & knowledge*.

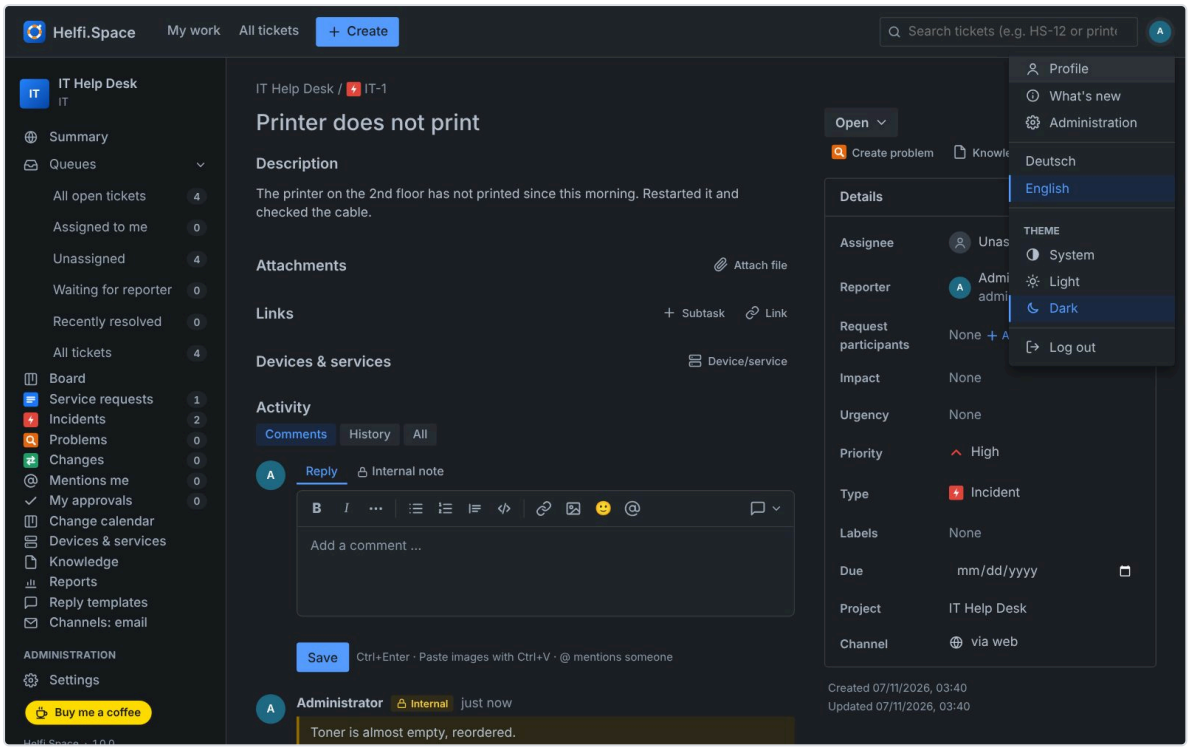
> The interface is available in German and English. Without a choice of your own it follows the browser language.

### Dark mode

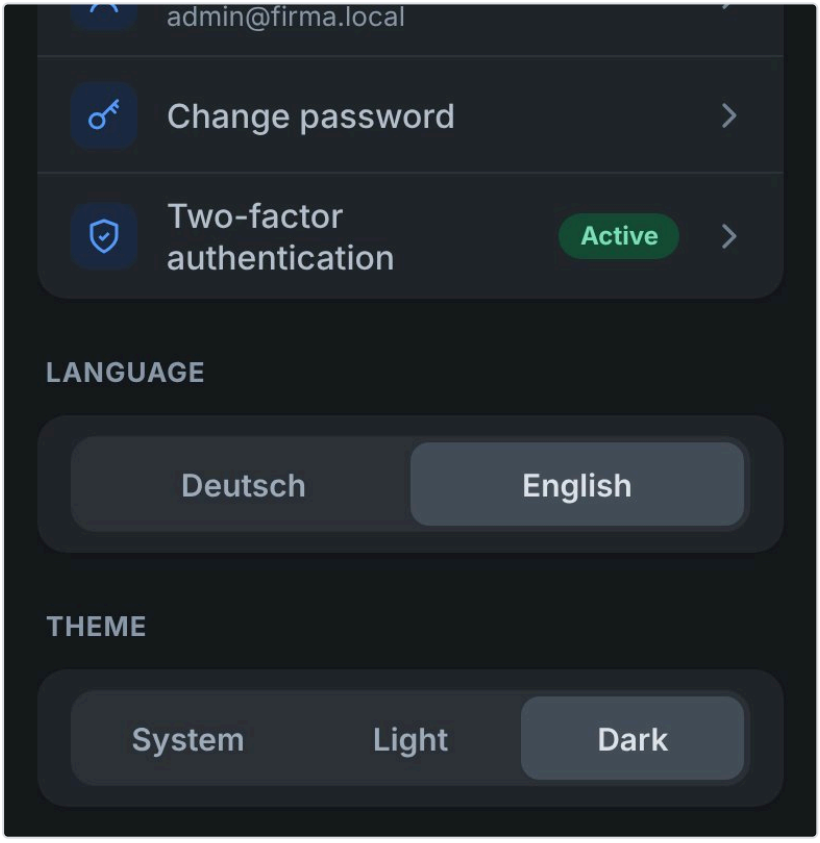
Under *Theme* in the user menu you choose how Helfi.Space looks:

- **System** (default): follows the setting of Windows, macOS or the phone – dark in the evening when the device switches.
- **Light** or **Dark**: always like that, whatever the device is set to.

The choice is stored with your account, so it also applies on your phone and on other PCs. On phones it is under *Me* → *Theme*. Emails and uploaded logos stay light so they look the same everywhere.




Ticket in dark mode, the user menu with “Theme” on the right

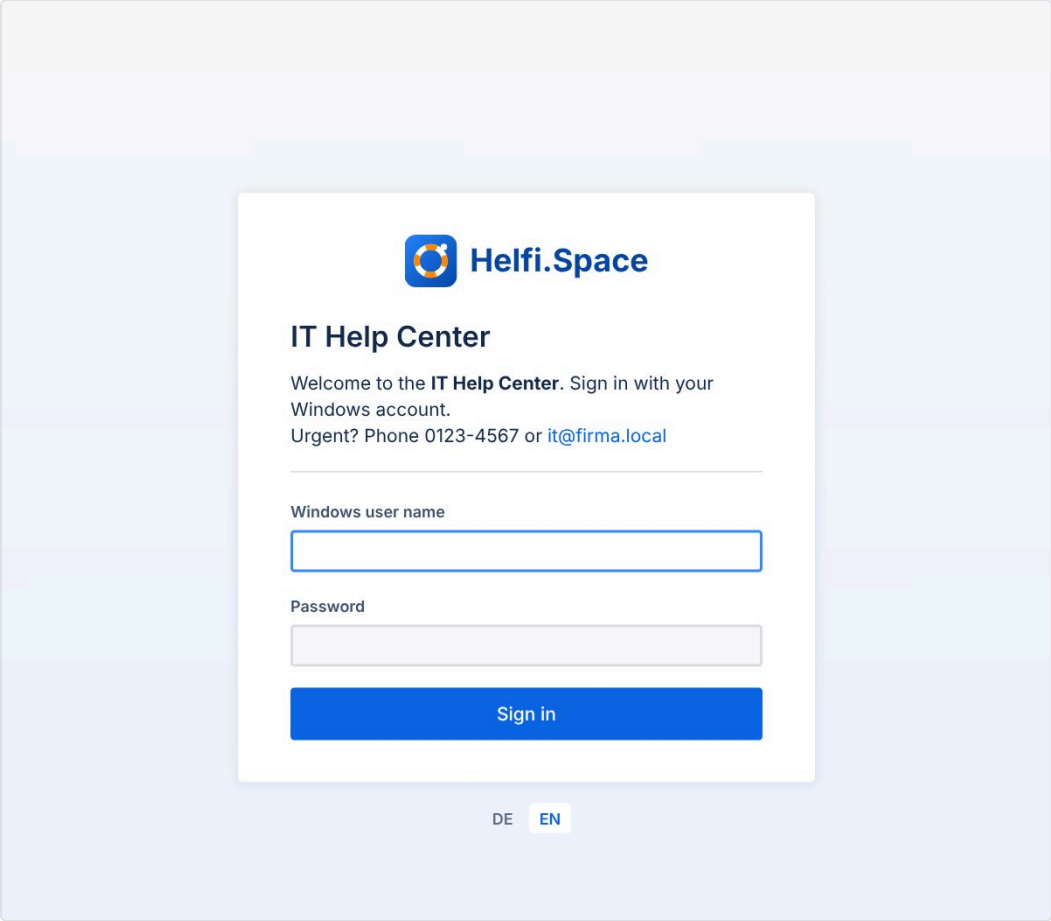



On phones: Me → Theme

# 2 For everyone: signing in and making requests ALL

## 2.1 Signing in

Open the address your IT gave you, e.g. `https://helfi.firma.local`. With Active Directory you sign in with your Windows account – as `m.muster`, `FIRMA\m.muster` or `m.muster@firma.local`. If single sign-on is set up, there is an additional button such as “Sign in with company account”, or you are redirected right away.



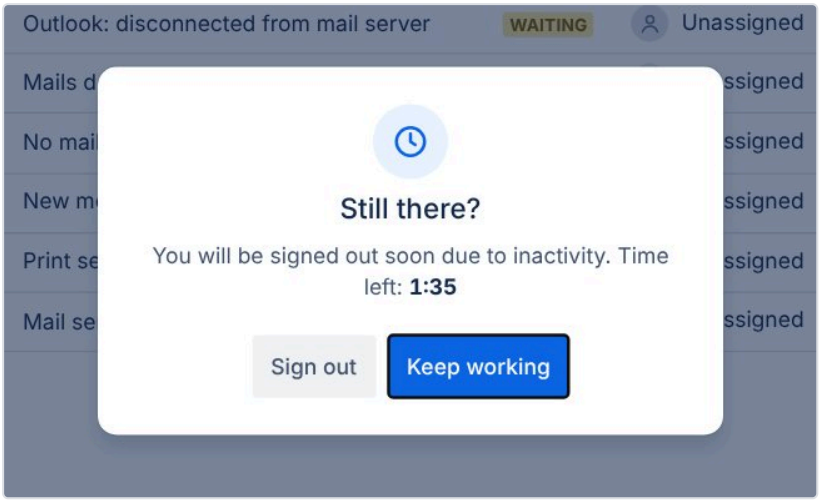

Sign-in page with a custom welcome text

**Without a company account:** if “Create account” appears below the sign-in, you can create an account yourself: enter name, email address and password and open the link in the confirmation email. If you have sent requests by email before, use the same address – they then appear in the account. You sign in with the email address. “Forgot password?” sends you a link to set a new password.

When you sign in **from outside the company network**, Helfi.Space may additionally ask for a 6-digit code from your authenticator app, depending on the settings (see 2.8).

**Automatic sign-out:** whoever does nothing for a while is signed out for security reasons. Two minutes before, Helfi.Space asks “Still there?” – *Keep working* simply continues. As long as you click or type,

you stay signed in.

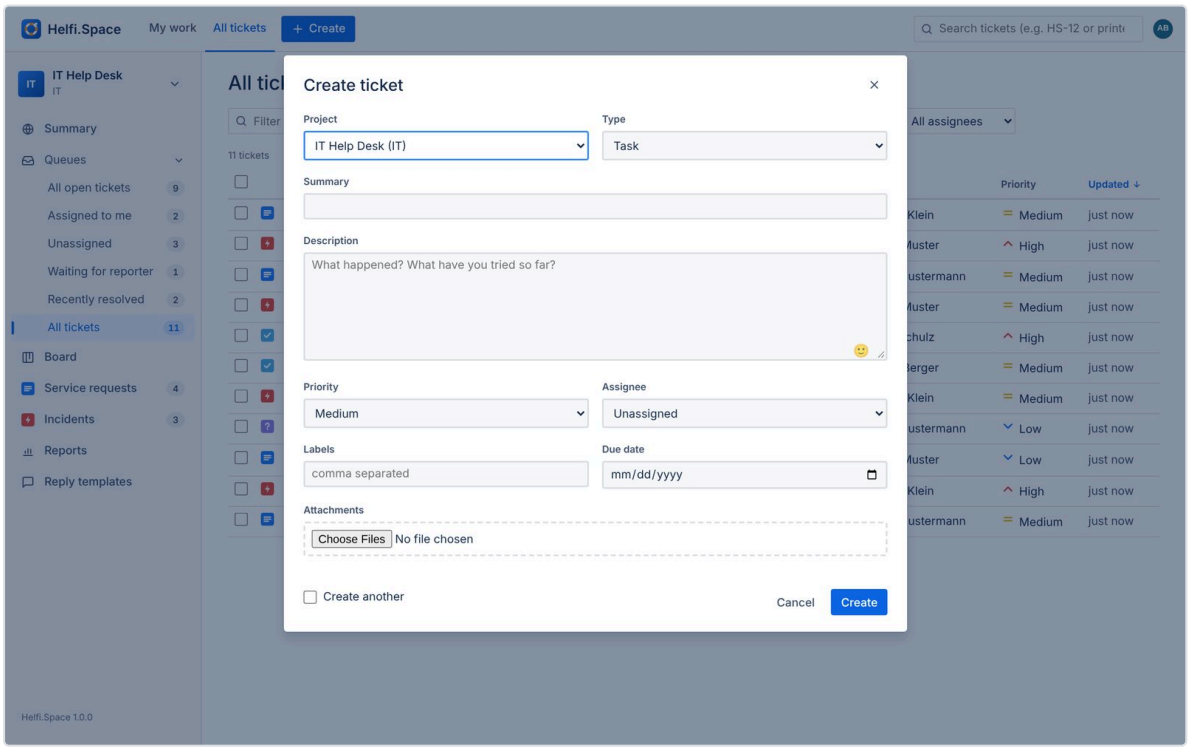



Two minutes before the automatic sign-out

## 2.2 Reporting a problem

There are two ways:

- **By email** to the help desk mailbox, e.g. `helpdesk@firma.local`. The subject becomes the summary, the text the description, attachments are included. People in *To* or *CC* automatically become participants. You receive a confirmation with the ticket key.
- **In the browser** via **Create** (if your IT allows it): enter summary, description, type and priority, optionally attach files.

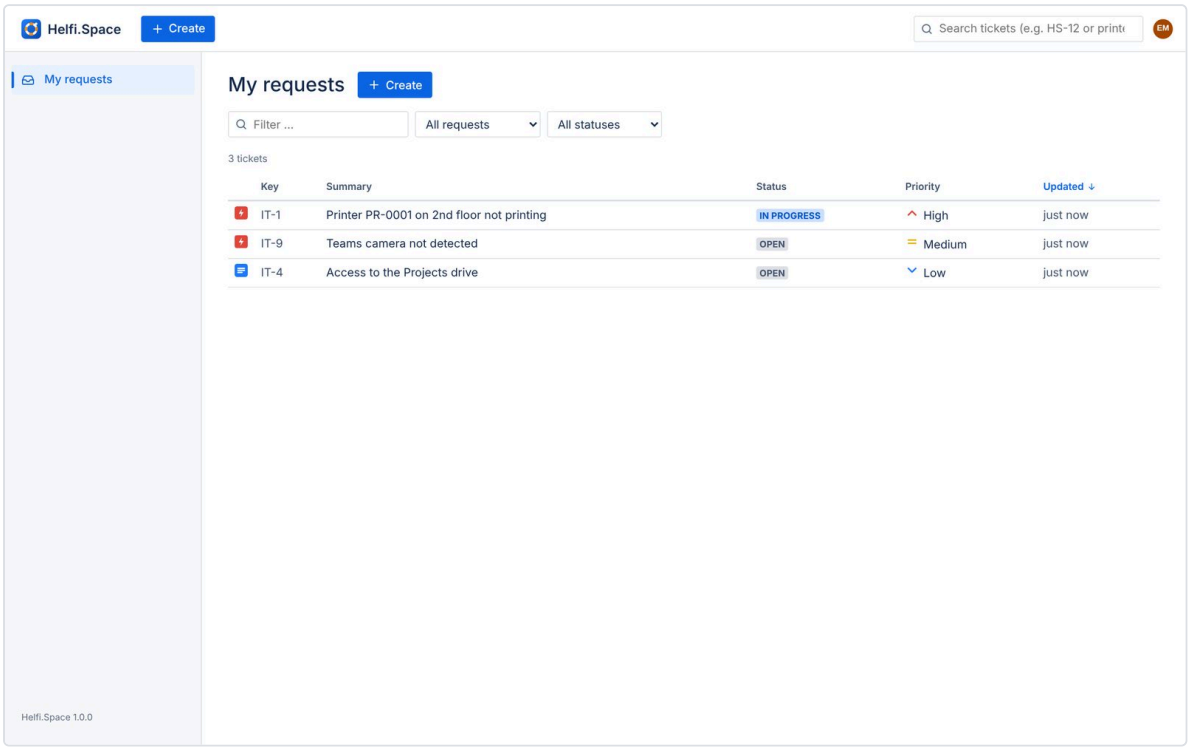



The “Create ticket” dialog

**Paste screenshots:** an image on the clipboard (e.g. taken with `Win`+`Shift`+`S`) can be pasted with `Ctrl`+`V` directly into the description or a comment.

## 2.3 My requests

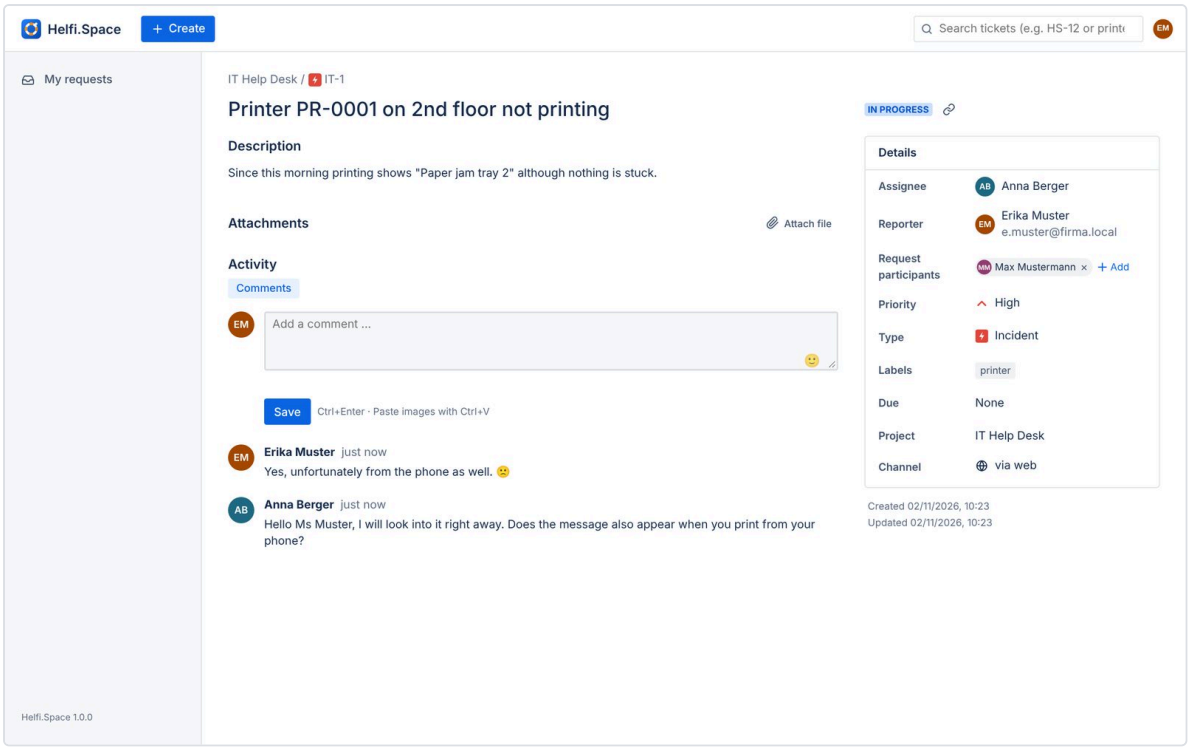
*My requests* lists all tickets you reported or participate in. Use the filters at the top to narrow them down by status or “I am a participant”.




My requests as seen by a user

## 2.4 Replies and files

Open a ticket and type into the comment field at the bottom; **Save** or `Ctrl`+`Enter` sends your reply to IT. Alternatively, simply reply to an email from Helfi.Space – the reply lands in the right ticket automatically and quoted history is cut off.




A ticket as seen by the reporter

Attach files under *Attachments* → *Attach file*. Images are shown directly, all other files are downloaded.

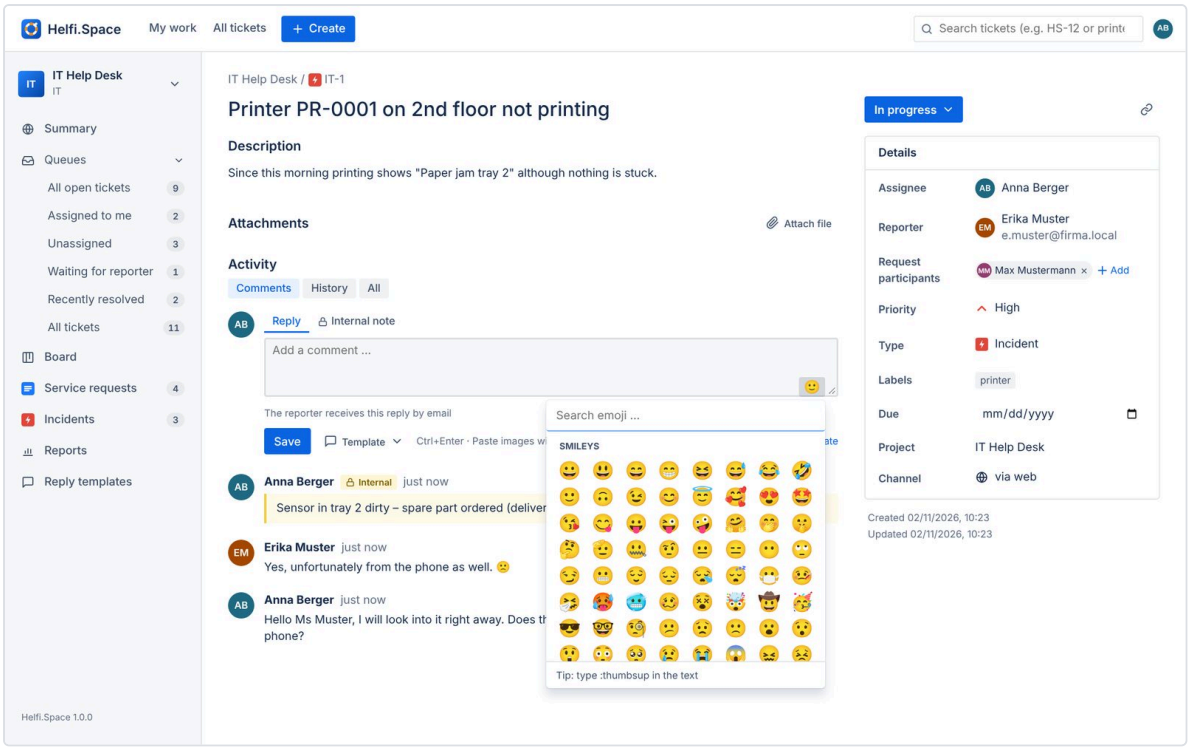
## 2.5 Request participants

Besides the reporter, further people can take part in a ticket, e.g. colleagues with the same problem. Under *Request participants* → *Add* you add colleagues who have an account (up to 20 per ticket): type a name and the suggestions appear below; pick with a click or `↑`/`↓` and `Enter`, `Esc` cancels.

- Participants see the ticket, can comment and attach files and receive the same emails as the reporter.
- They do not see internal notes and cannot change status, assignee or title. Of the reporter they only see the name, not the email address and logon name.
- Participants can remove themselves; the reporter and agents can remove them too.
- **Agents** can also enter any email address and find people directly in **Active Directory** (marked "AD"), by email address too. The person is taken over with name and email address from AD – they do not need to have signed in before.

## 2.6 Emojis

- 🙂 **button** in the toolbar of the comment field (in other text fields at the bottom right): picker with search (e.g. "thumbs" or "printer"), groups and "Recently used".
- **Short names:** `:thu` suggests 👍 (arrow keys and Enter or Tab to accept, Esc to close). `:tada:` turns into 🎉 right away.
- **Classic smileys** are converted as soon as a space follows: `:)` → 🙂, `;)` → 😉, `:D` → 😄, `:(` → 🙁, `:P` → 😛, `:O` → 😮, `<3` → ❤️, `(y)` → 👍. `Ctrl`+`Z` undoes the conversion right away.

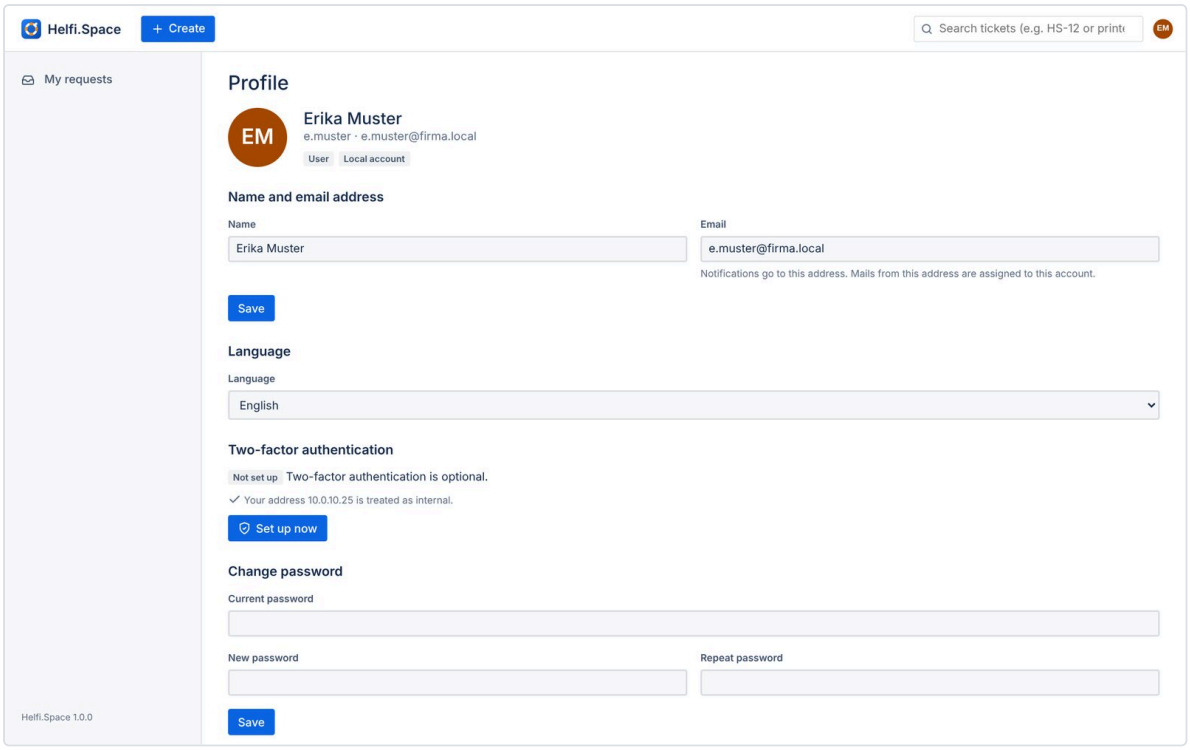



Emoji picker in the comment field

## 2.7 Profile, language and password

Click your picture at the top right → *Profile* to reach:

- **Name and email address** – editable for local accounts (a new email address requires your current password). For AD and SSO accounts both come from the company account.
- **Language** of the interface and of emails sent to you.
- **What's new** (also in the menu at the top right): what changed in each version. After an update Helfi.Space shows the news once by itself. Clicking the version number at the bottom of the sidebar also leads there, “Helfi.Space” next to it opens the website `helfi.space`.
- **Two-factor authentication** setup and management.
- **Change password** (local accounts only; change AD passwords in Windows as usual).




A user's profile

## 2.8 Setting up two-factor authentication

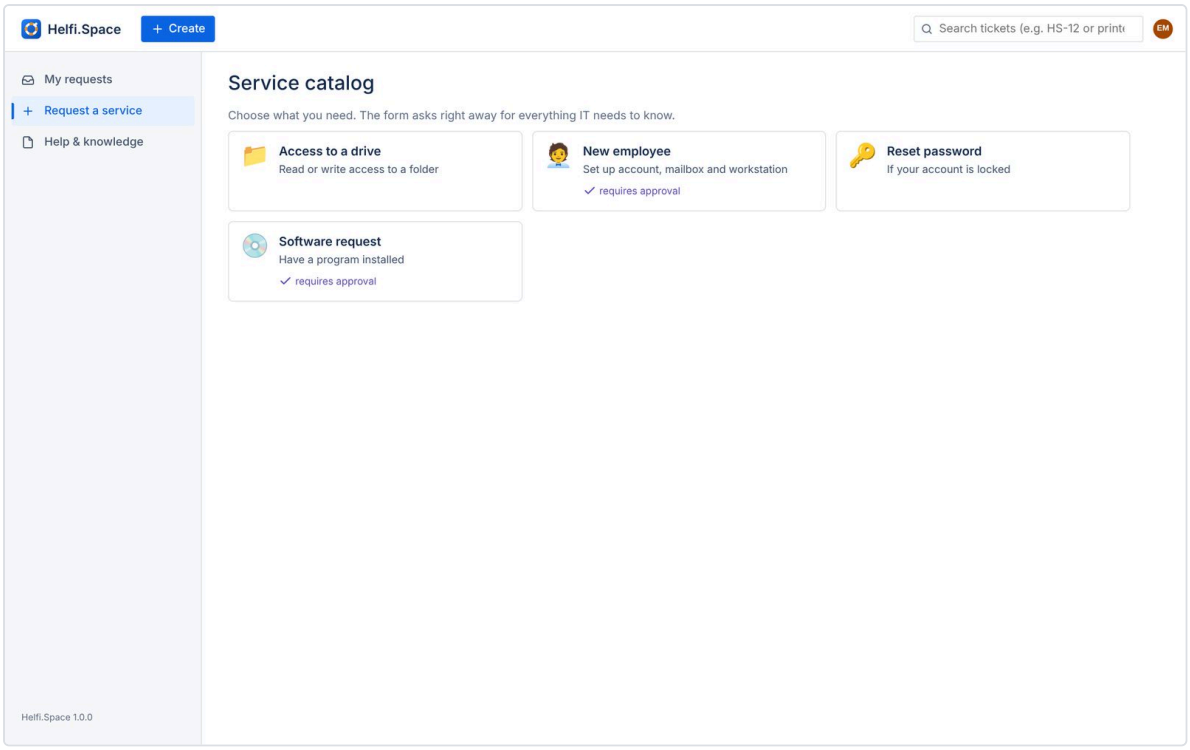
When your IT requires it (or voluntarily): *Profile → Two-factor authentication → Set up now*.

1. Open an authenticator app on your phone (e.g. Microsoft Authenticator, Google Authenticator, Aegis, 2FAS).
2. Scan the QR code.
3. Enter the 6-digit code shown and click **Activate**.
4. Keep the **10 recovery codes** safe (print them or store them in a password manager). Each works exactly once if your phone is not at hand.

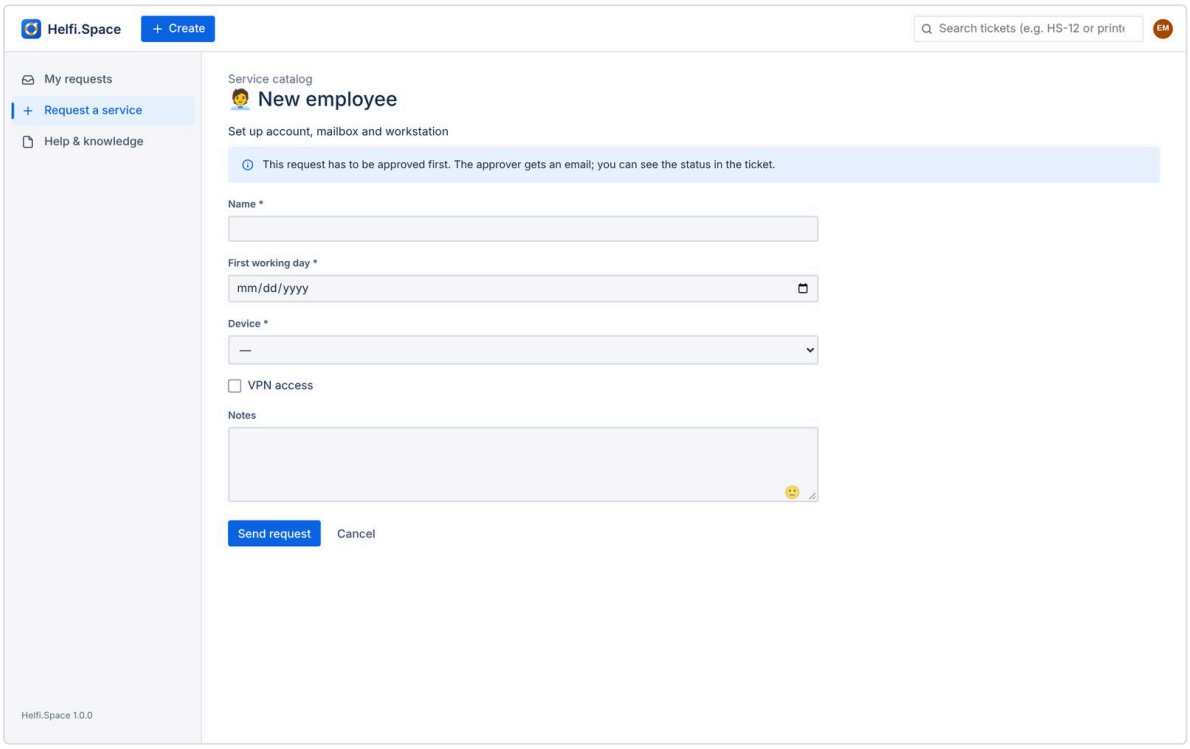
> For security reasons, first-time setup normally only works **inside the company network** (except after an SSO sign-in whose second factor was checked by the identity provider). This way nobody with a stolen password can register their own second factor from outside. **Lost your phone?** Sign in with a recovery code or ask IT to reset your 2FA.

## 2.9 Requesting a service

For frequent requests there are ready-made forms, e.g. "New employee" or "Software request": *Request a service* in the sidebar. Clicking a tile opens the form; required fields are marked with *. After sending you land in the new ticket.




Service catalog as a user sees it

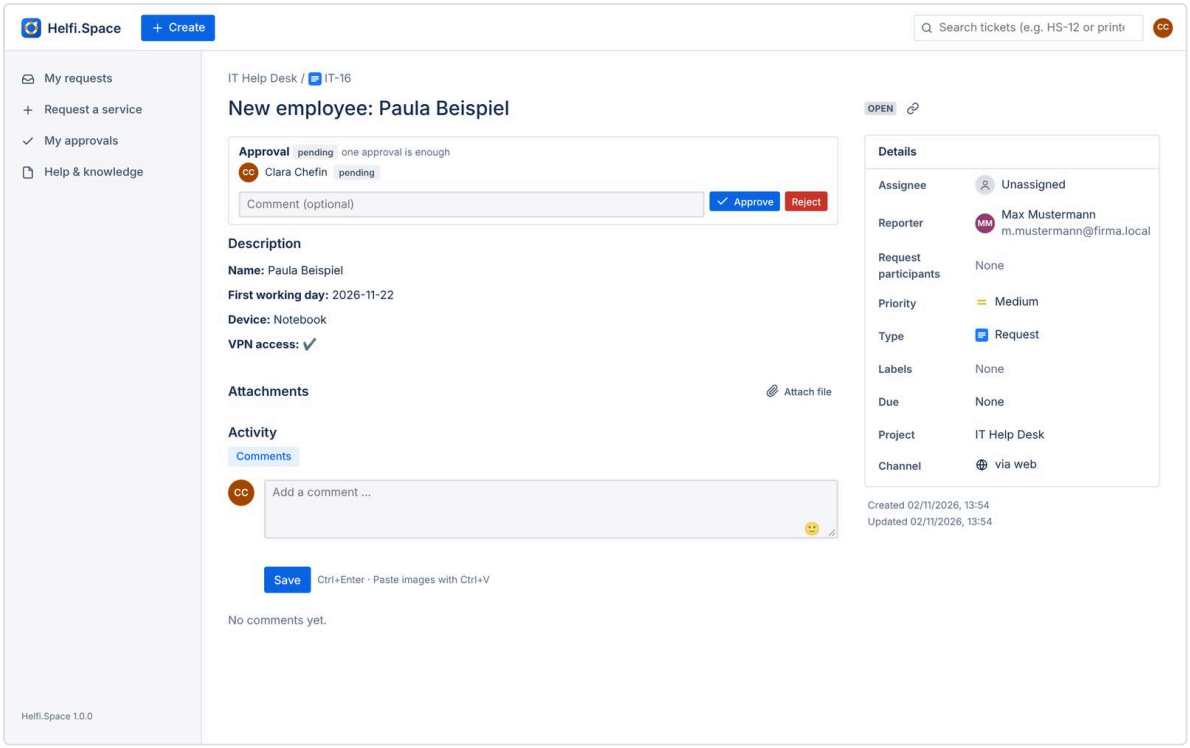



Form “New employee” – this request needs approval

If an offer says “requires approval”, someone has to agree first, e.g. your manager. You can see the status at the top of the ticket.

## 2.10 Approving

Approvers get an email and see *My approvals* in the sidebar. In the ticket one click on **Approve** or **Reject** is enough, optionally with a comment. The decision is sent to the reporter as a reply; a rejection closes the request. Approvers without the agent role only see the requests they decide on, and only the reporter's name.

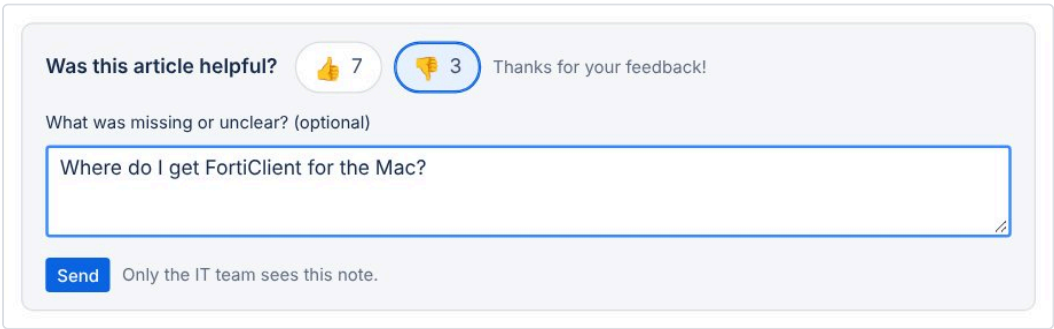



Approval in the ticket as the manager sees it

## 2.11 Help & knowledge

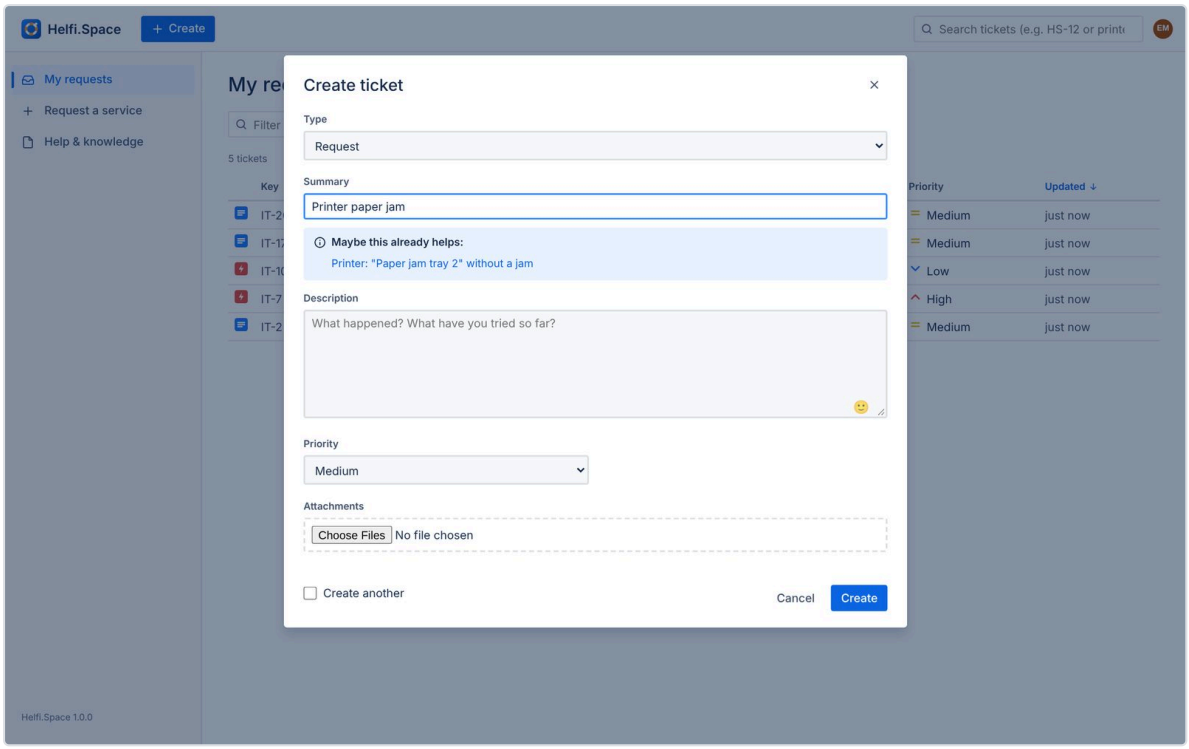
Under *Help & knowledge* you find guides from IT, e.g. "Set up VPN when working from home". While you report a problem, Helfi.Space already suggests matching articles ("Maybe this helps already") as soon as you type the summary. A click opens the article without losing your report.

**Was the article helpful?** Click 👍 or 👎 below the article – a second click withdraws the vote. After 👎 you can briefly say what was missing or unclear; only the IT team reads it.




"Not helpful" – with a short note for the IT team

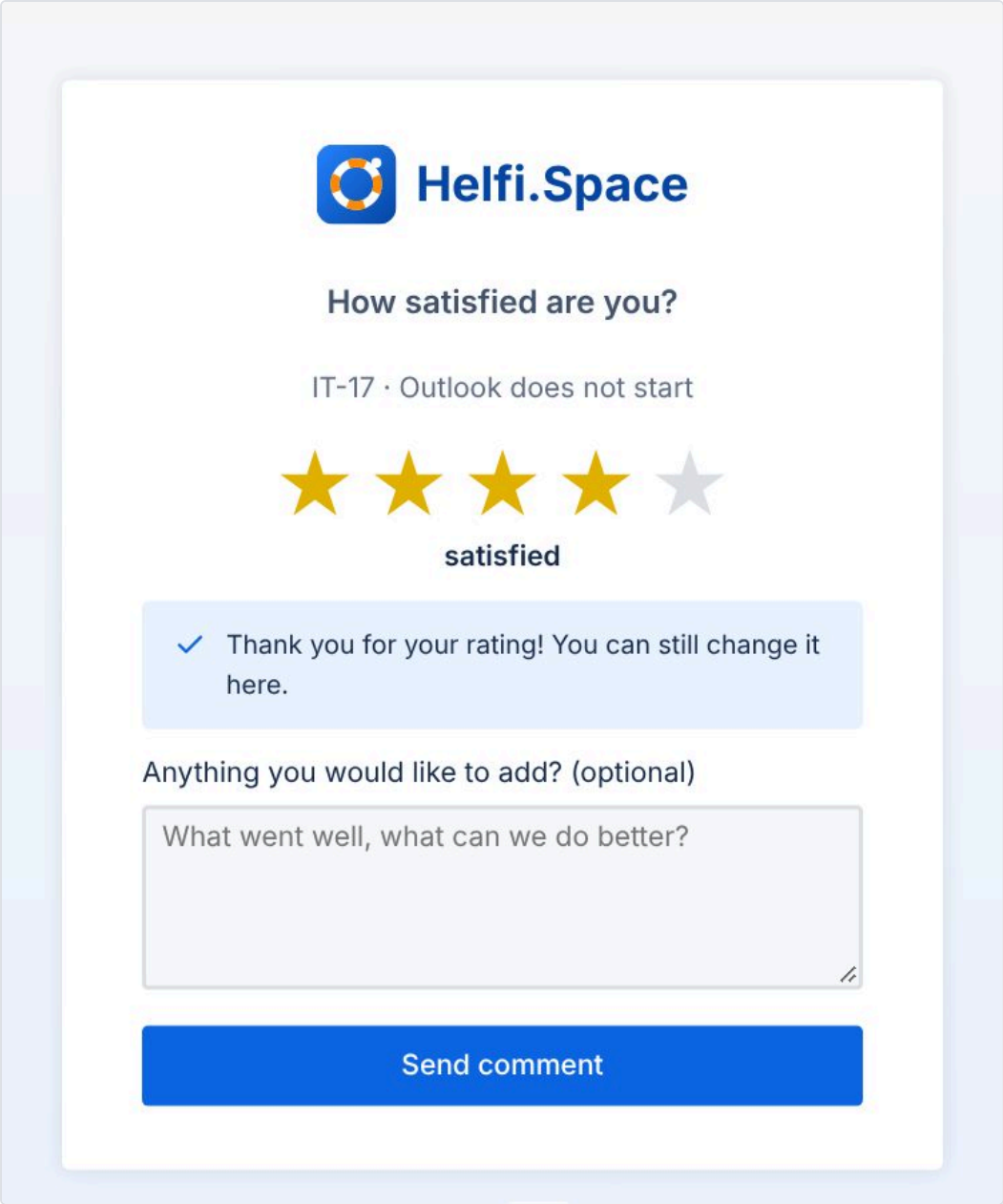
Below each article you can write a **comment** if something is missing or out of date – IT gets notified. **Print / PDF** prints the article or saves it as a PDF.




Knowledge suggestion while reporting

## 2.12 **Rating**

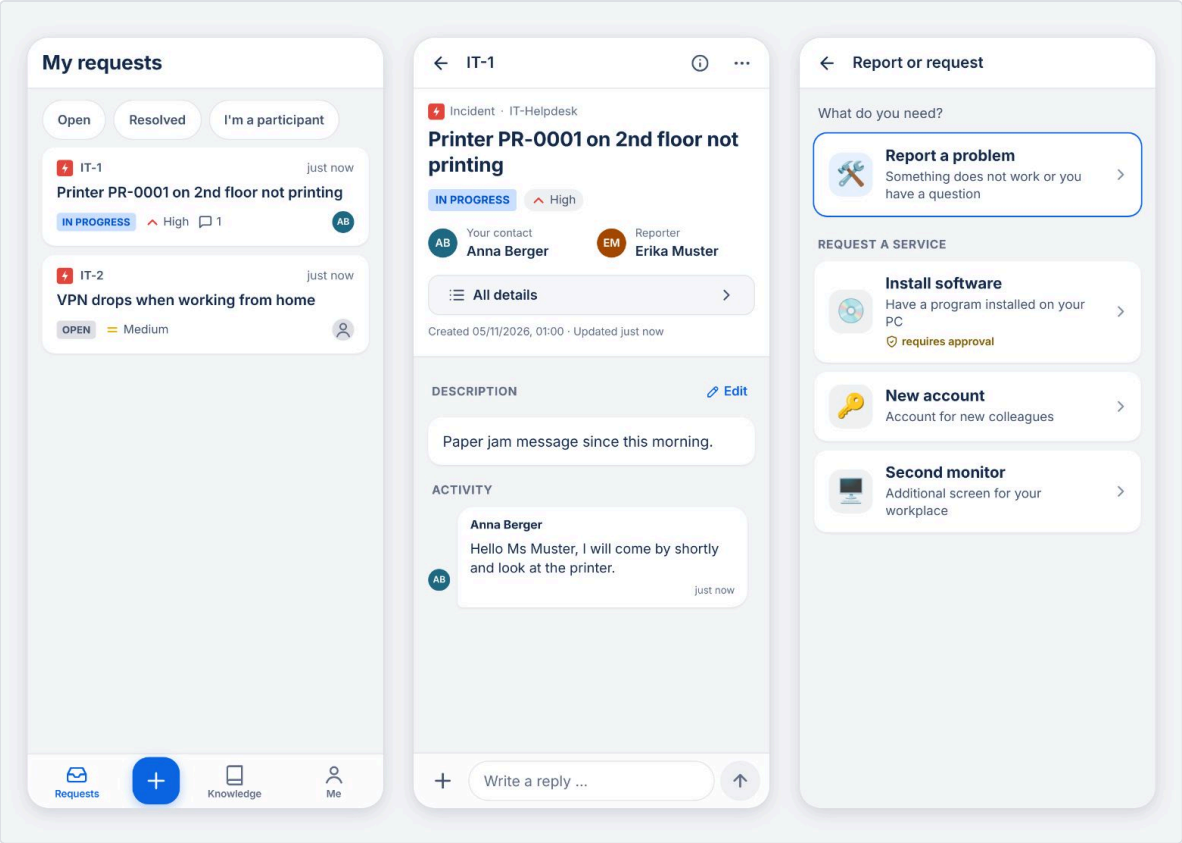
When your ticket is done, IT may ask you for a short rating. The “Resolved” email contains five links from “very satisfied” to “very dissatisfied”: one click is enough, no sign-in needed. On the page you can still change the stars and optionally write a comment. When signed in, you can also rate directly in the ticket.




Rating page after one click in the email

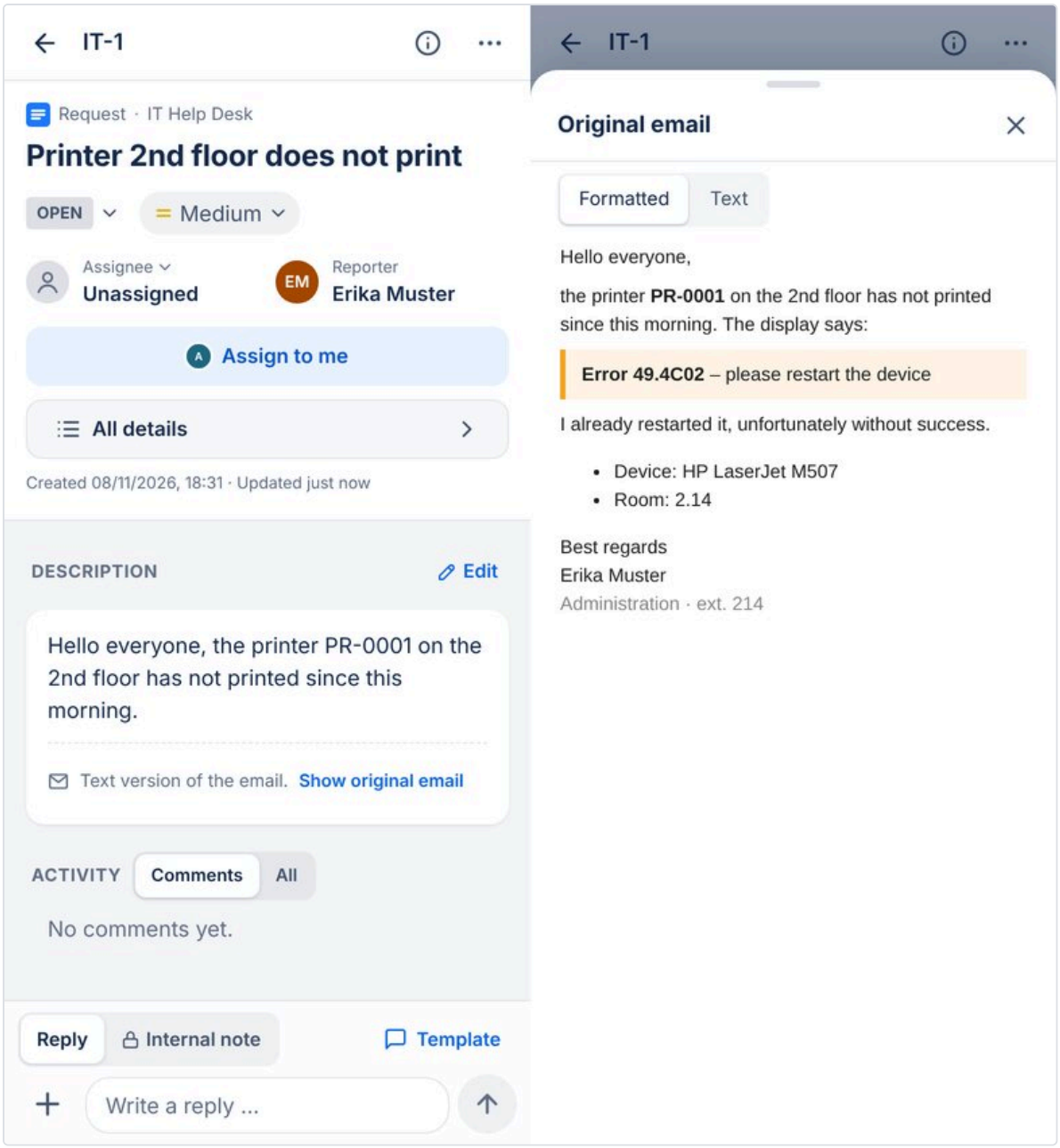
## 2.13 On a smartphone

On a phone Helfi.Space opens automatically in its own interface that works like an app. At the bottom there is a bar with **Requests** (agents: *My work* and *Tickets*), the round **+**, **Knowledge** and **Me**. Pages slide in from the right; go back with the arrow at the top left, a swipe from the left edge or the back key. Pull lists down to refresh them.




My requests, a ticket with the reply bar and the + with the service catalog

- **Report a problem:** + → *Report a problem*. Enter a title and description, use **Take photo** to open the camera or **File** to attach pictures and documents. While you type the title, Helfi.Space shows matching help articles – maybe the solution is already there. *Submit* creates the request and opens it. What you have not sent yet is kept as long as the tab is open.
- **Request a service:** under + the offerings of the service catalog appear as tiles. Required fields are marked with *; if an offering needs approval, the form says so.
- **Ticket:** the comments look like a chat. Write your reply at the bottom, attach photos or files with +. Agents add status changes and assignments with *All* next to “Activity”; the phone remembers the choice. With **@** in the reply bar agents mention colleagues; under *Tickets* the *Mentions* chip shows the tickets they were mentioned in. Change title and description of your own request with *Edit* next to "Description".
- **Knowledge:** search, categories, *Recently viewed*, *Most read* and the page tree. Long articles have a *Contents* list at the top, wide tables scroll sideways. Below each article you can write a comment.
- **Me:** name, email address and password (local accounts), language, theme (light/dark), two-factor authentication, *Install app*, what's new and – only in the phone browser – *Desktop view*. The user menu of the desktop view (*Phone view*) brings you back.
- **Original email:** if a request or reply came by email, it says “Text version of the email. *Show original email*” below it. This opens the email as it was written in a window from the bottom – with *Formatted*/*Text* to switch. Swipe down, × or back closes it.




Original email on the phone: note below the description and the email in its window

**Images** – attached photos and pictures in the description, replies or knowledge articles – also open in a window inside the app, not on a new page. A tap zooms to the original size (to pan around), *Download* saves the file; back, ✕ or swiping down closes it.

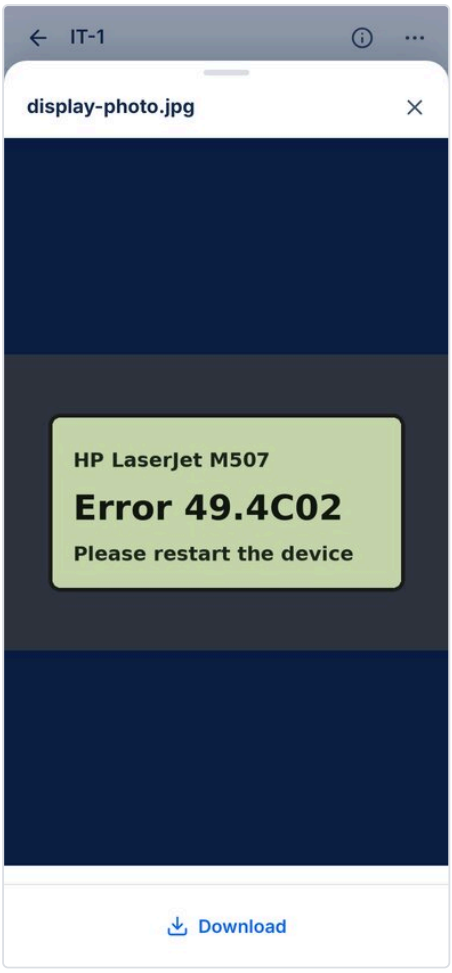
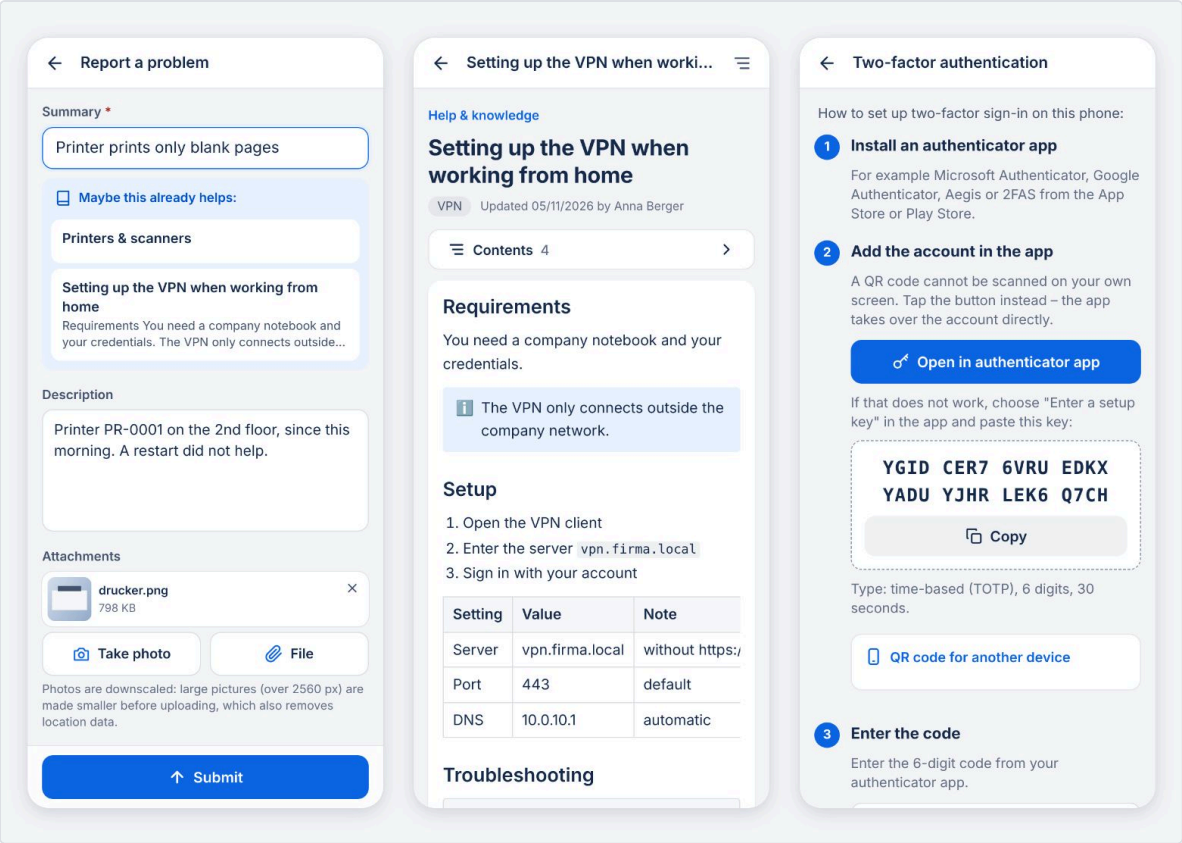



Image in the app's window




Reporting a problem with a photo and knowledge suggestions, a help article and setting up two-factor authentication on the phone

**Setting up two-factor authentication on the phone:** a phone cannot scan a QR code on its own screen. So under *Me → Two-factor authentication → Set up now* tap **Open in authenticator app** – the app takes over the account directly. If that does not work, copy the key shown and paste it in the app under "Enter a setup key". The QR code is still there for a second device. Then enter the 6-digit code and copy, share or save the recovery codes. If 2FA is required for your account, this works the same way right during sign-in.

**Install as an app:** *iPhone/iPad:* open Helfi.Space in Safari, tap *Share* at the bottom, *Add to Home Screen → Add*. *Android:* in Chrome tap ⋮ at the top right → *Install app* (or *Me → Install app → Install now*). Helfi.Space then starts full screen with its own icon and updates itself; after an update "New version – Reload" appears briefly. Without a connection the app shows a notice with *Try again*. Installed, Helfi.Space always stays in the phone view – links to desktop pages lead back into the app too, because there is no address bar and no back button there.

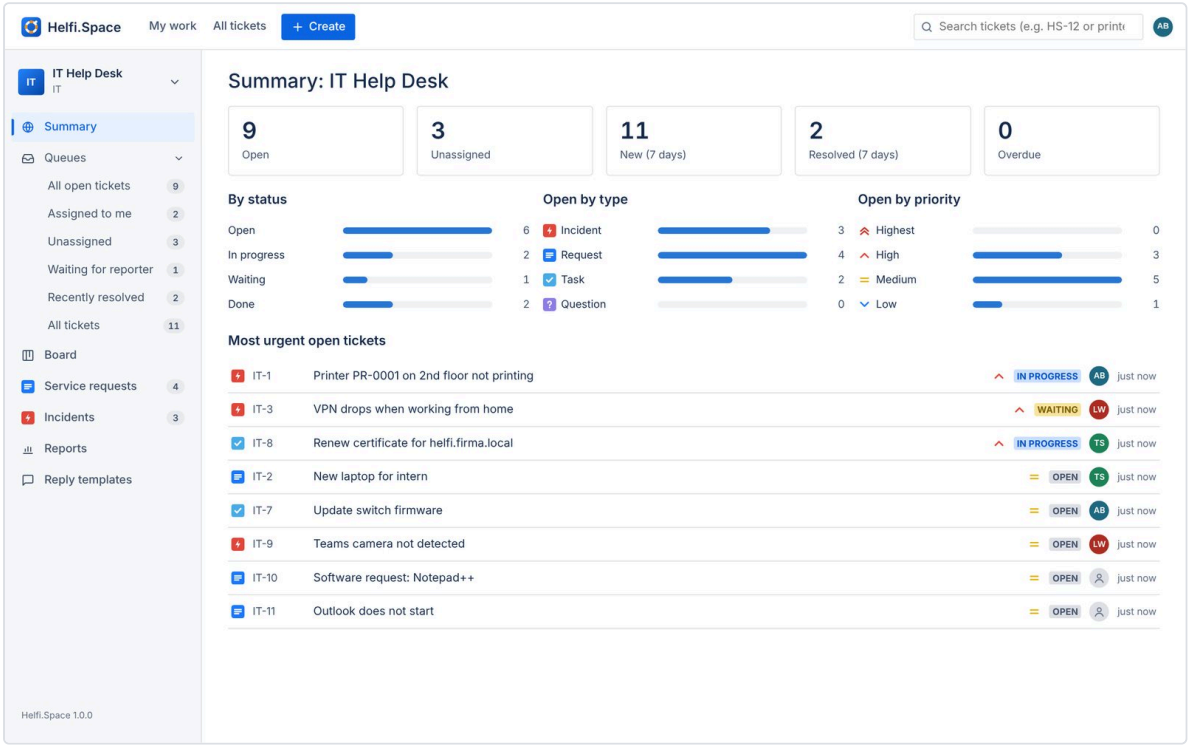
**Privacy:** photos are uploaded without the location where they were taken, and large photos are downscaled on the phone already. Unsent drafts are deleted when you sign out. Tickets, attachments and replies are not stored on the device, only the program files. Nothing is loaded from the internet.

# 3 For agents AGENTS

## 3.1 Queues and summary

For the selected project, the sidebar shows fixed **queues** with live counts: *All open tickets*, *Assigned to me*, *Unassigned*, *Waiting for reporter*, *Recently resolved* (7 days) and *All tickets*. Below them *Service requests* and *Incidents* (open requests and incidents) and *Mentions me* (see Mentioning colleagues). Switch the project at the top of the sidebar.

The **summary** shows key figures, distributions by status, type and priority and the most urgent open tickets.




Project summary

*My work* in the header collects everything assigned to you, everything unassigned and everything you reported yourself.

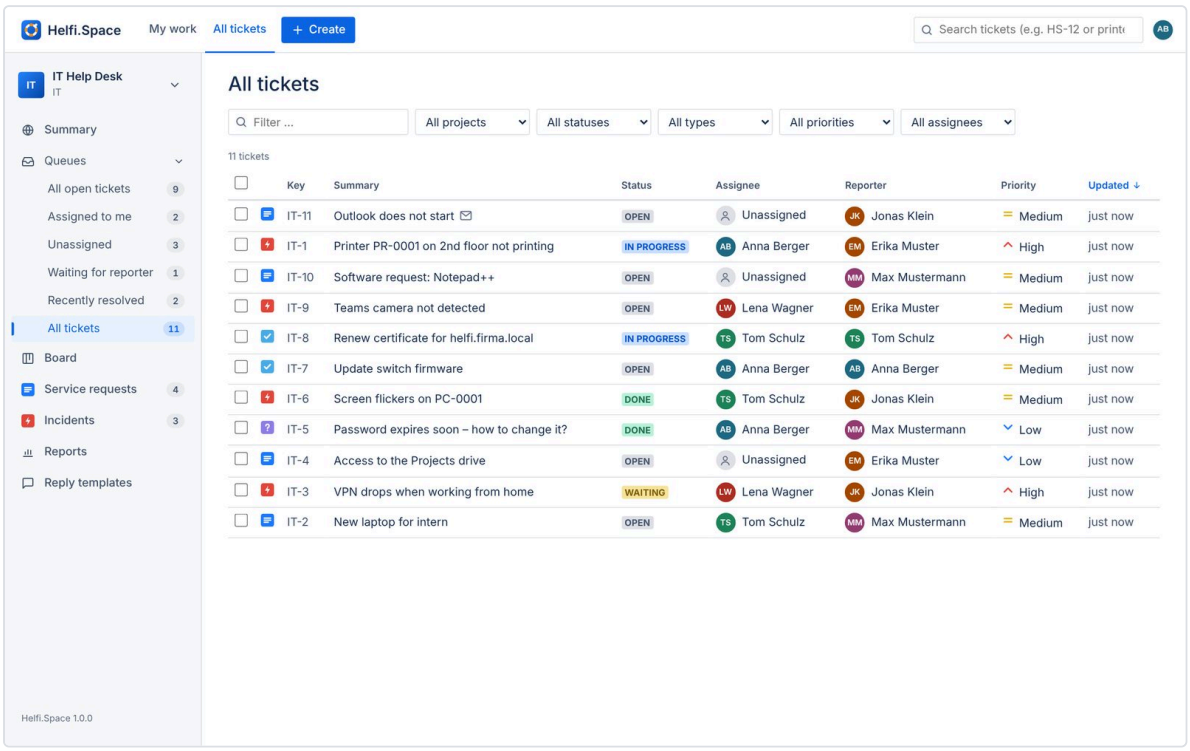
## 3.2 Board

The board shows the tickets in four columns by status. Drag a card into another column to change its status, or within a column to set the order. At the top you can filter by text, assignee (avatars), type or “Only mine”.

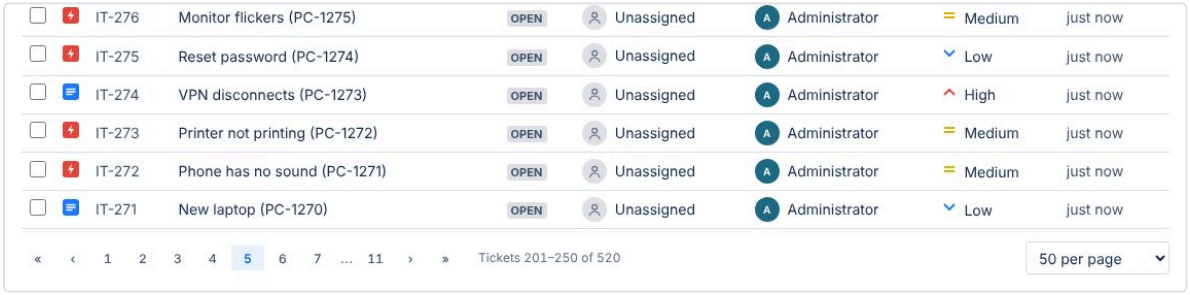
## 3.3 Ticket list and filters

*All tickets* shows a sortable list. Filters for project, status, type, priority, assignee and – with a project selected – component can be combined; clicking a column header sorts. The page address contains the filters – bookmark or share it.

**Pages:** long lists (search results too) are shown page by page. Above the table you see e.g. “Tickets 101–200 of 7956”, below it you turn pages with `« ‹ 1 … 5 6 7 … 80 › »`. Next to it you choose 50, 100 (default), 250 or 500 tickets per page – the browser remembers the choice. The page is part of the address ( `?page=3` ), so the back button and bookmarks work; another filter, a new search or sorting starts on page 1 again. *Select all* in the multiple selection takes the tickets of the current page. A search with more than 5000 matches shows the best 5000 – narrow it down then. On phones the list loads the next page while scrolling, with no upper limit.



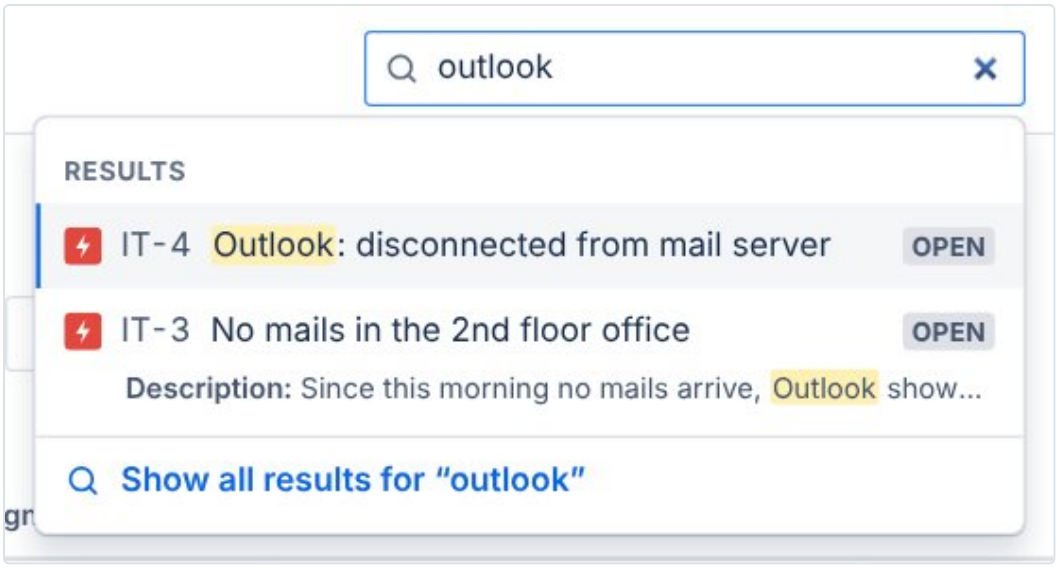

Ticket list with filters




Pages below the list with the number per page

## Search

The search box at the top searches the full text of all tickets: title, description, replies, labels and attachment names. Agents also find internal notes and the people involved (also by email address and logon name). Users without the agent role find tickets by title, description, public replies, attachment names, labels and the names visible in the ticket – not by email addresses or logon names.

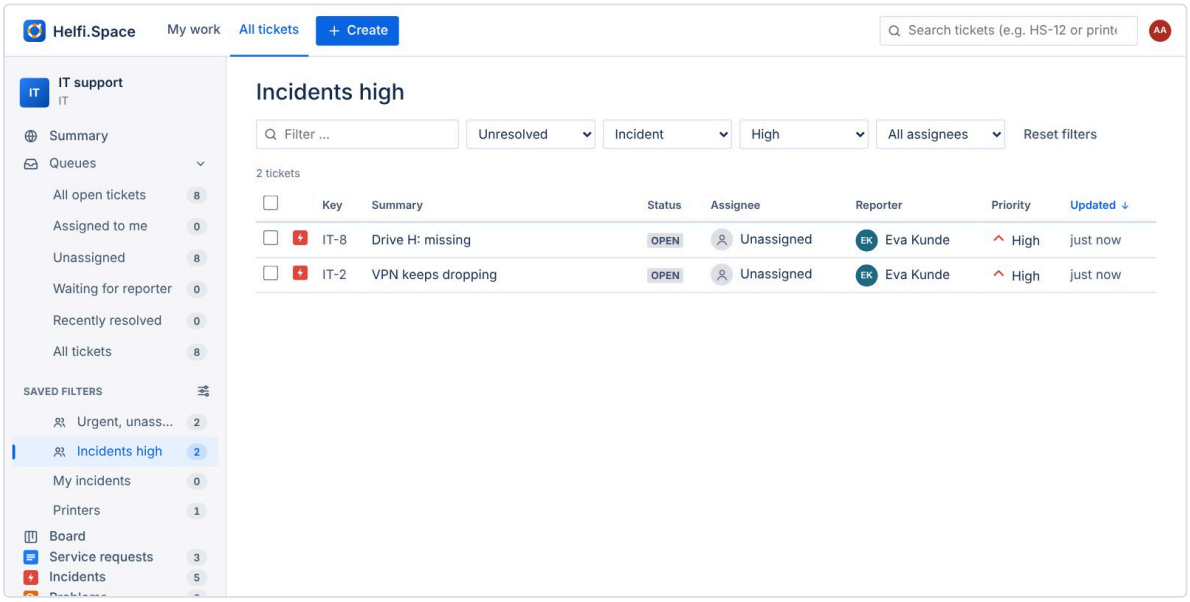



Results while typing – with where they matched; ↑/↓ and Enter open the ticket

- **Results while typing:** from two characters up to eight tickets appear with key, title, status and where they matched. Open a ticket directly with ↑/↓ and Enter or a click. Enter without a choice or *Show all results* opens the ticket list with all results. The empty field shows the recently opened tickets. On phones the *Tickets* tab works the same way.
- **Word parts** match: “printer” also finds “networkprinter”.
- **Several words** must all appear, in any order. `"exact phrase"` in quotes, `-word` excludes.
- Results are sorted by **relevance**, title matches first. Each result shows where it matched with the words highlighted.
- A ticket number jumps straight to the ticket.

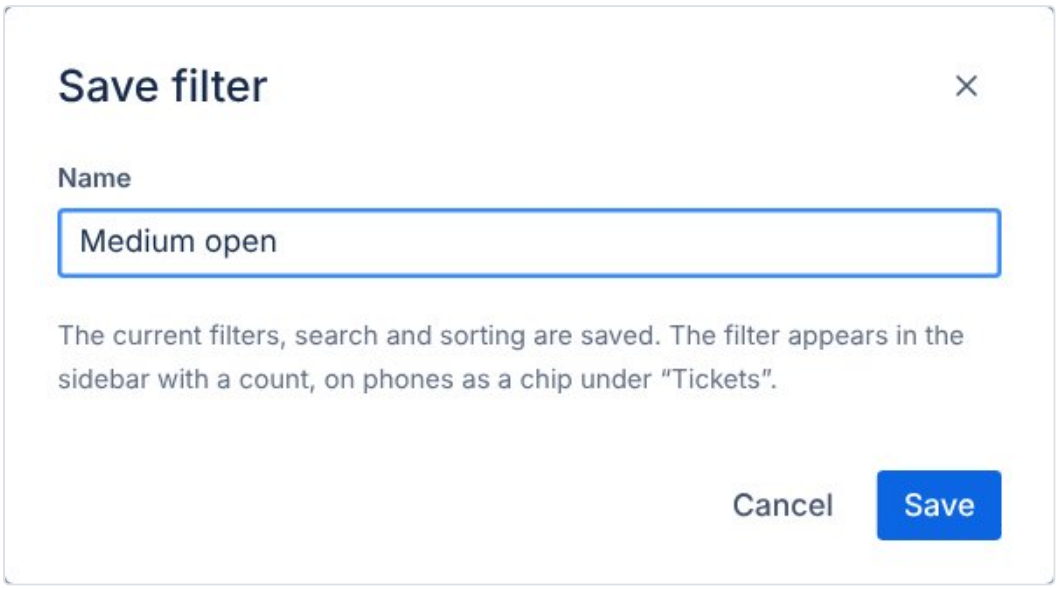
## 3.4 Saved filters and custom queues

Set filters, search and sorting in the ticket list and choose **Save filter**. The filter appears on the left under *Saved filters* with its current count; the list then carries its name.

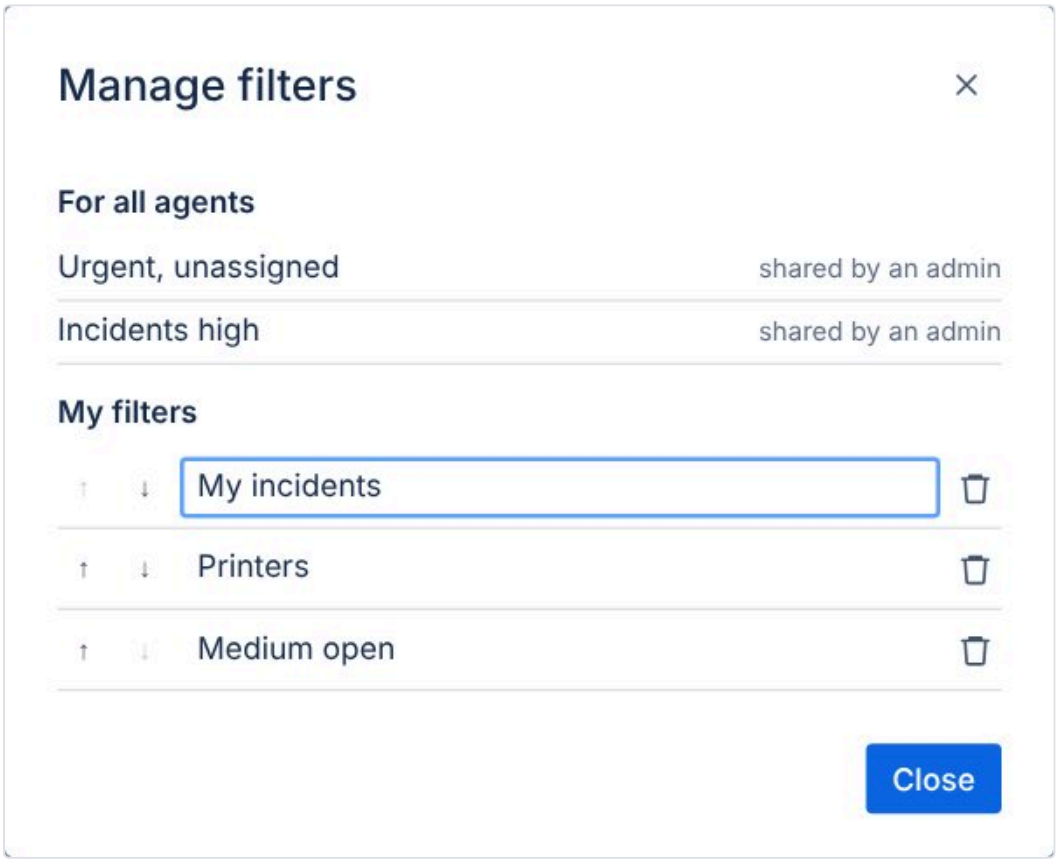



Saved filters in the sidebar – the people icon marks those shared with everyone

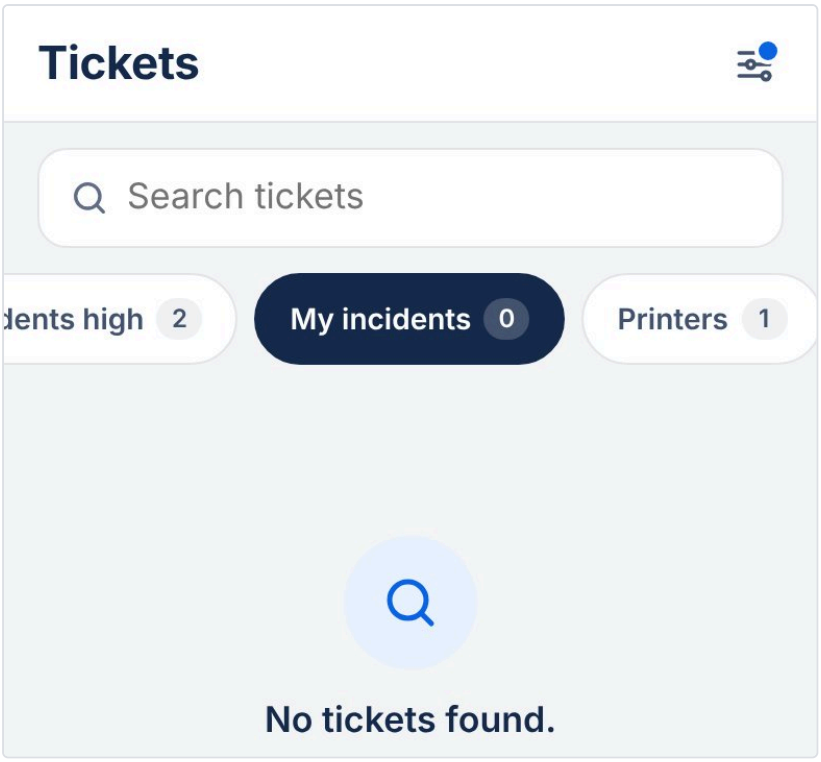
- **Personal** filters are visible to you only (up to 50).
- ADMIN **Share with all agents:** tick it when saving – the filter appears as a queue for every agent. “Assigned to me” then refers to each person, so a “My urgent ones” queue fits everyone.
- **Managing:** the icon next to *Saved filters* opens the list to rename, sort (↑/↓) and delete. Only admins change shared queues.
- **On phones** the filters are chips with a count after the queues in the *Tickets* tab.




Saving a filter (admins also see “Share with all agents”)

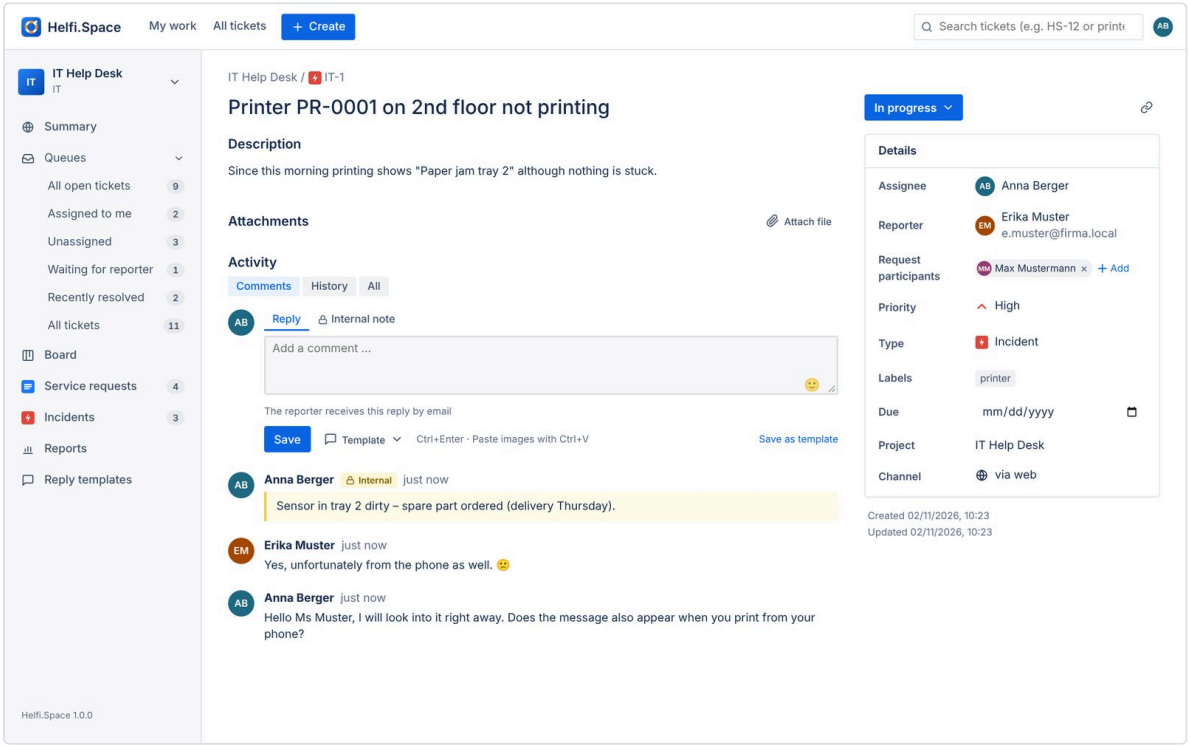



Managing filters




On a phone as a chip

## 3.5 Working on a ticket




Ticket view: description and activity on the left, status and details on the right

- **Status** via the button at the top right (e.g. “Open” → “In progress”).
- Change **assignee**, **priority**, **type**, **component**, **labels**, **due date** and **project** directly in the details. “Assign to me” takes one click or the `i` key.

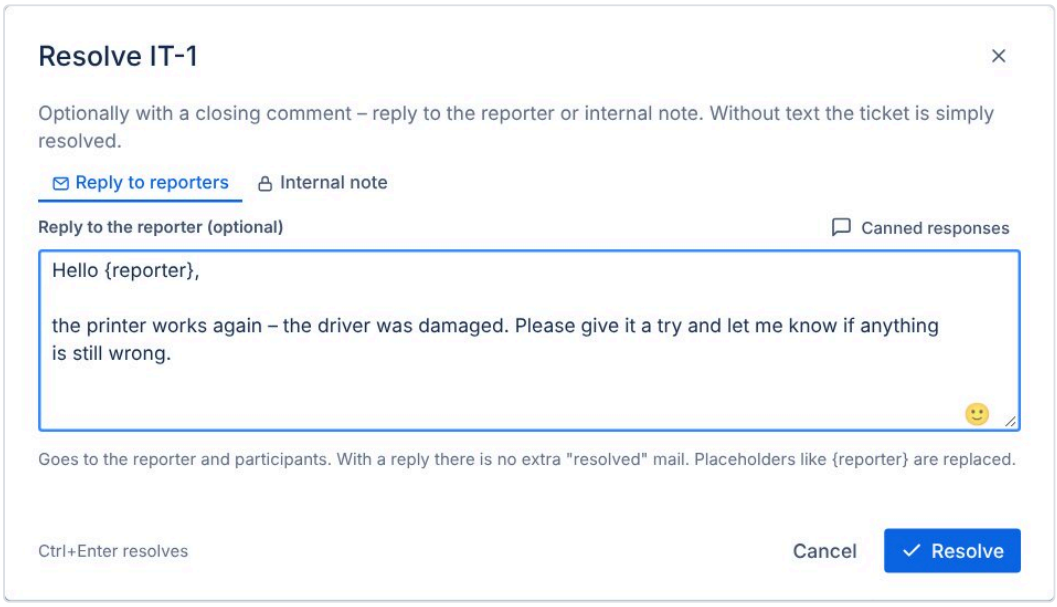
- Click **title** and **description** to edit them.
- **Changing the project:** the ticket gets a new key in the new project; the old one stays in the history. Replies to old emails are still assigned correctly.
- **Activity:** tabs *Comments*, *History* (every change with person and time) and *All*. For a changed description, **Show change** shows the differences and the previous version.
- **Copy link** via the chain icon at the top right.

## 3.6 Resolving with a closing comment

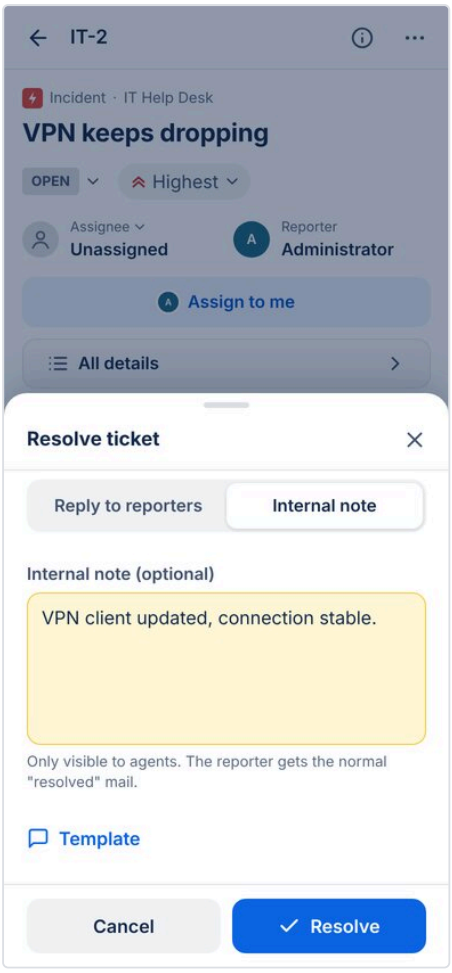
When you choose *Done* in the ticket's **status menu**, a window opens for a **closing comment** – write the comment and close the ticket in one step:

- **Reply to reporter** (default): goes to the reporter and participants. There is then no extra "Resolved" mail – your reply already informs. Reply templates and placeholders like `{reporter}` work.
- **Internal note**: only visible to agents; the reporter gets the normal "Resolved" mail.
- Clicking *Resolve* without text simply closes the ticket. `Ctrl`+`Enter` in the text field resolves right away, *Cancel* leaves everything as it was.

Several tickets are resolved via the multiple selection in the ticket list (*Resolve*), on phones via *Status → Done* – both also with a reply or an internal note.




Resolving with a reply to the reporter

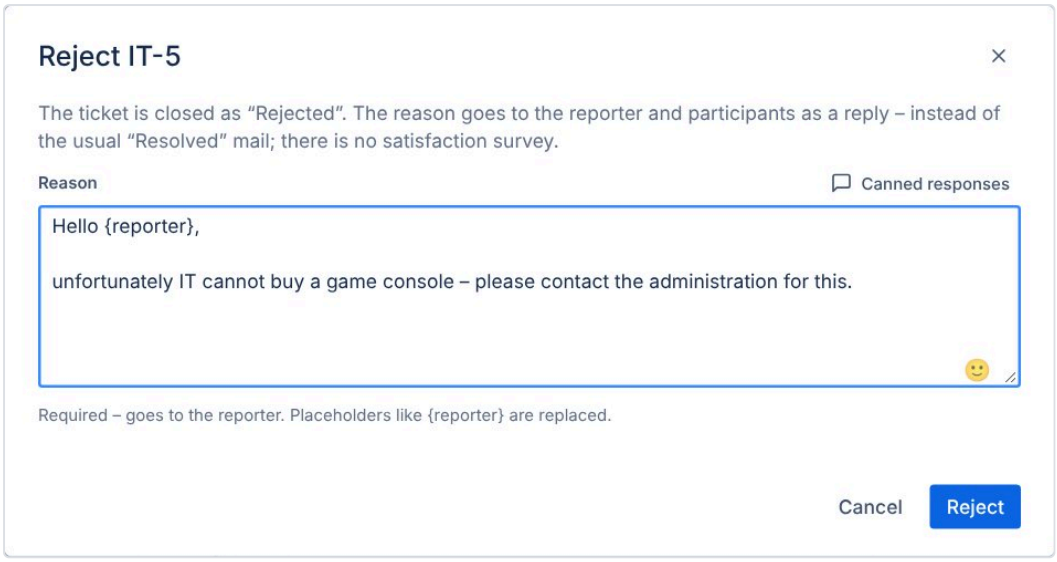



On the phone: resolving with an internal note

## 3.7 Rejecting and duplicates

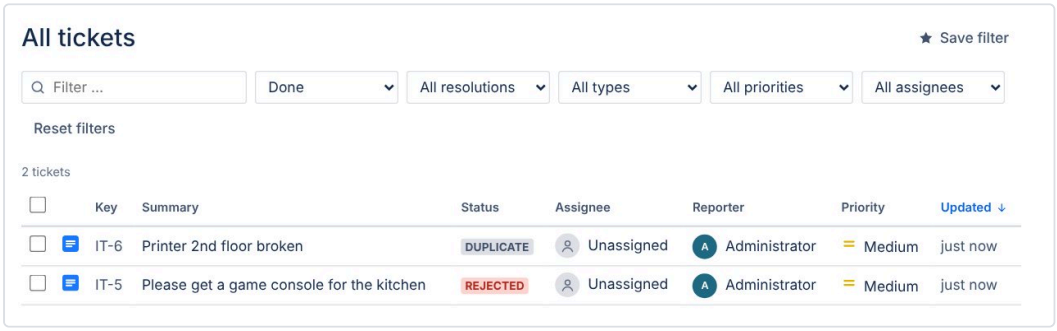
Not every ticket ends with a solution. Besides *Done* there are therefore two more ways to close – both in the ticket's **status menu** (on phones under *Status*):

- **Reject …** – the request is deliberately not carried out. The **reason** is required and goes to the reporter and participants as a reply; reply templates and placeholders like `{reporter}` work. No extra “Resolved” mail and no satisfaction survey is sent.
- **Close as duplicate …** – the request is already handled in another ticket. Choose the *original ticket*; both are linked, and the reporter gets a short note with the original's number (or your own text). With *Add reporter and participants to the original* they keep following the original – only if it contains nothing confidential.




Rejecting a ticket with a reason

The ticket then shows *Rejected* or *Duplicate* instead of *Done* – in lists, under *My requests* and on phones –, duplicates with a link to the original. In the ticket list you filter by *Resolution* once the status *Done* is chosen; with the multiple selection you reject several tickets at once. A reply mail to it does not reopen the ticket but becomes an internal note – reopen it yourself if needed. If the reporter writes in the web interface, it reopens as usual. In the reports, rejected tickets and duplicates are counted separately and not towards the avg. resolution time.

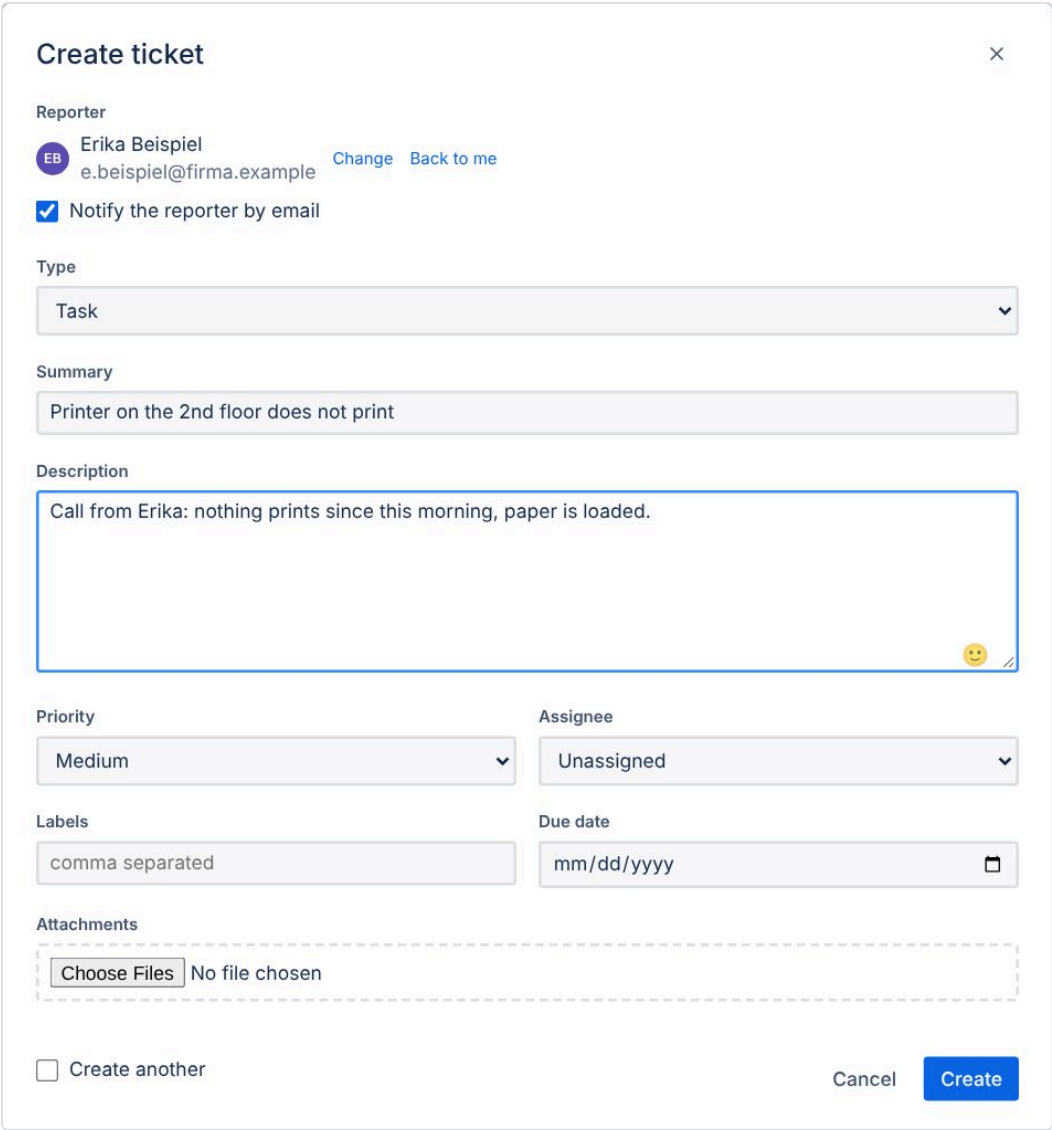



Done tickets with their resolution

## 3.8 Entering tickets for others

When a request comes in by phone, in passing or to another mailbox, enter the ticket right away on behalf of that person. It then appears under their *My requests* – so they get used to the ticket system.

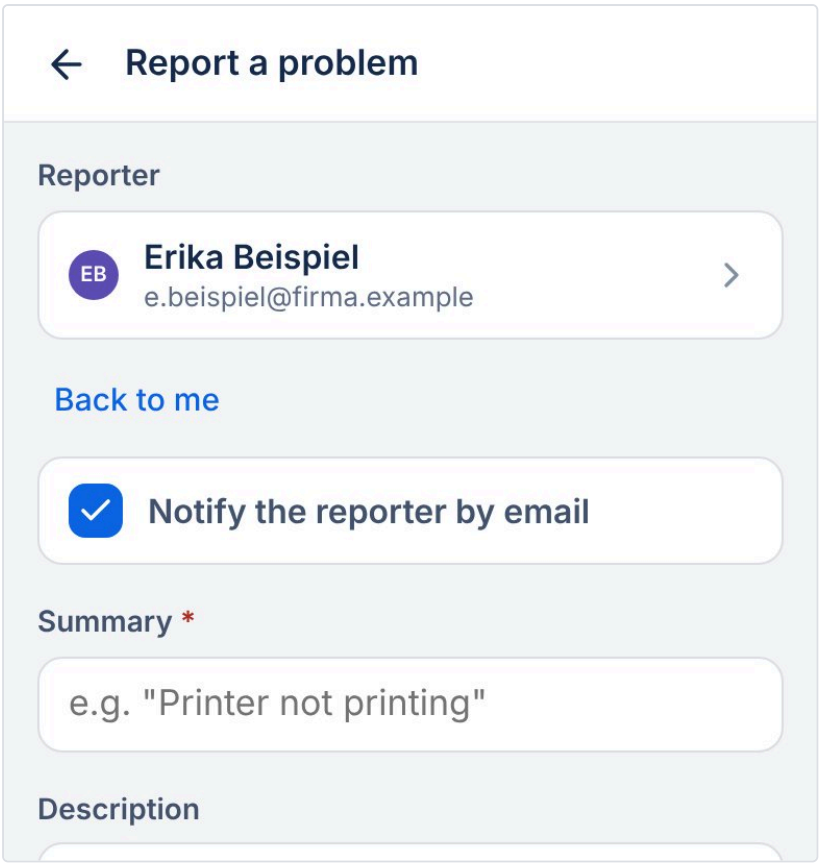
1. Open **Create**. The **reporter** is shown at the top, “Me” by default.
2. Click *Enter for someone else* and type a name, user name or email address. From 3 characters, people from AD appear as well (marked “AD”). An email address without an account is created as an email contact.
3. **Notify the reporter by email** is ticked: the person gets an email with the ticket key and link – with an account also a hint that they can report new requests themselves next time.
4. Paste the text from the email or note and click **Create**.




Entering a ticket for a colleague

The history shows who created the ticket for whom. *Back to me* undoes the choice.

**Change later:** in the details under *Reporter* click *Change reporter*, pick the person and confirm. The previous reporter keeps access as a request participant (except agents and pure email contacts). If a former reporter later replies to an old email, it only becomes an internal note. The same applies to mails from agent addresses matched only via the ticket number in the subject – reply to the reporter in the ticket. On phones: *All details → Reporter*.

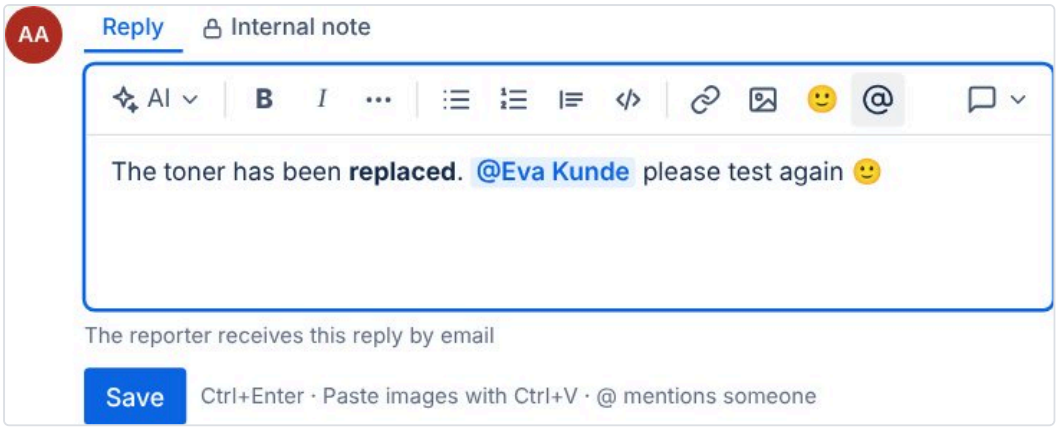



On phones: choosing the reporter when creating

## 3.9 Formatting replies

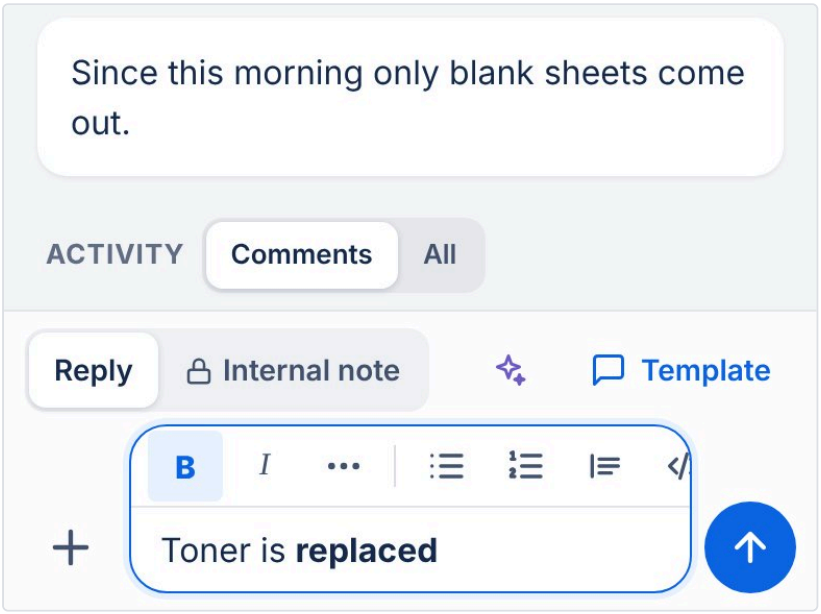
Above the comment field there is a toolbar – in tickets, in comments on knowledge articles and on phones:

- **AI** (if set up): suggest a reply, polish, shorten, translate.
- **Formatting:** bold (`Ctrl`+`B`), italic (`Ctrl`+`I`), under … underline, strikethrough and code; bullets, numbered list, quote and code block. You see the formatting right away while writing. Shortcuts work too: “- ” or “1. ” at the start of a line starts a list, “> ” a quote.
- **Link** (`Ctrl`+`K`; also ticket numbers such as IT-42 or KB-3), **image** (or Ctrl+V), **emoji** (also “:)” or “:thumbsup:”) and **@** to mention someone.
- On the right: **Canned responses** to insert – *Save as template* is in the same menu.
- `Ctrl`+`Enter` saves.




Comment field with toolbar: bold, mention and emoji

Reporters and participants get the reply formatted by email (with a readable text part for programs without HTML). On phones the bar appears above the field as soon as you write and can be swiped sideways.




On a phone: format bar while writing

## 3.10 Replies and internal notes

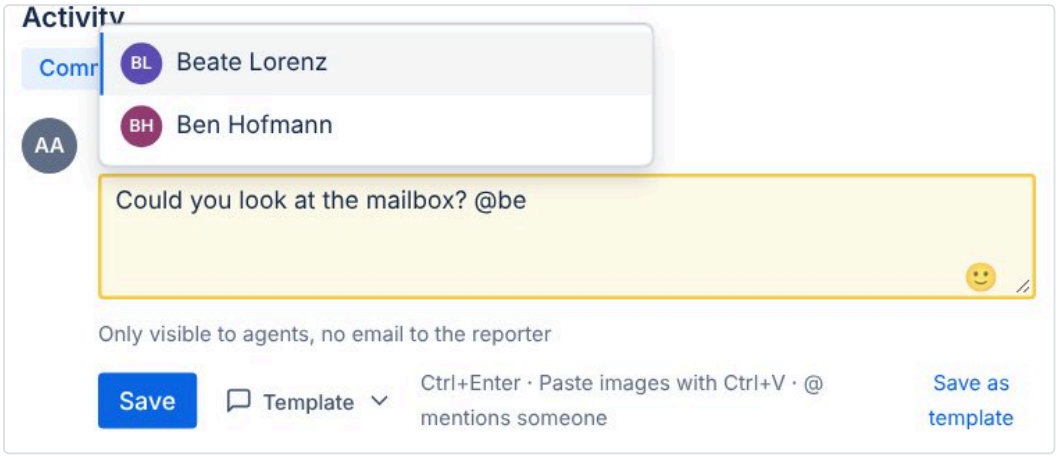
Above the comment field choose between **Reply** and **Internal note**:

- **Replies** are emailed to the reporter and all participants, each in their own language.
- **Internal notes** (yellow background) are only visible to agents. They are never sent by email, and pasted images stay internal as well.

> Emails coming from an agent's address never count as agent replies (sender addresses can be forged). Agents therefore always reply via the web interface. Email replies to assignment, SLA or approval mails sent to agents become internal notes and do not reopen the ticket.

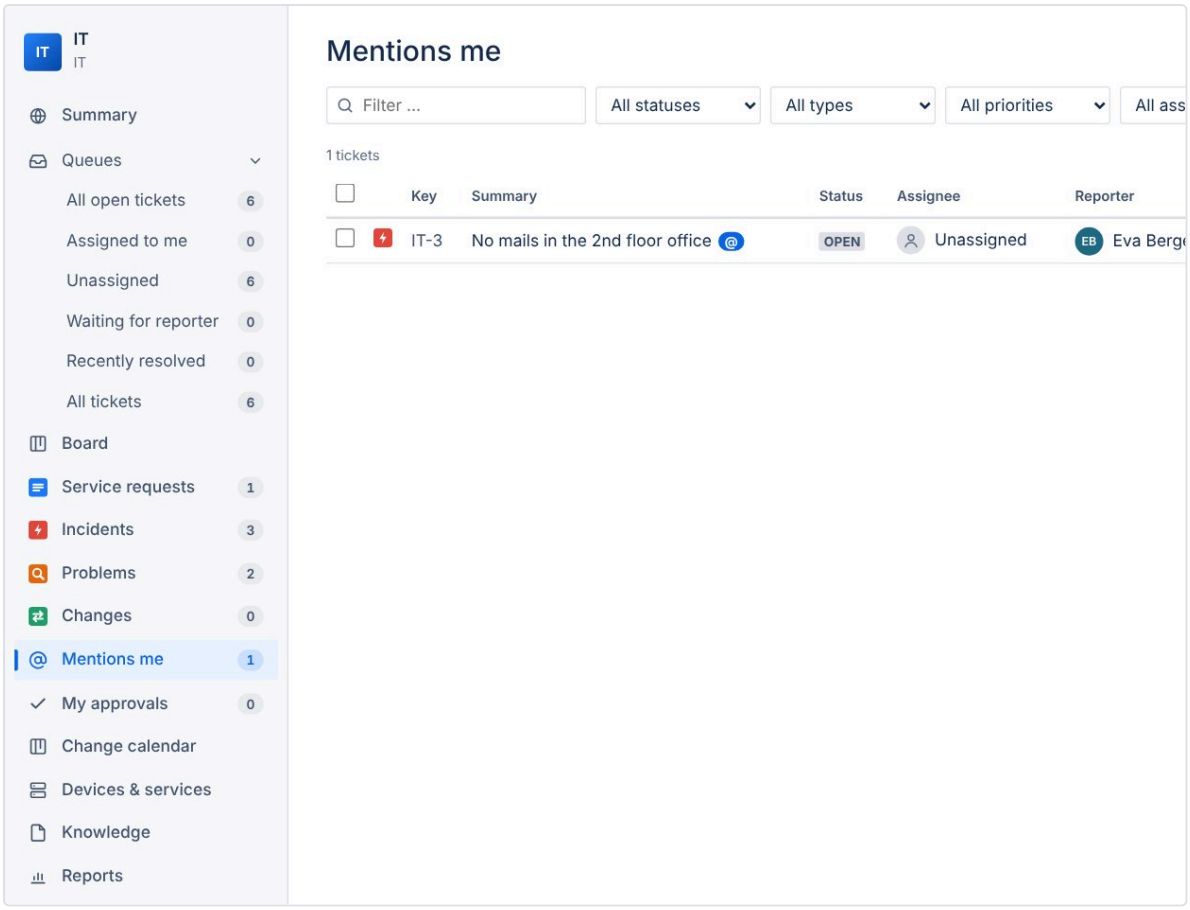
## 3.11 Mentioning colleagues (@)

When you need someone in a ticket, mention them: type **@** and a few letters in the reply field, then pick the person with a click, ↑/↓ and Enter or Tab; Esc closes the list. The name is highlighted in the comment, your own mentions in colour.

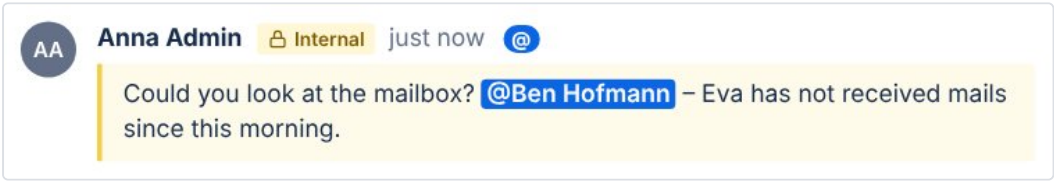



"@be" in an internal note – the list shows matching agents

- **Who you can mention:** in an *internal note* only agents. In a *reply* also the ticket's reporter, participants and approvers. People who cannot see the ticket are not in the list.
- **What happens:** the person mentioned gets an email from the ticket system with your name, the comment and the link. Reporters and participants who get the reply by email anyway do not get a second one.
- **Mentions me:** the sidebar lists all tickets in which you were mentioned; the number counts those you have not opened yet. In lists an **@** marks new mentions, in the ticket the comment in question. Opening the ticket marks the mention as seen.
- A mention does not make anyone a participant. To keep someone informed, add them under *Request participants*.



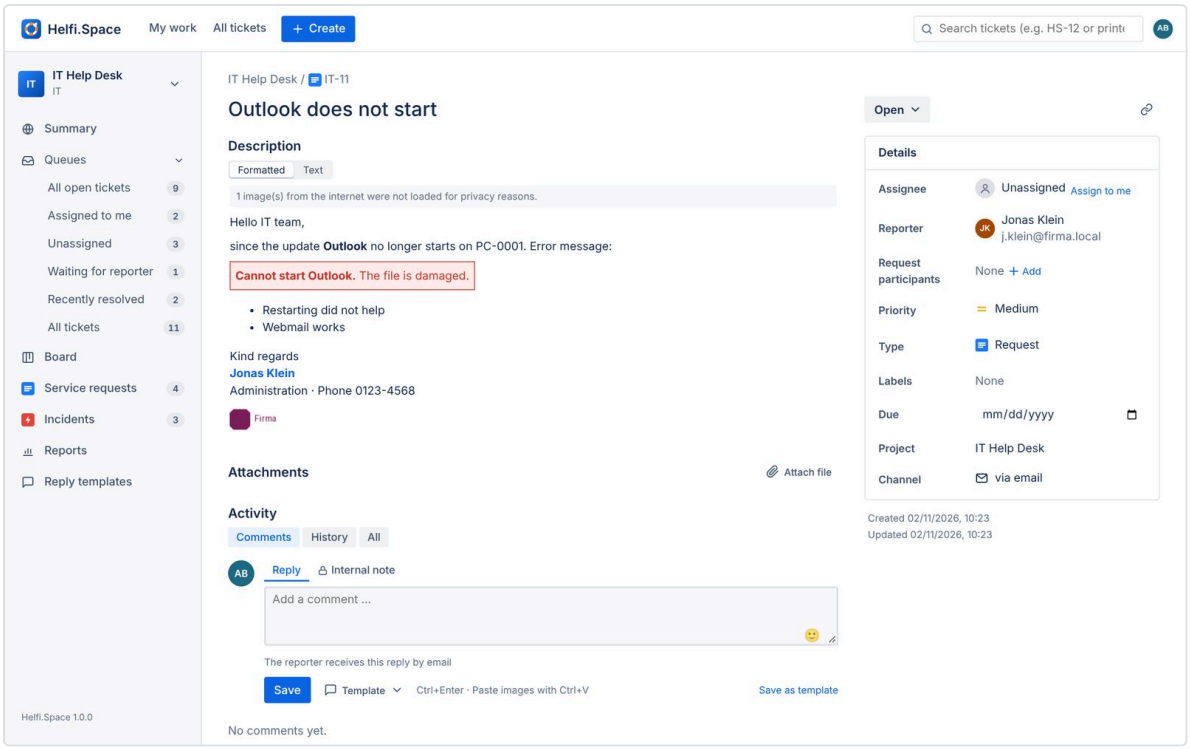

"Mentions me" with its count; the @ marks a new mention




The comment with your mention in the ticket

## 3.12 HTML emails

When a ticket or a reply arrives as an HTML email, Helfi.Space shows it **formatted** – with tables, bold text, colours, signature and embedded images. The **Formatted | Text** switch changes to the text version; the browser remembers the choice.

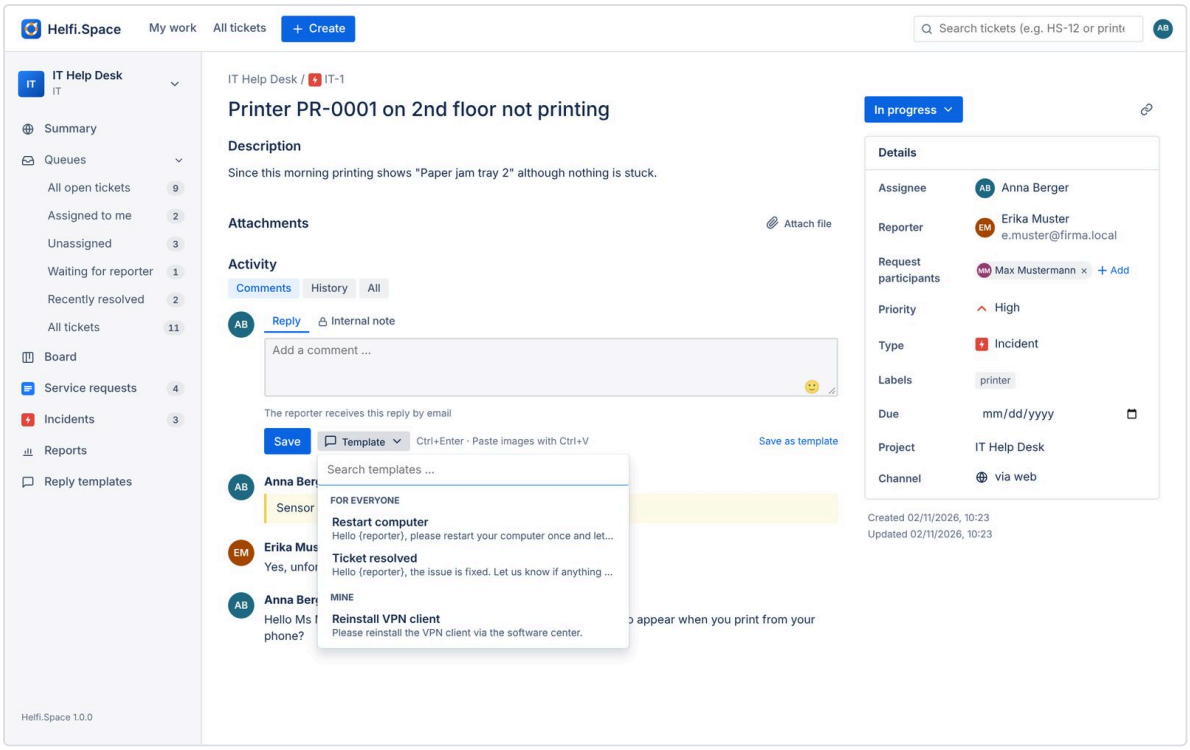



Formatted view of an HTML email

- Images from the internet (often tracking pixels) are never loaded for privacy reasons; a notice states how many there were.
- In replies the quoted history is collapsed (“Show quoted text”).
- Editing always uses the text version. After a description has been edited, only the text applies.

## 3.13 Reply templates

For replies that are always the same: click **Template** in the comment field, type to search, insert with Enter or a click. The text lands at the cursor position and can still be adjusted before saving.



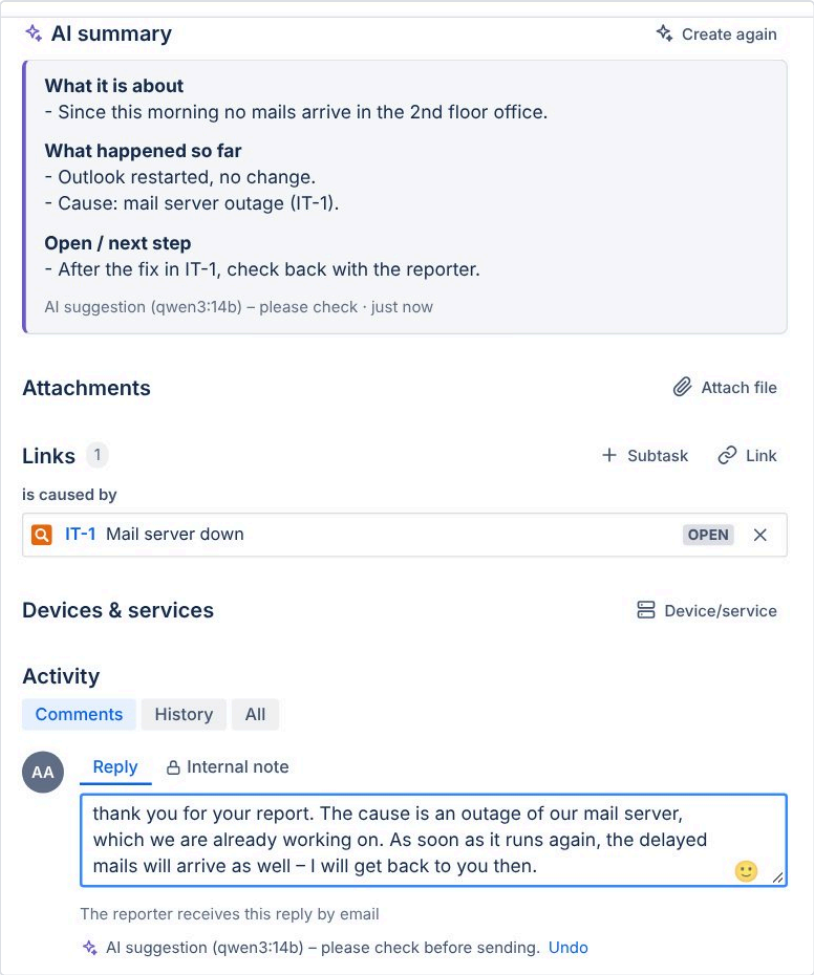

Choosing a template: “For everyone” at the top, your own below

- **Placeholders** are replaced on insertion: `{reporter}`, `{ticket}`, `{title}`, `{agent}`.
- **New template:** write the text in the comment field and choose *Save as template*. Personal templates are managed in your *Profile*; templates for everyone are maintained by admins under *Administration → Reply templates*.
- **Take over examples** ADMIN: under *Administration → Reply templates*, *Examples* creates ten frequent replies for everyone – from “Confirm receipt” to “Solved – please confirm” –, in German or English and with placeholders. Adapt them as you like afterwards.

## 3.14 AI assistance

If an admin has set up AI assistance, agents get these helpers – depending on which functions are switched on. Everything the AI returns is a **suggestion**: nothing is sent or saved automatically, and requesters see none of it. The only exception, if explicitly set: assigning the component automatically.

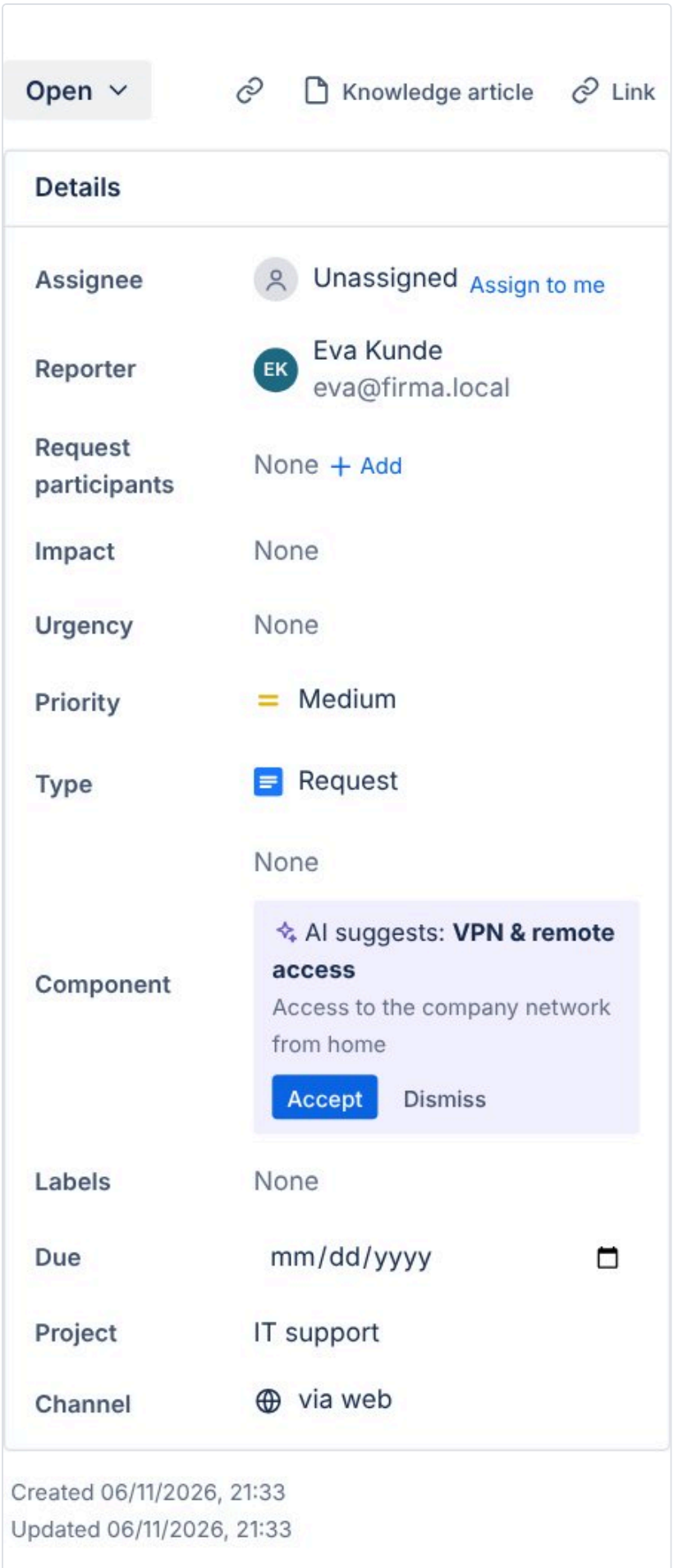
- **AI summary** below the description: *Summarize* creates an overview “What it is about – What happened so far – Open / next step”. It is kept; if the ticket changes, a note says so and *Create again* fetches a current one.
- **AI** in the reply field (the sparkle icon on phones): *Suggest a reply* drafts an answer from the ticket history. If you have already typed notes, the AI builds the reply from them. With text in the field there are also *Polish text*, *Shorten* and *Translate* (German/English).
- The suggestion replaces the text in the field; *Undo* brings back the previous text. Helfi adds the salutation and closing itself when sending the mail – please read and adjust the suggestion before saving.




AI summary and a reply suggestion built from notes – marked, with “Undo”

### Detecting the component

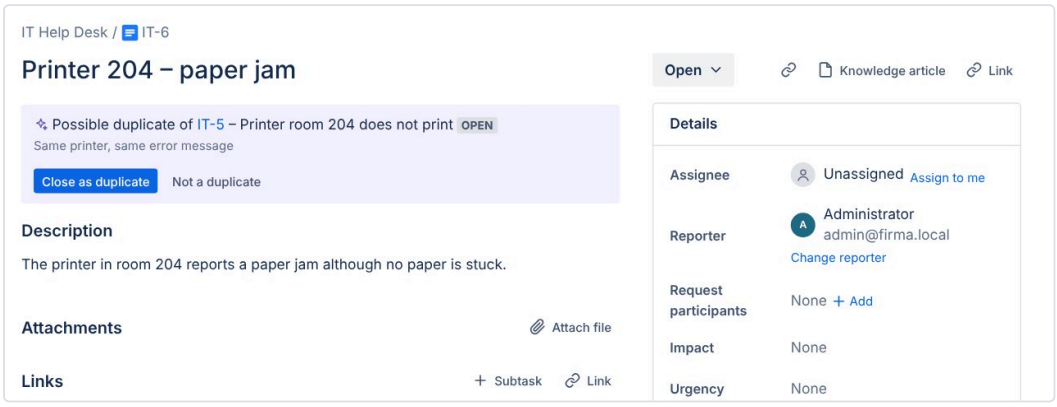
If *Detect component* is on, the AI looks at new tickets that no component rule matches and works out which component fits. In *Suggest* mode the details show “AI suggests: …” with a short reason: **Accept** sets the component (and gives the ticket to the responsible person if nobody is assigned yet), **Dismiss** hides the suggestion. On phones the suggestion is under *All details*. If you pick a different component yourself instead, that counts as a correction. In *Assign automatically* mode the AI sets the component directly; the history says “by AI”.


Open
Knowledge article
Link
Details
Assignee
Unassigned
Assign to me
Reporter
EK
Eva Kunde
eva@firma.local
Request participants
None
+ Add
Impact
None
Urgency
None
Priority
Medium
Type
Request
None
Component
AI suggests: VPN & remote access
Access to the company network from home
Accept
Dismiss
Labels
None
Due
mm/dd/yyyy
Project
IT support
Channel
via web
Created 06/11/2026, 21:33
Updated 06/11/2026, 21:33

AI suggestion for the component

### Detecting duplicates

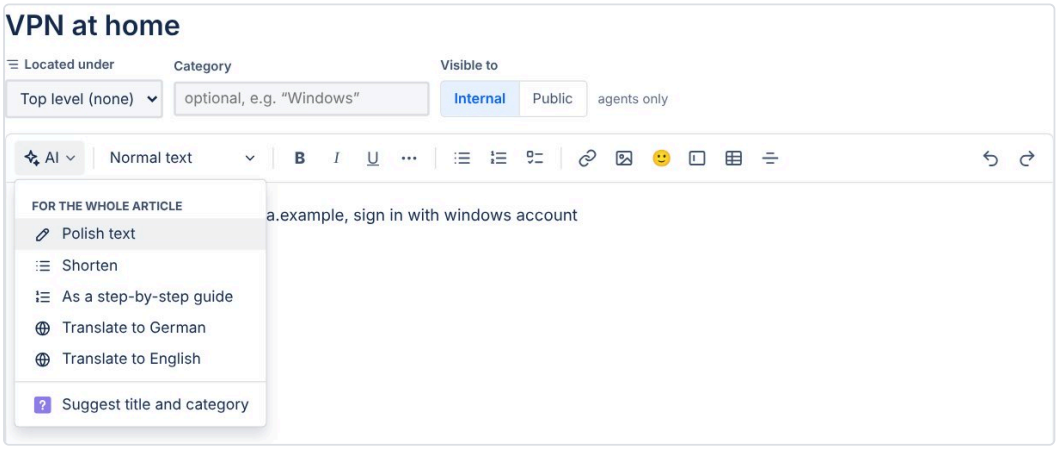
If *Detect duplicates* is on, Helfi.Space checks every new ticket: first it searches similar **open** tickets of the same project from the last 30 days without AI; only if there are any does it ask the AI whether the new ticket reports the same. The ticket then shows “Possible duplicate of IT-12” with a reason. **Close as duplicate** opens the dialog with the original filled in (see Rejecting and duplicates), **Not a duplicate** hides the hint. The AI never closes anything itself; reporters do not see the hint.




Hint about a possible duplicate

### Writing knowledge articles

In the article editor **AI** is the first button of the toolbar. It works on the selection – without one on the whole article – and keeps the formatting: *Polish text*, *Shorten*, *As a step-by-step guide* (notes become numbered steps), *Translate into German/English* and *Suggest title and category* (to accept or adjust). For a new article from a ticket there is also *Write a draft from IT-42*: problem – cause – solution from the ticket history, without names, addresses, computer names and ticket numbers. *Undo* brings back the previous text; please check the draft before publishing.

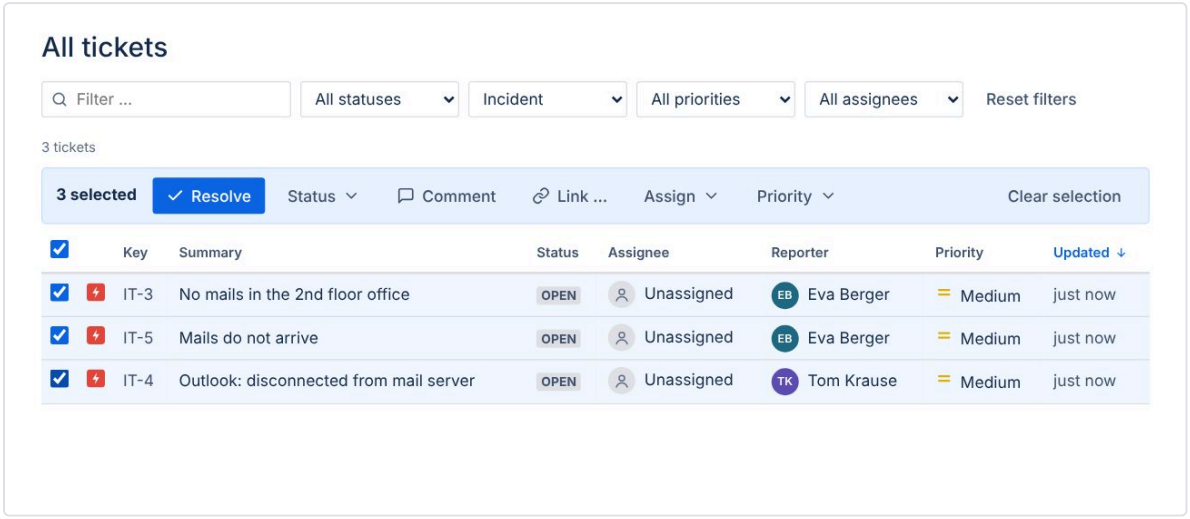



AI in the article editor

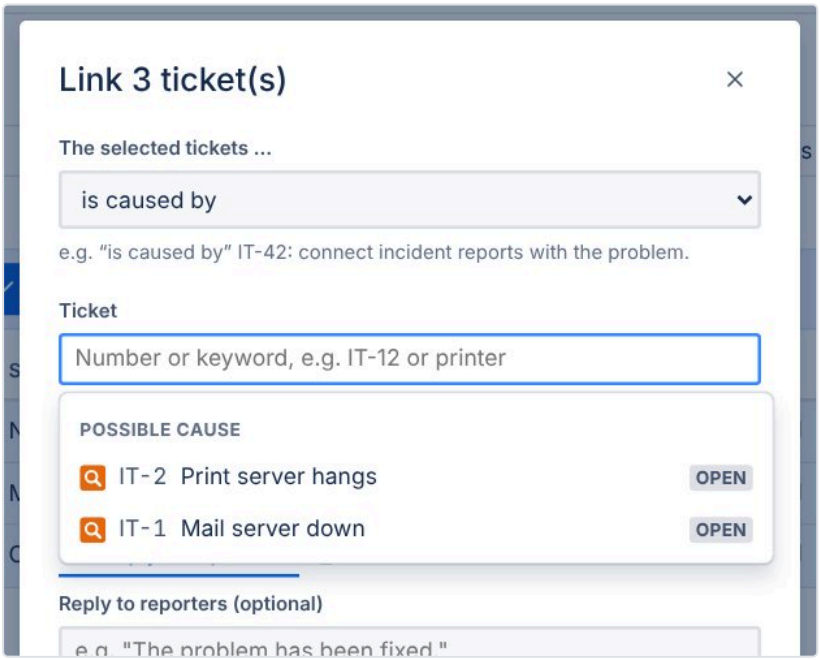
## 3.15 Several tickets at once

In *All tickets* select tickets with the check boxes (Shift+click for a range, `x` for the highlighted ticket). The action bar offers:

- **Resolve** and **Status** – each with an optional comment: as *Reply to reporters* (mail to reporter and participants) or as *Internal note*, also with a template and placeholders per ticket.
- **Comment** – reply or internal note for all selected tickets without changing the status.
- **Link …** – connect all selected tickets with one ticket, e.g. “is caused by IT-42”; optionally with status and comment right away (“the cause is being fixed in IT-42”).
- **Assign**, **Priority** and **Move** to another project.
- Set or remove the **component** – once all selected tickets belong to one project. Nobody is assigned automatically; whoever is responsible sees the tickets through their filters.




Three incidents selected – action bar with Resolve, Status, Comment and Link

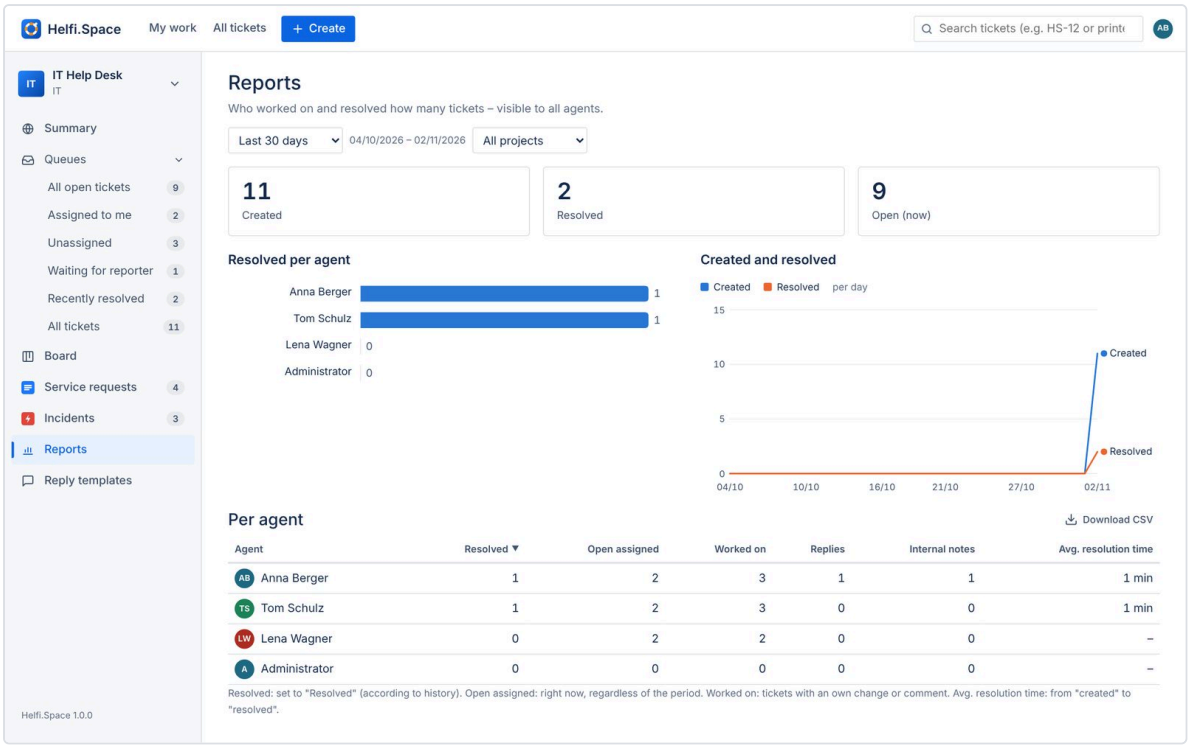



Link all selected incidents with the problem at once – with suggestions for the possible cause

## 3.16 Reports

*Reports* shows, for a period (7, 30, 90 days, this year or custom) and optionally one project: created, resolved and open tickets, “Resolved per agent”, the “Created and resolved” trend and a table per agent (resolved, open assigned, worked on, replies, internal notes, average resolution time). The table

can be sorted and downloaded as CSV. Once components are used, *By component* shows the tickets of the period per component, how many are open, rejected or duplicates, and the average resolution time (without rejected ones and duplicates).




Reports




Reports → By component

## 3.17 Keyboard shortcuts

? shows all shortcuts. They never fire while you are typing in a field.

| Where | Key | Action |
|---|---|---|
| everywhere | `c` | create ticket |
| | `/` | search |
| | `g` then `b` / `i` / `m` | board / all tickets / my work |
| ticket | `i` | assign to me |
| | `a` | choose assignee |
| | `m` | comment |
| | `e` / `l` | edit title / labels |
| | `.` | actions with search (status, priority, type, template ...) |
| | `Ctrl`+`Enter` | save comment |
| lists | `j` / `k` | next / previous ticket |
| | `o` / `Enter` | open highlighted ticket |
| | `x` | select highlighted ticket |

`Enter` on a focused link or button activates it. `Enter` only opens the row highlighted with `j`/`k` when the focus is not on a control.

## 3.18 Tickets are never deleted

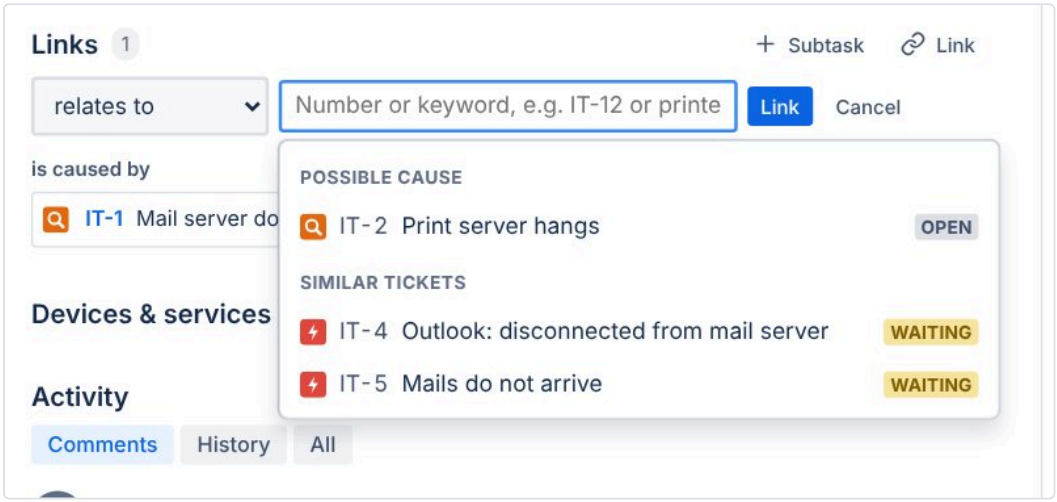
Tickets are records: who reported what and when, and what was done? That is why they **cannot be deleted**, not even by administrators. Finished work is set to “Done”. Against spam, use a sender rule “Ignore” (5.7) – then no ticket is created in the first place.

# 4 IT service management (ITIL) AGENTS

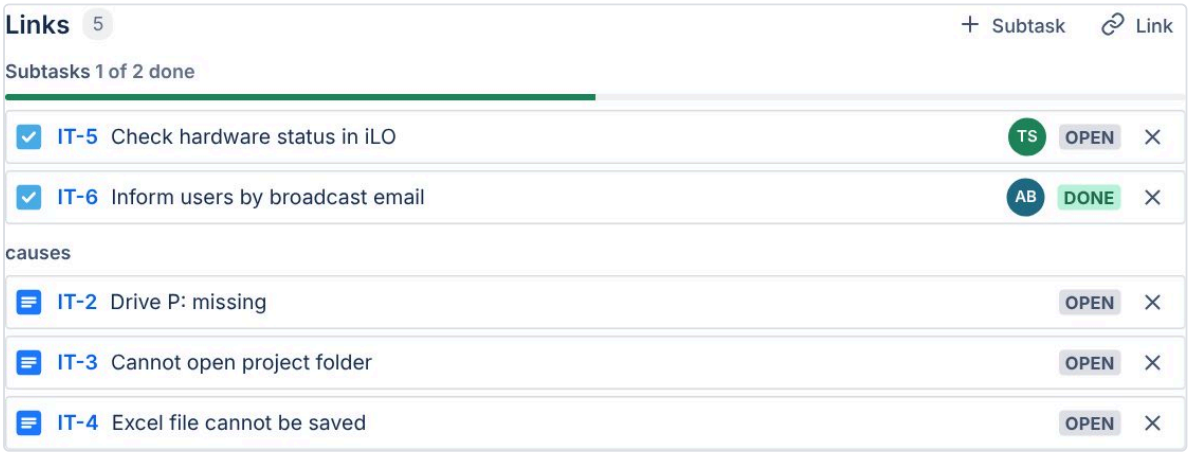
Helfi.Space covers the most important ITIL processes without forcing you to use all of them: incidents and major incidents, service levels, problems, changes, knowledge, service catalog, devices and services, and satisfaction. Whatever is not switched on or set up stays invisible.

## 4.1 Links and subtasks

In the *Links* section of a ticket – faster via the **Link** button at the top next to the status, on phones under *Details* – you connect tickets. Search the target ticket by number or keyword; while the field is empty, Helfi suggests *possible causes* (open problems, major incidents) and *similar tickets*. Kinds: “relates to”, “duplicates” (optionally resolving the duplicate right away), “blocks”, “causes” and “is subtask of”. **Subtask** directly creates a task with an assignee; a bar shows how many are done.




Linking: suggestions while the field is empty – possible cause and similar tickets

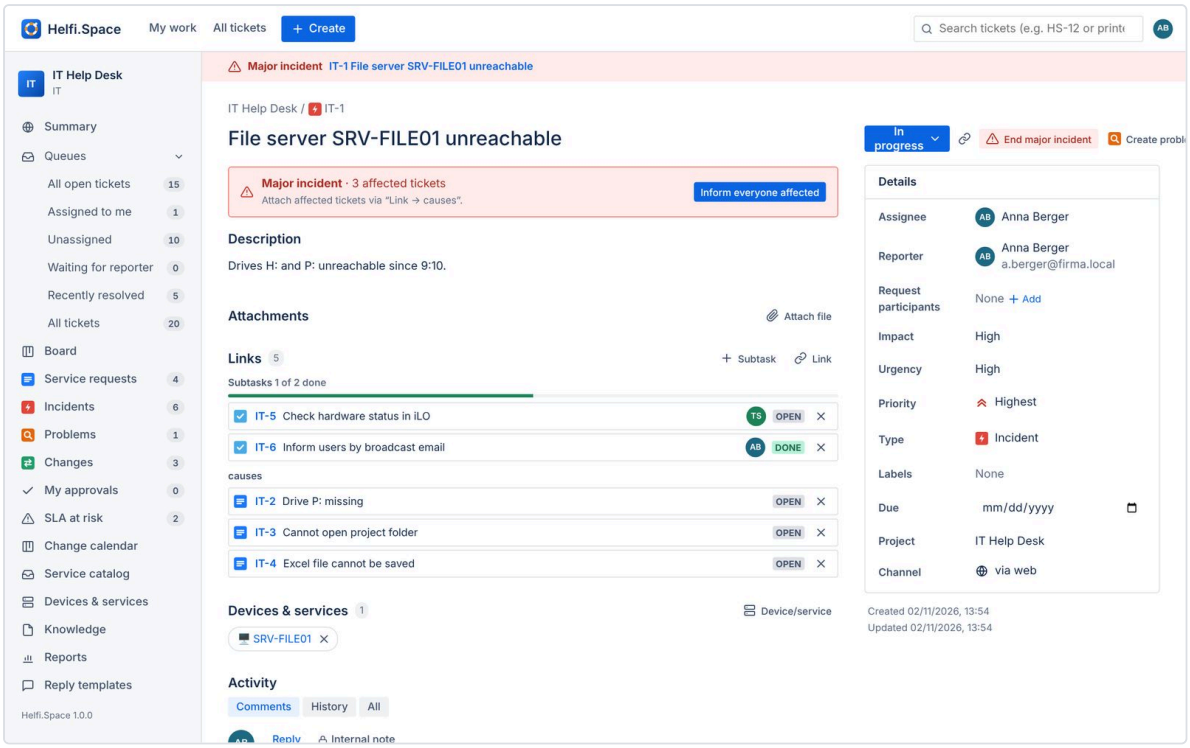



Subtasks with progress and caused incidents

## 4.2 Impact, urgency and major incident

In the details you set **impact** (how many are affected?) and **urgency** (how fast must it be fixed?). When both are set, the priority follows the ITIL matrix, e.g. high/high = Highest.

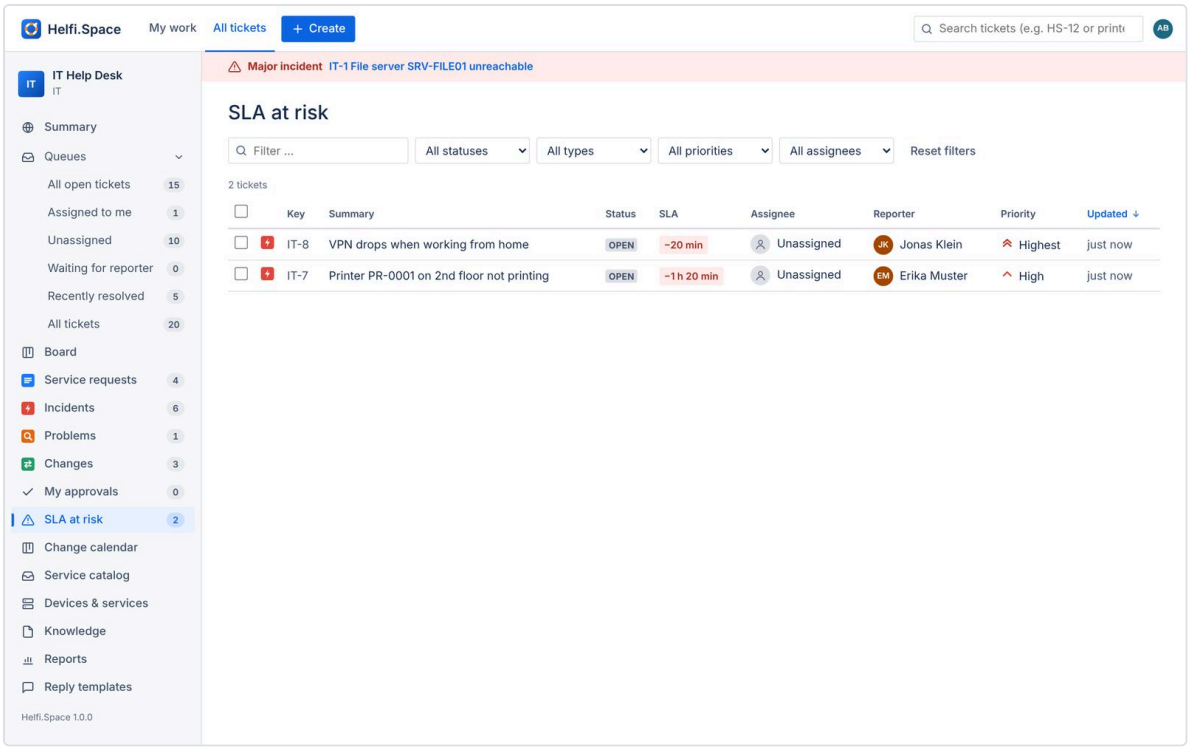
Mark a severe incident affecting many people as a **major incident**. All agents then see a red bar at the top. Link individual reports with “causes”. **Inform everyone affected** sends a message to the reporters and participants of all open linked tickets; with “resolve” they are resolved together.




Major incident with subtasks, affected tickets and device

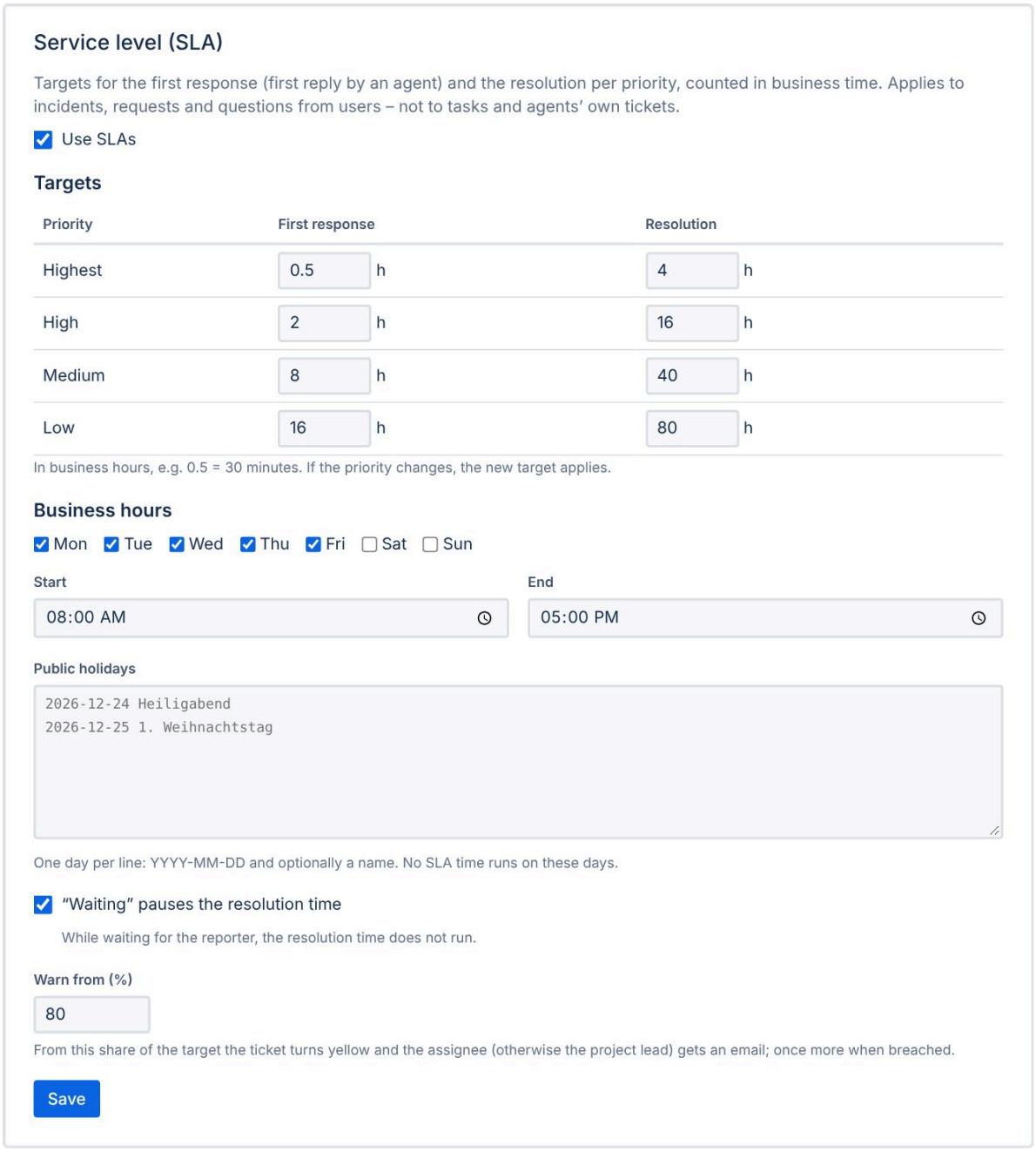
## 4.3 Service levels (SLA)

With SLAs switched on, every ticket from users has two clocks: **first response** (until IT’s first public reply) and **resolution**. Both only run during business hours; holidays and, optionally, the status “Waiting” pause them. At 80 % of the time the ticket turns yellow and the assignee (otherwise the project lead) gets an email; when the target is exceeded it turns red. The queue *SLA at risk* shows all tickets running short.




Queue “SLA at risk” with remaining time

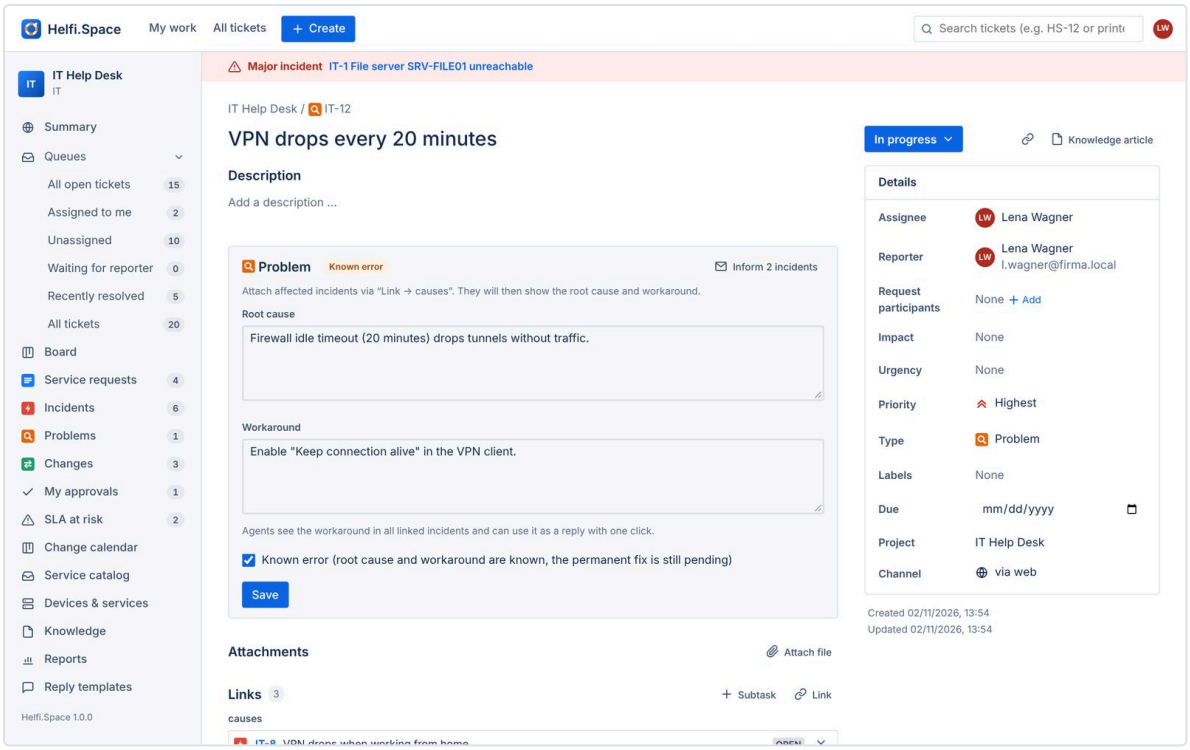
ADMIN Targets per priority, business hours, holidays and warning threshold under *Administration → Service level*. Default: off.




Administration → Service level

## 4.4 Problems and known errors

When the same incidents pile up, there is usually a common cause. **Create problem** in the ticket creates a problem and links the incident. Attach further incidents with "causes". In the problem you record the **root cause** and **workaround**. Marked as a **known error**, agents see the workaround in all linked incidents and can send it as a reply with one click. The tab *Known errors* under *Knowledge* lists them all.



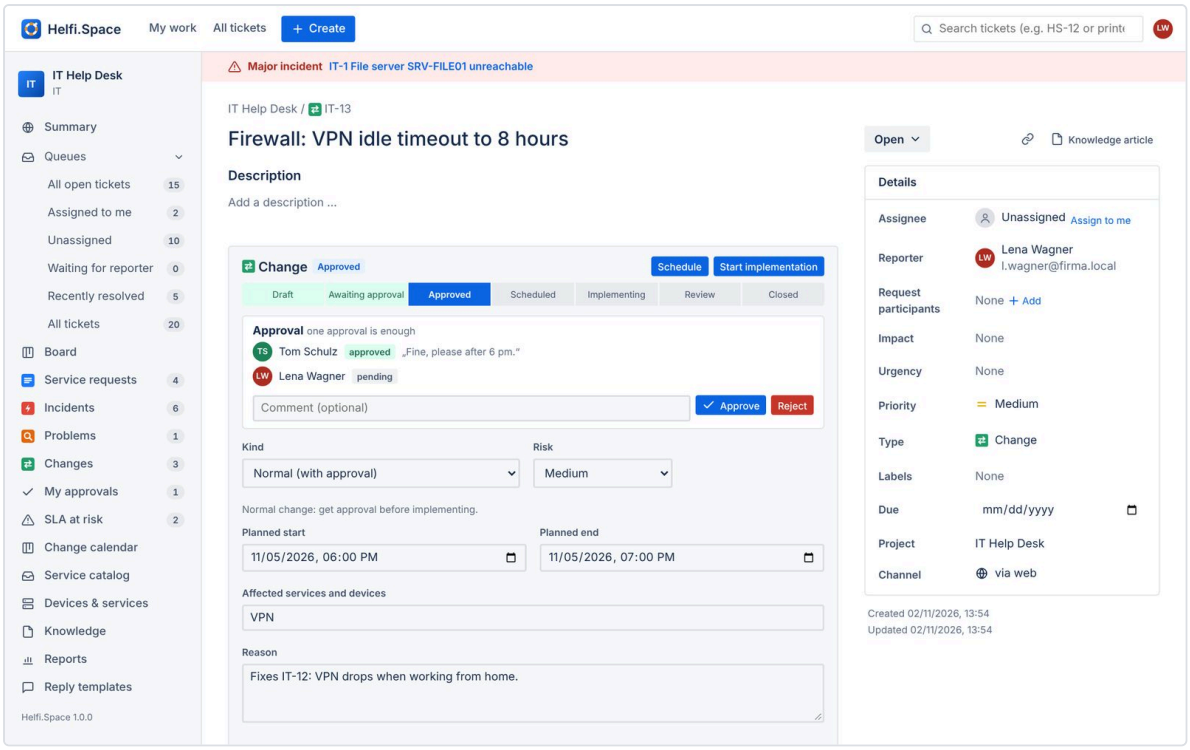

Problem with root cause, workaround and “known error”

## 4.5 Change management

Planned changes to IT (firewall rule, update, move) are tickets of type **Change**. There are three kinds:

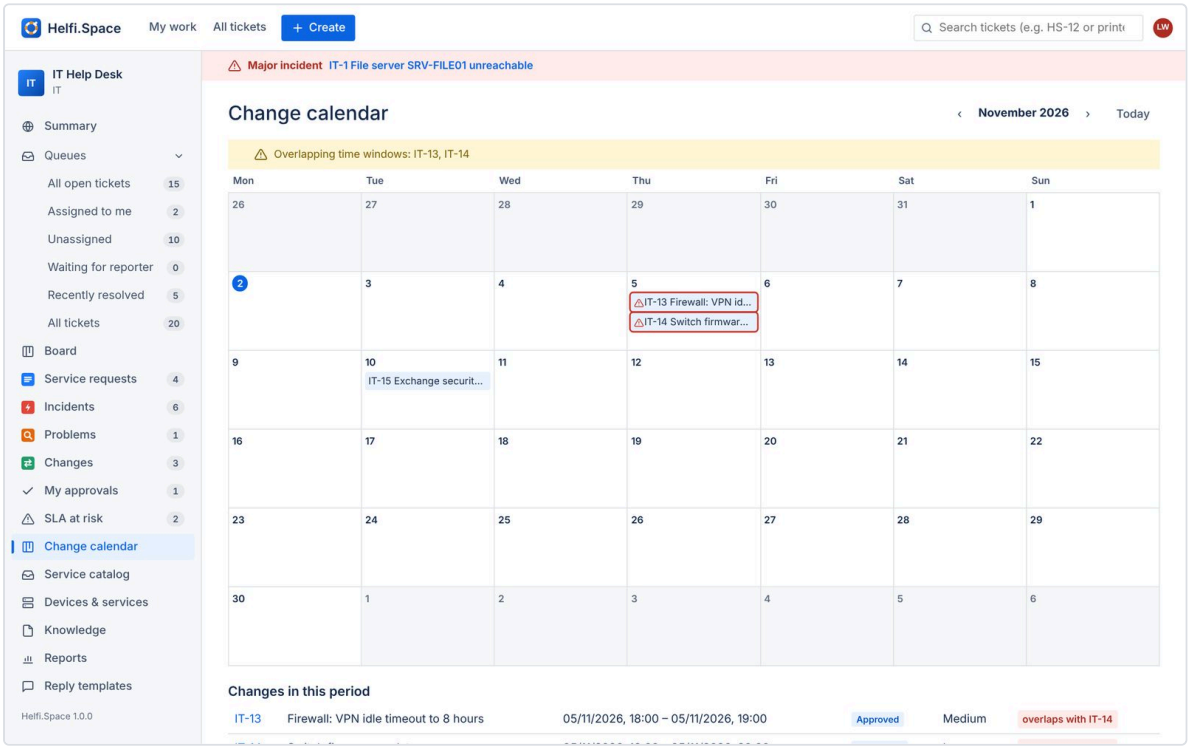
- **Normal:** needs approval.
- **Standard:** pre-approved routine based on a template.
- **Emergency:** implement immediately, approve afterwards.

The *Change* section guides you through the phases Draft → Awaiting approval → Approved → Scheduled → Implementing → Review → Closed. Helfi.Space names missing details (reason, time window, implementation and backout plan) at the respective step. Approvers decide in the ticket or under *My approvals*. When closing you choose the outcome; for “With issues” and “Rolled back” a review is required.



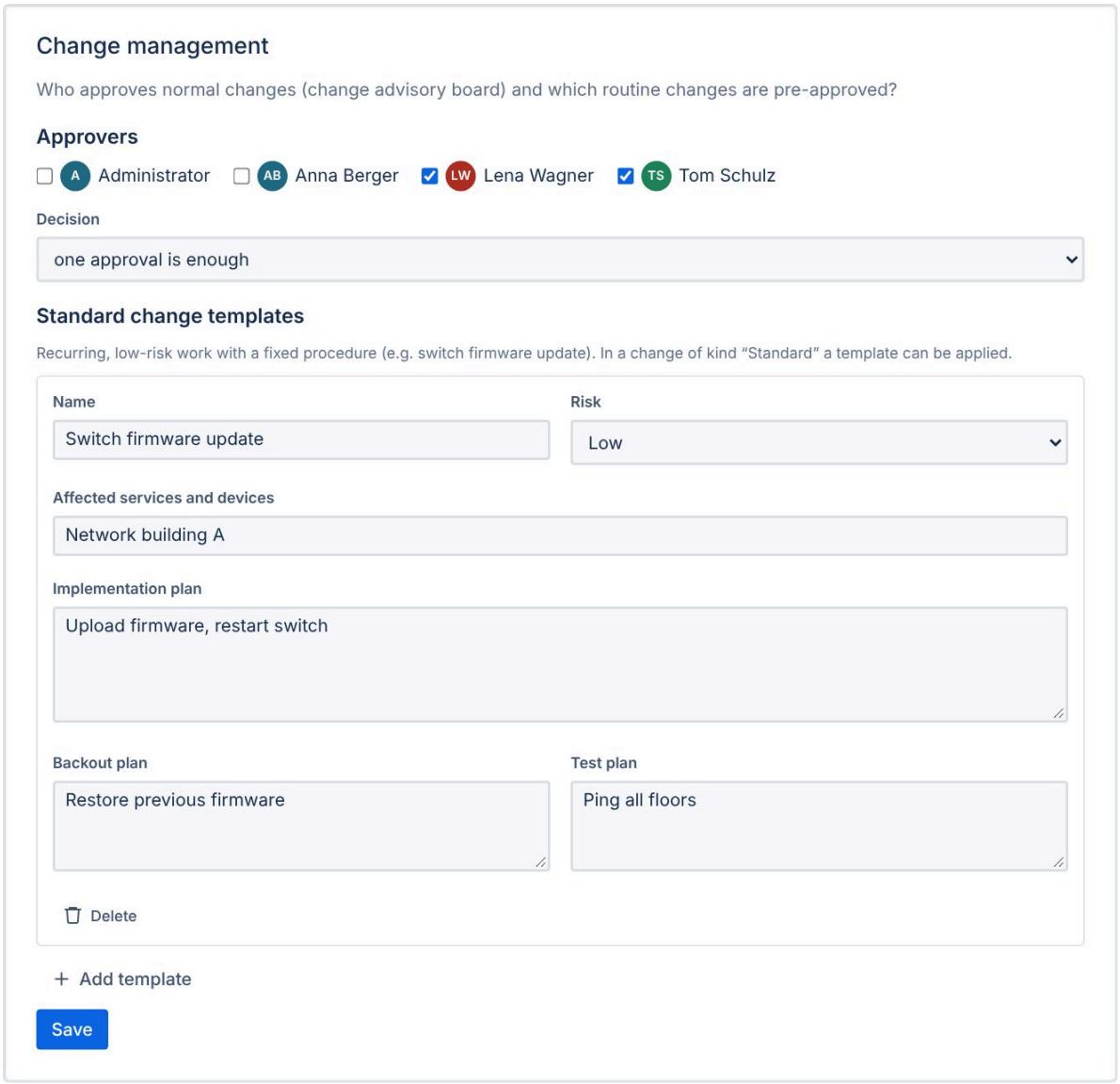

Change with phases, approval and planning

The **change calendar** shows all planned changes in the month; overlaps are marked red.




Change calendar with an overlap

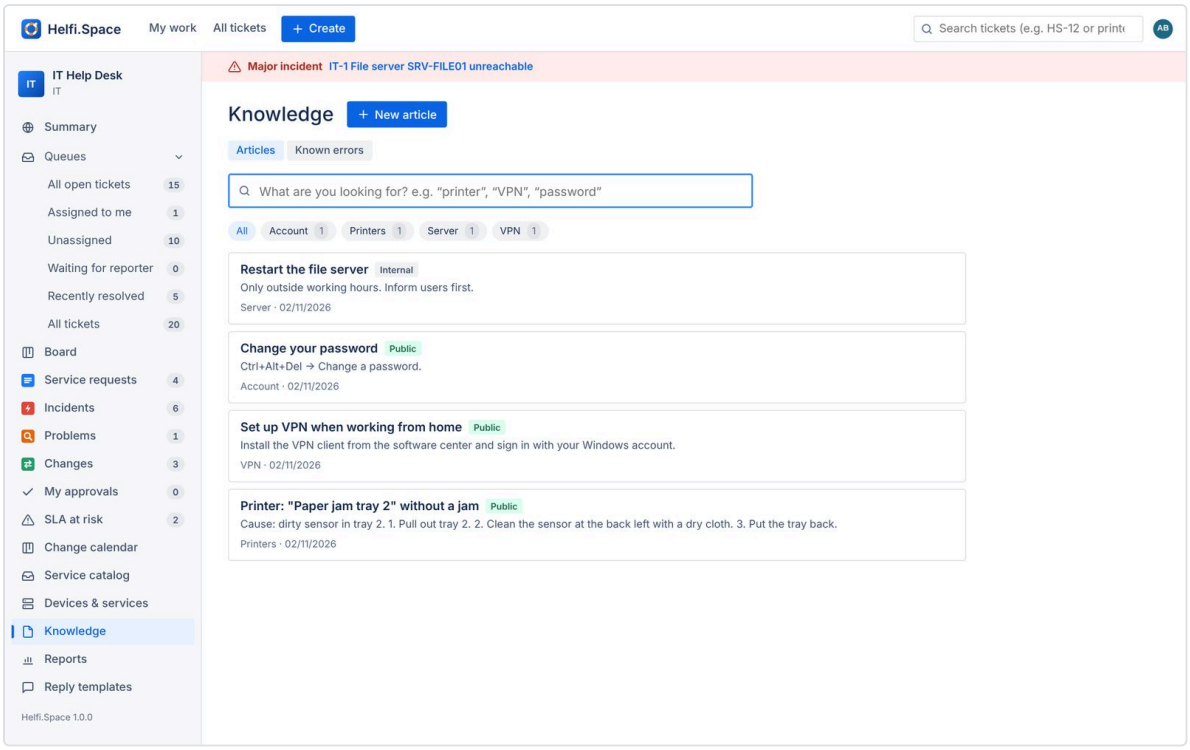
ADMIN Approvers, rule (“one approval is enough” or “everyone must approve”) and templates for standard changes under *Administration → Change*.




Administration → Change

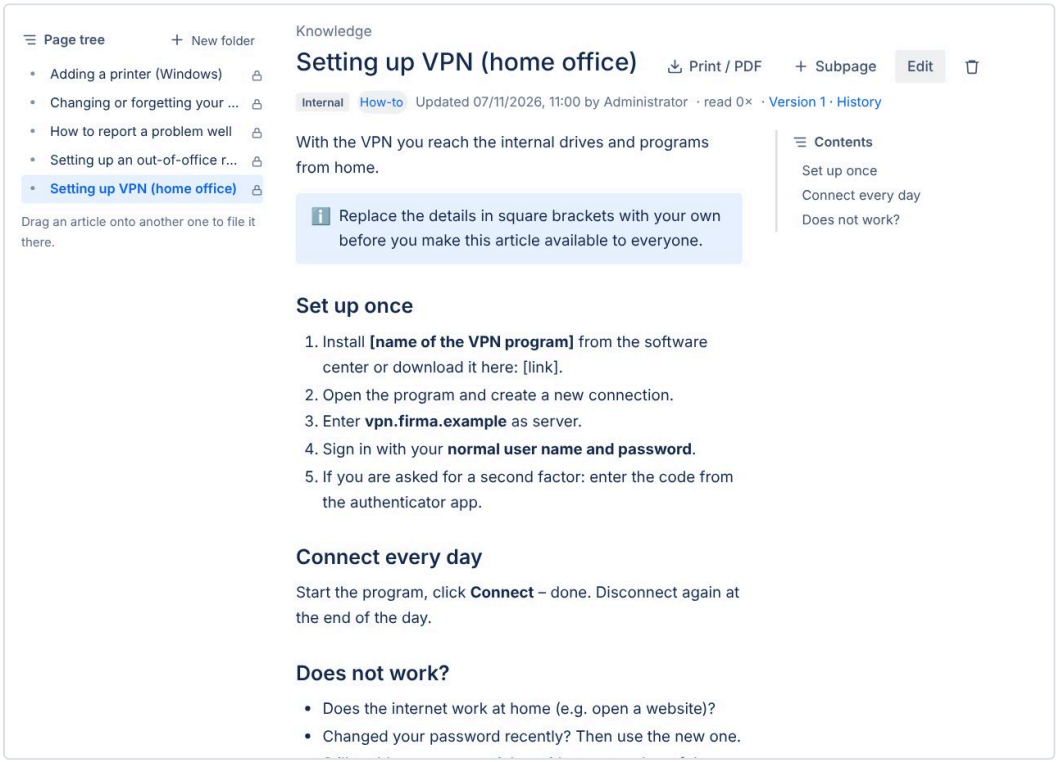
## 4.6 Knowledge base

*Knowledge* collects guides and solutions. Articles are **internal** (agents only) or **public** (everyone signed in, also suggested while reporting). Search ranks title matches higher. From a solved ticket, **Knowledge article** at the top of the ticket creates a draft; remove personal details before publishing. From three headings on, the article view shows a table of contents on the right.




Knowledge base with categories

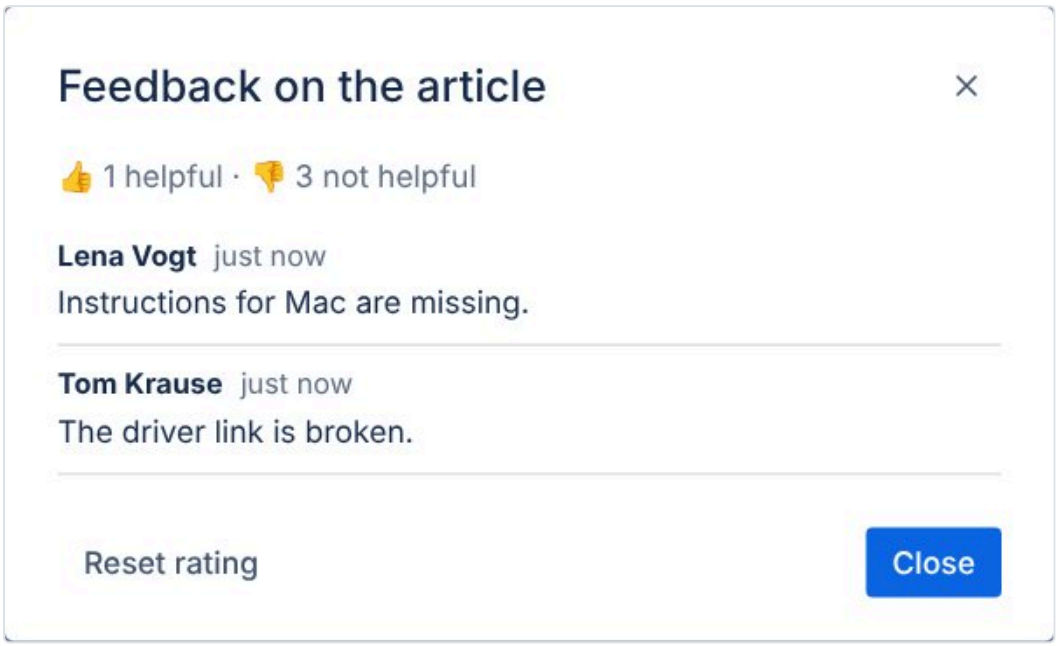
**Example articles:** if the knowledge base is still empty, *Examples* takes over ready sample articles – such as "How to report a problem well", "Setting up VPN" or "Changing or forgetting your password". They show how a good article is built and are created **internal**: replace the details in square brackets with your own and then make them public.




A taken-over example article

### Rating and feedback

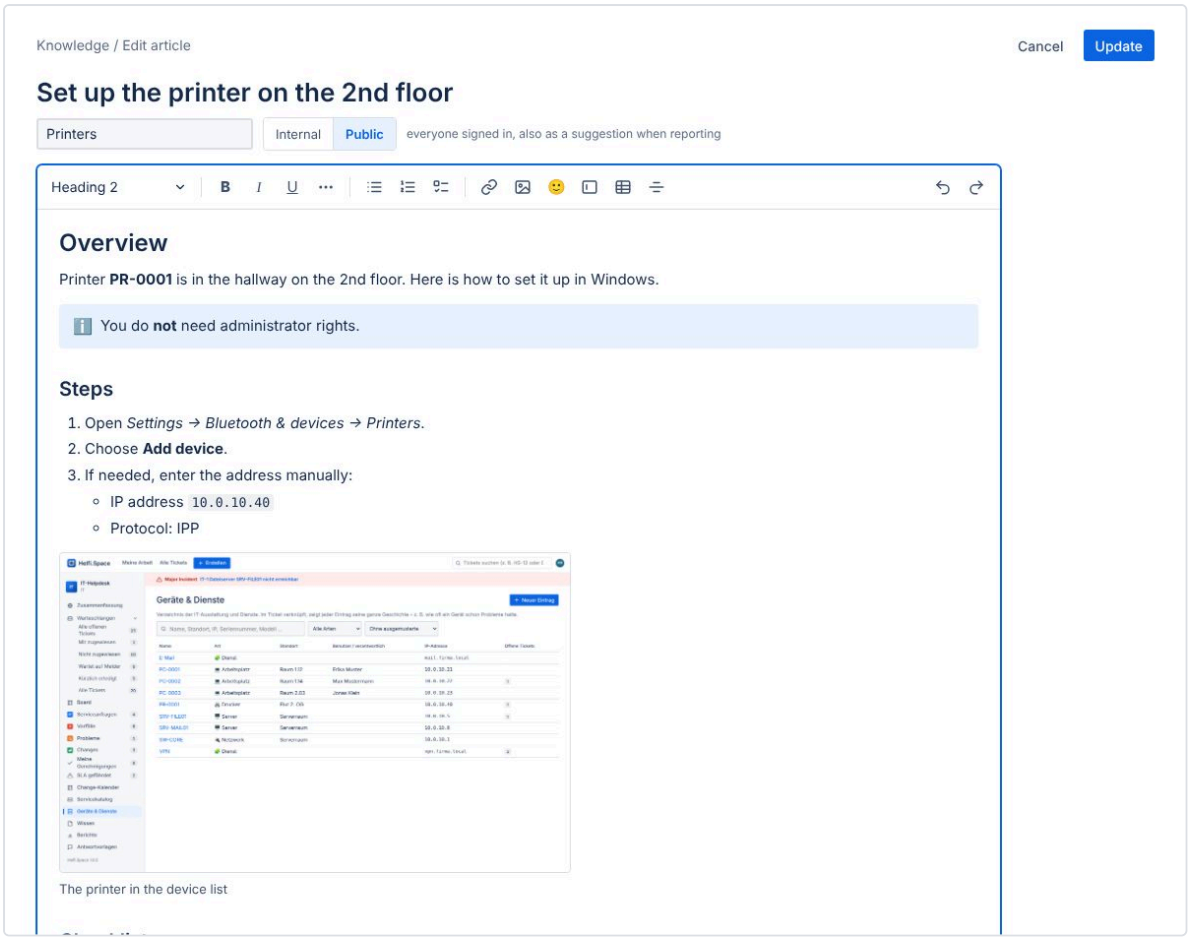
Below every article there is “Was this article helpful?” with 👍 and 👎. The article list shows the rating per article; the *Needs rework* filter lists articles with 👎 votes, most first. On the article, *Feedback* opens the notes on what was missing. Once you have reworked the article, *Reset rating* sets the votes back to zero – the notes stay under “Earlier feedback”.




Feedback on an article

### The editor

You write articles directly in their final look. The toolbar stays at the top while scrolling and offers text style (normal, heading 1–3, quote, code block), bold, italic, underline, under **…** also strikethrough and inline code, bulleted list, numbered list, checklist, link, image, emoji, info panel (note, tip, warning), table and divider.



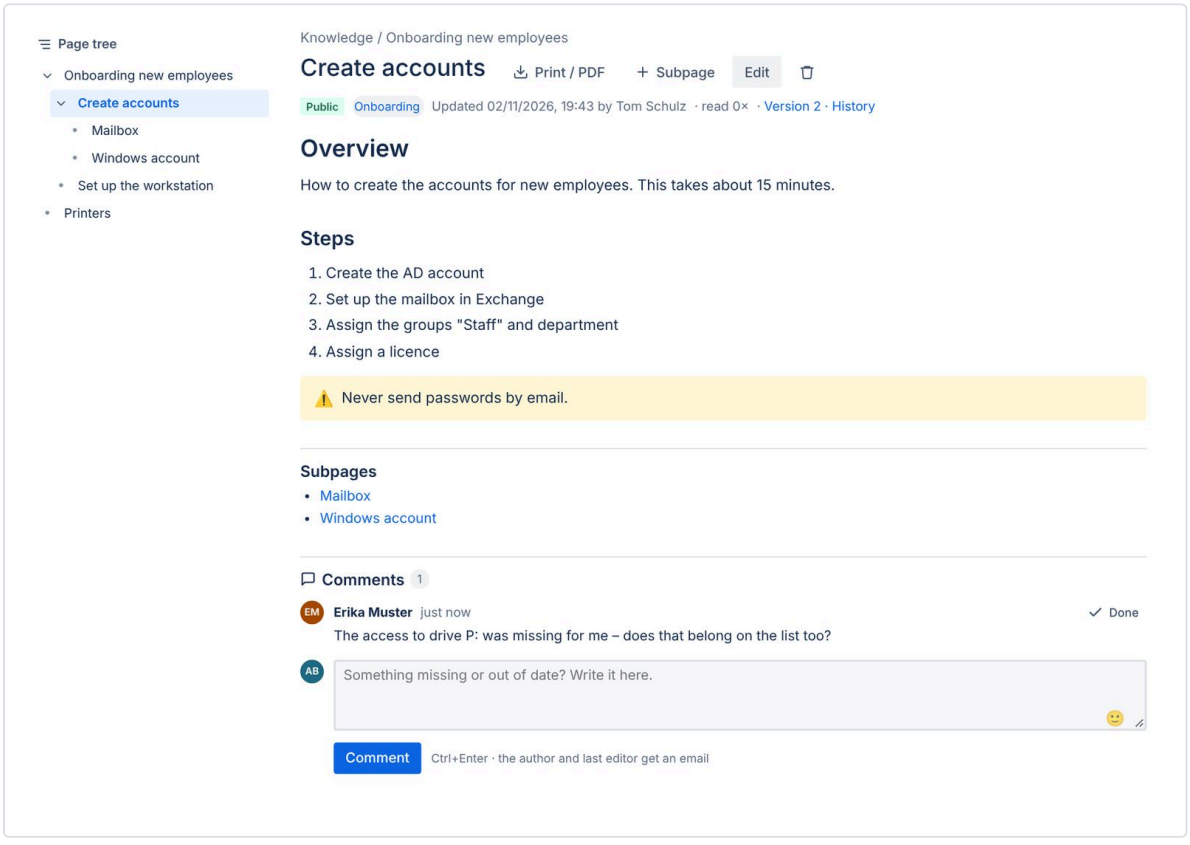

Editor with info panel, nested list and image

- **Images:** with `Ctrl`+`V` from the clipboard, by drag & drop or with the image button. Clicking an image sets its size (small, medium, full width) and a caption.
- **Pasting from Word, web pages or emails:** headings, lists, tables, bold text and links are kept. Images from external web addresses are not taken over; copy them individually as an image.
- **Links** (`Ctrl`+`K`) point to web addresses, email addresses, tickets ( `IT-42` ) or other articles ( `KB-3` ).
- **Tables:** `Tab` jumps to the next cell; in the last cell it adds a new row. Add or delete rows and columns with the table button.
- **Draft:** the browser keeps saving unsaved changes. Next time you open the article you can restore them. Leaving the page with unsaved changes asks first.

| Type | Result |
|---|---|
| `#`, `##`, `###` + space | Heading 1–3 |
| `-` or `1.` + space | Bulleted / numbered list (`Tab` indents) |
| `[]` + space | Checklist (click the box to tick it) |
| `>` + space | Quote |
| ``` ``` + Enter | Code block (Enter on an empty last line leaves it) |
| `---` + Enter | Divider |
| `Ctrl`+`B` / `I` / `U` / `E` | Bold / italic / underline / inline code |
| `Ctrl`+`Alt`+`1`…`3`, `0` | Heading 1–3, normal text |
| `Ctrl`+`S` | Save |

## Page tree

Articles can sit below other articles, for example “Onboarding → Create accounts → Windows account”. In the editor, **Located under** sets where the article sits; on an article, **Subpage** creates an article right below it. On a PC, simply drag articles in the page tree onto another one or onto “top level”. **New folder** in the page tree header creates a short collection page without text of its own, e.g. “Windows” or “Microsoft Office”. The start page shows the page tree and recently changed articles; each article shows its path, the tree on the left and its subpages at the end.

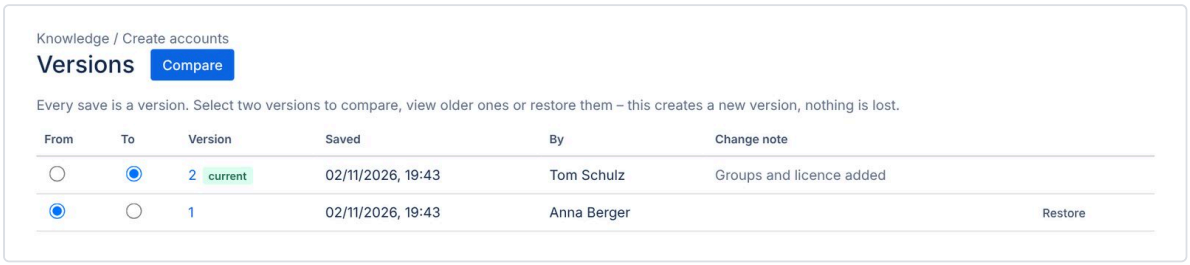



Article with page tree, subpages and a comment

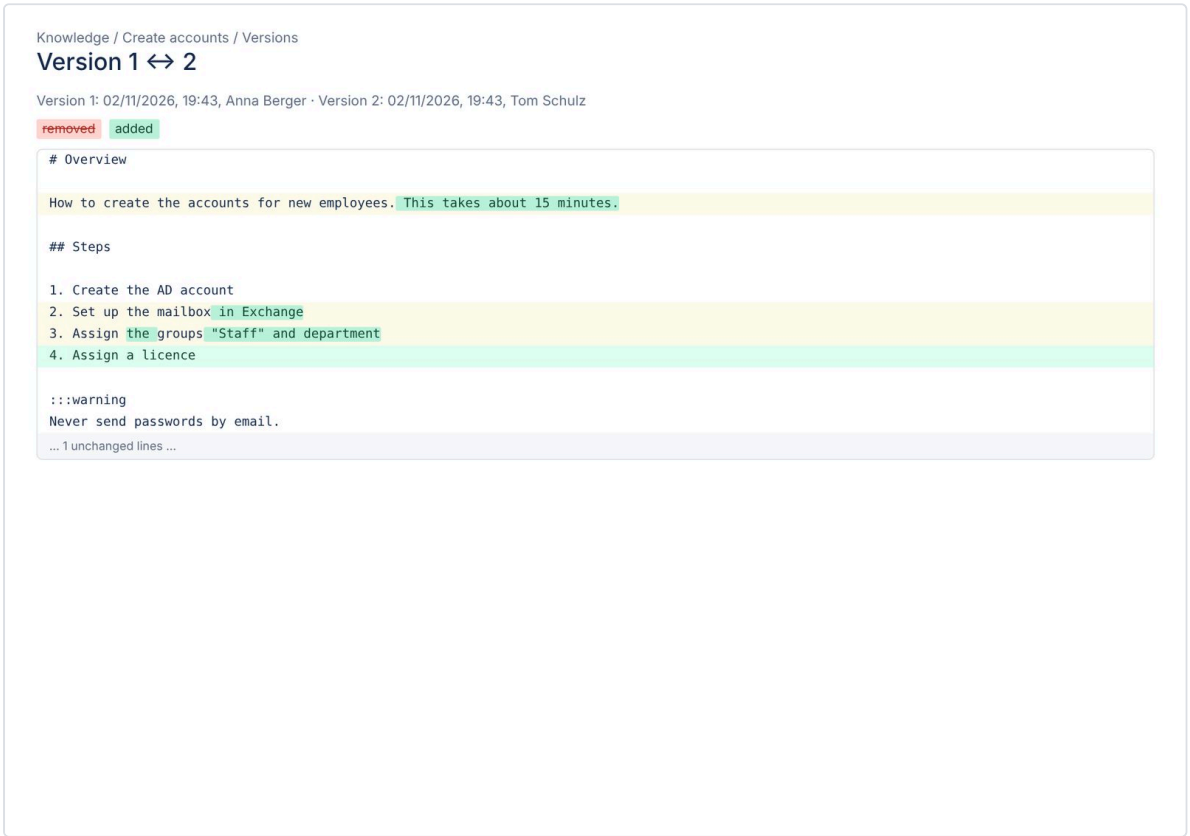
- Users only see public pages. If a public page sits below an internal one, it moves up a level for users.
- If you delete an article with subpages, they move up one level. Moving creates no new version.

### Versions

Every save with a change is a version. In the editor header you can add a short **change note**. *Version n · History* on the article lists all versions. Select two and **compare** them: removed text is red and struck through, new text green. You can view older versions and **restore** them; this creates a new version, the history stays complete.



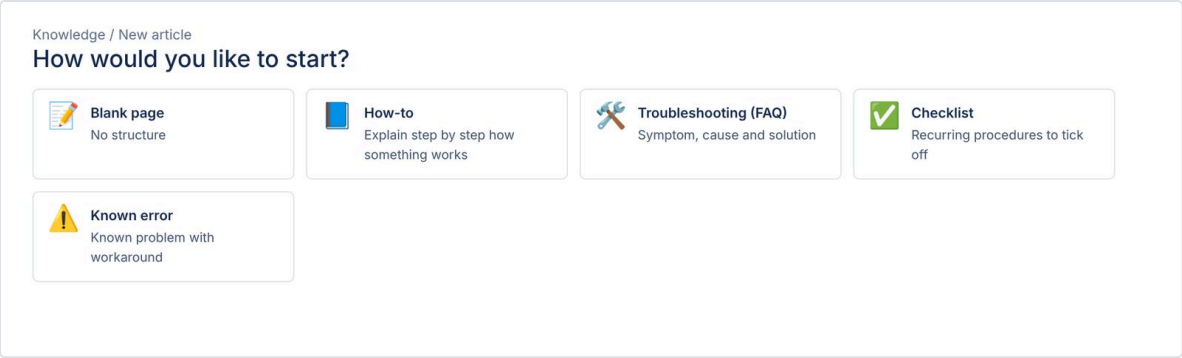

Versions of an article




Comparing two versions

### Templates

**New article** first asks how you want to start: blank page, *How-to*, *Troubleshooting (FAQ)*, *Checklist*, *Known error* or a template of your own. Save your own templates in the editor with the copy icon (*Save as template*) and manage them under *Knowledge → Templates*.




Choosing a template for a new article

### Comments

Anyone who can see an article can comment below it – including users on public articles. The author and last editor get an email. Agents mark comments as **done**; done comments collapse. You can delete your own comments. In the article list, 💬 shows the number of open comments.

### Print and PDF

**Print / PDF** on the article opens the print dialog; choose “Save as PDF” there. Only title, path, date and content are printed, without page breaks inside images, tables or info panels. If the article has subpages you can print **with all subpages**: you get one document with a table of contents, each page starting on a new sheet – for example a complete onboarding handbook.

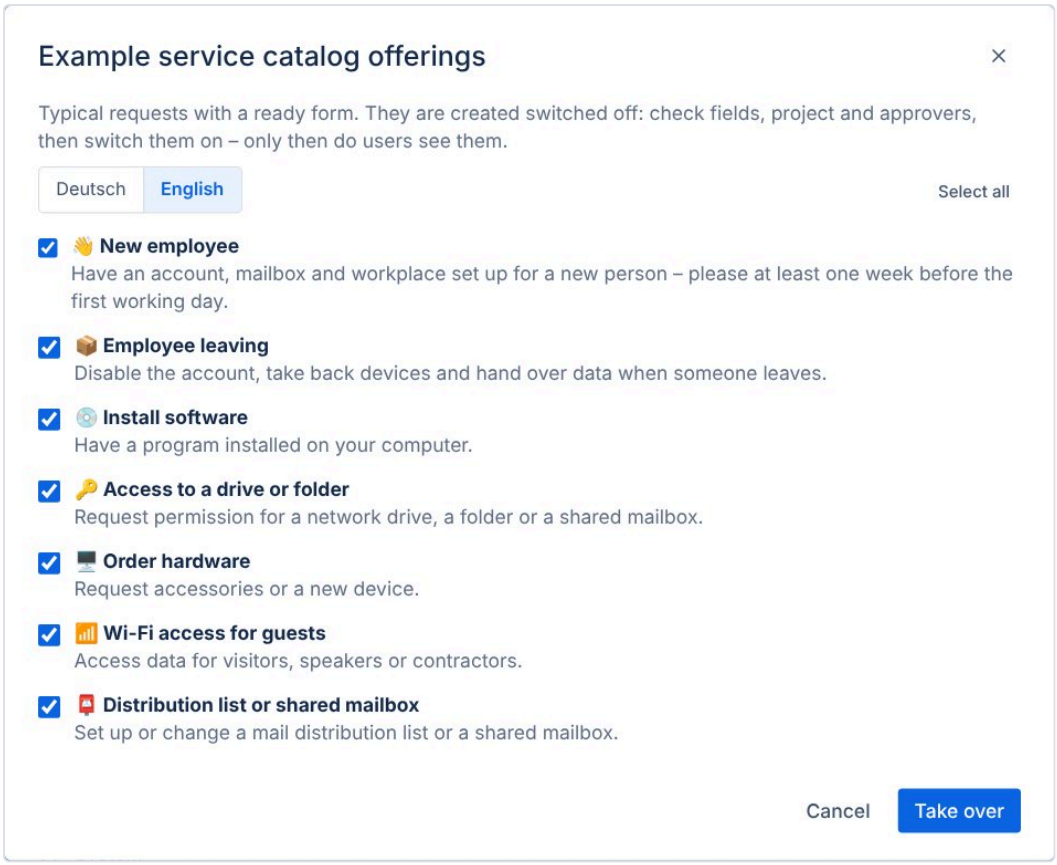
## 4.7 Setting up the service catalog ADMIN

Under *Administration → Service catalog* you create offers: name, emoji, description, project and form fields (text, multi-line, choice, check box, date; each optionally required). Optionally choose approvers, also without the agent role, and whether one or all approvals are needed. Requests become tickets of type *Request*; the answers appear in the description. How users see the catalog is described in 2.9.




Administration → Service catalog

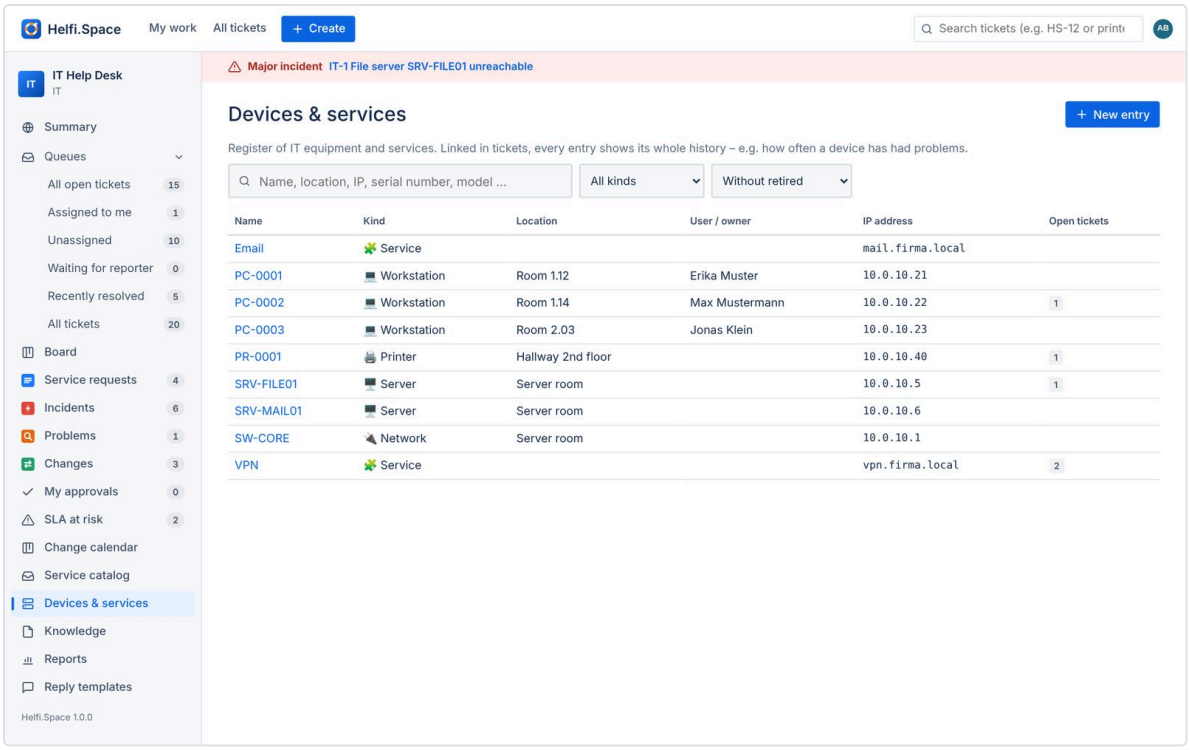
**Take over examples:** *Examples* creates typical offers with a ready form, e.g. “New employee”, “Install software” or “Wi-Fi access for guests” – in German or English. They start **switched off**, so users only see them once you have checked fields, project and approvers and switched them on.



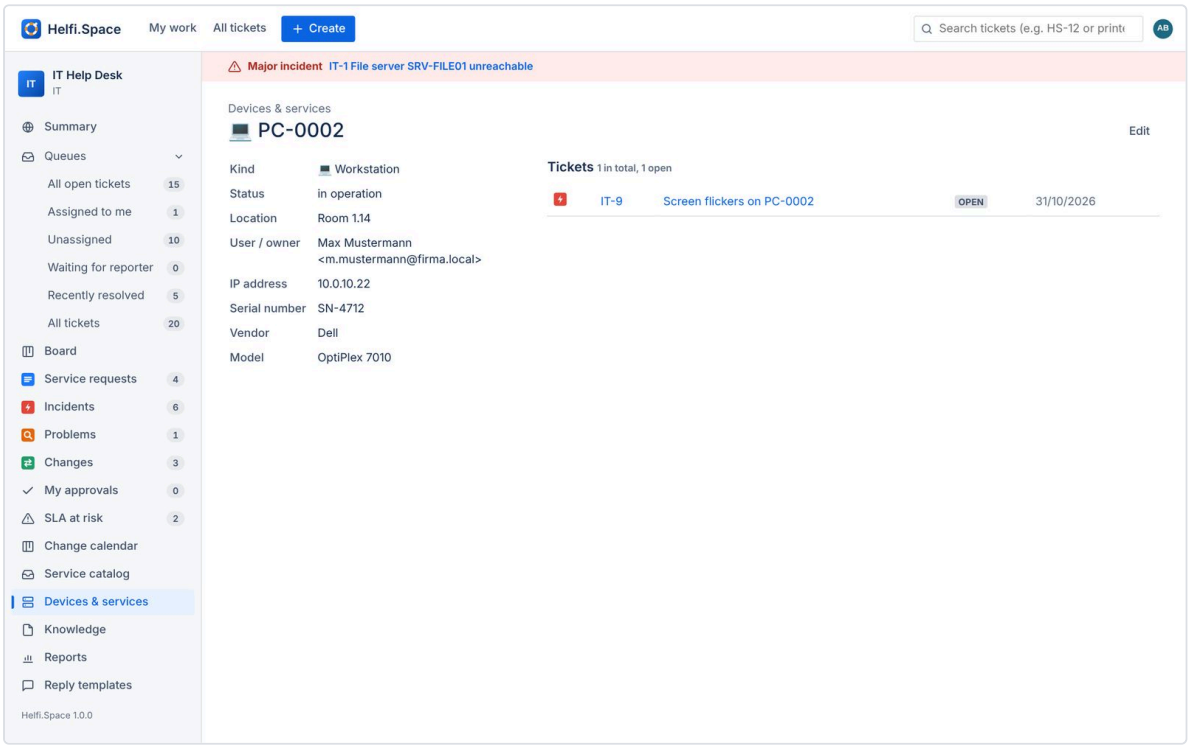

Choosing example offers

## 4.8 Devices & services

A lean register of IT equipment and services (CI register). Each entry has a kind, status (in operation, in maintenance, retired), location, user, IP address, serial number, vendor and model. In a ticket you link affected devices and services in the *Devices & services* section. For changes these are the affected services. An entry's detail page lists all its tickets, so recurring faults stand out.




Devices & services with open tickets per entry

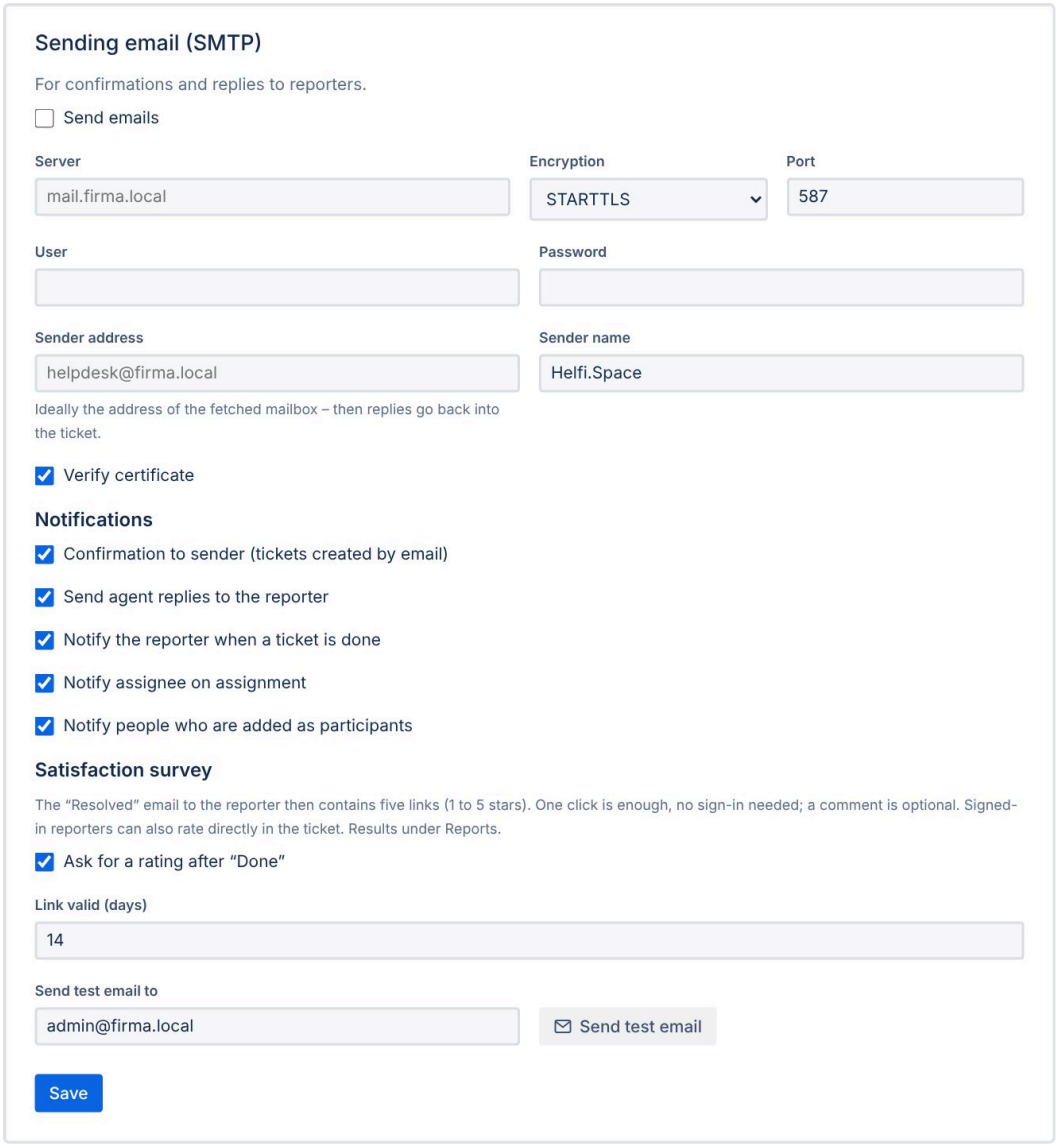



A device with its ticket history

ADMIN **CSV import** from your inventory tool: the first line contains the column names (name/hostname, type, location, IP address, serial number, vendor, model, user, status – German or English). Values like “Notebook” or “Switch” are mapped to the matching kind. Existing names are updated. Only administrators can delete an entry, and only while no tickets are linked; otherwise set it to “retired”.

## 4.9 Satisfaction survey ADMIN

Switch it on under *Administration → Email → Satisfaction survey*. The “Resolved” email to the reporter then contains the rating links (2.12). Each link contains a random token; only its hash is stored. The link is valid for a configurable number of days and only shows the ticket key and title. If a ticket is reopened and resolved again, there is a new survey. Agents see the rating at the top of the ticket.



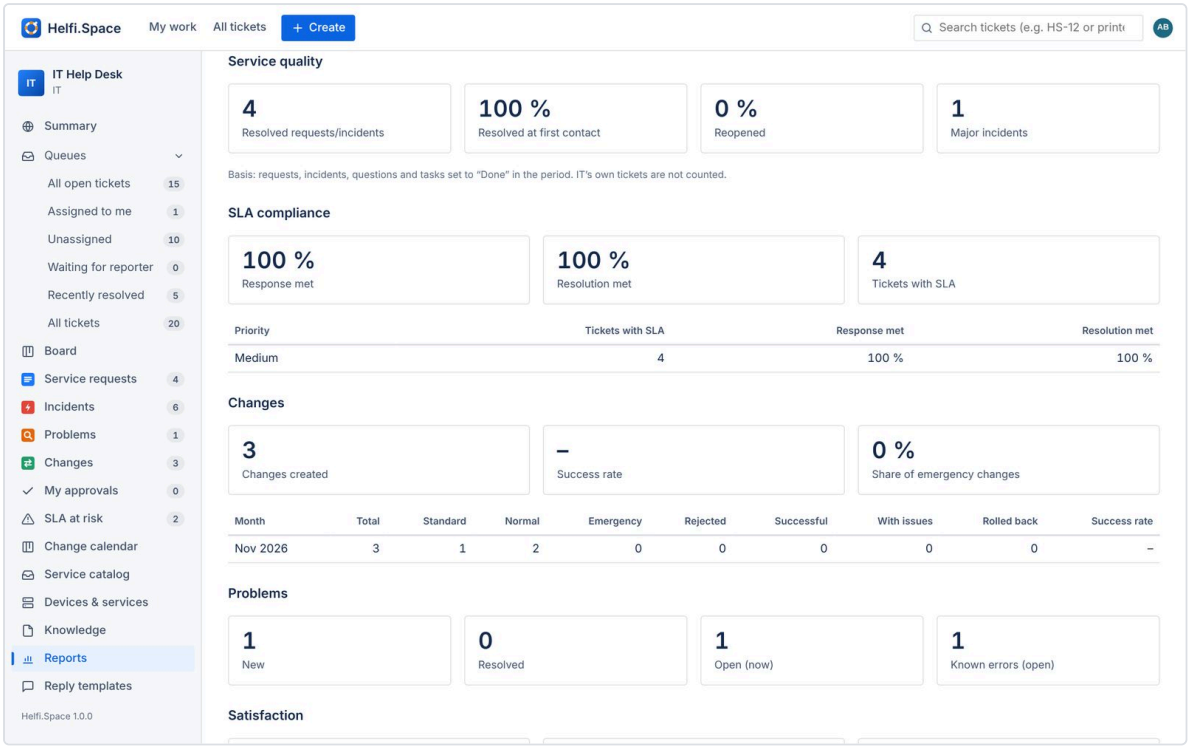

Administration → Email with the satisfaction survey

## 4.10 ITIL key figures

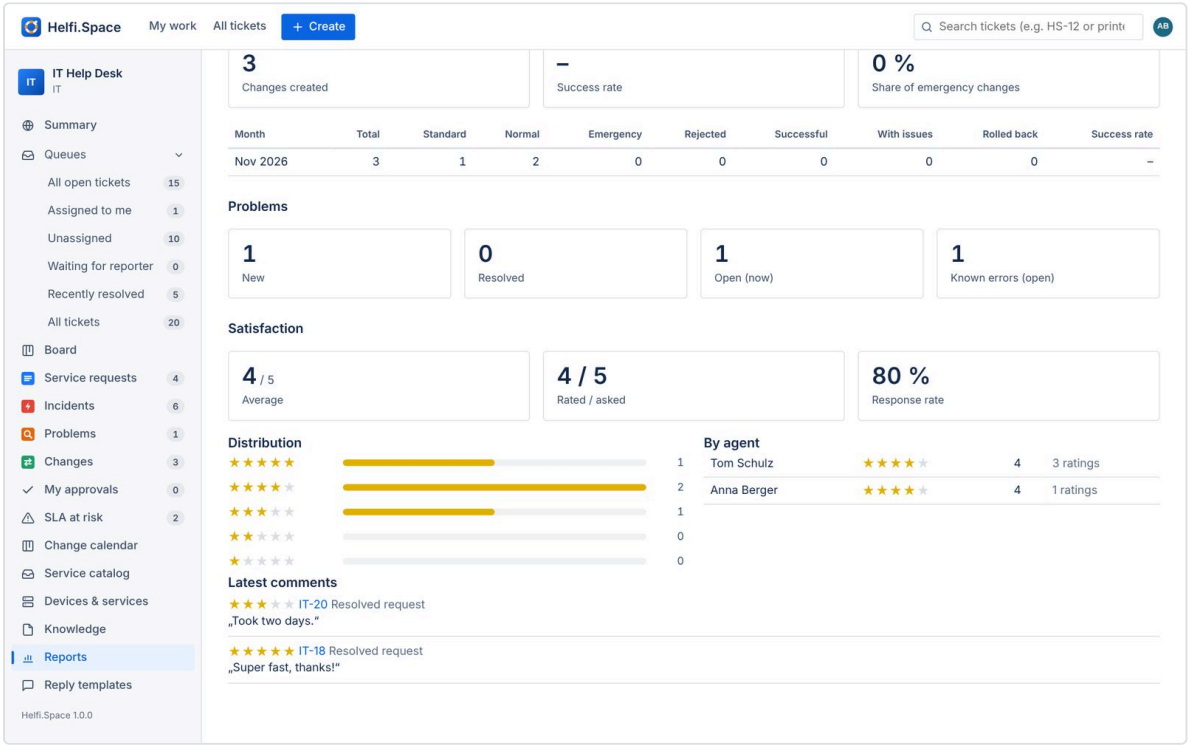
Under *Reports*, the agent statistics (3.9) are followed by further figures for the same period:

- **Service quality:** resolved requests and incidents, resolved at first contact (at most one reply, no follow-up question, never reopened), reopen rate, major incidents.
- **SLA compliance** for response and resolution, overall and per priority.
- **Changes** per month: kinds, rejected, outcomes, success rate and share of emergency changes.
- **Problems:** new, resolved, open, known errors.

- **Satisfaction:** average, response rate, distribution of stars, values per agent and the latest comments.




ITIL key figures in the report




Satisfaction in the report

# 5 For administrators: setup ADMIN

## 5.1 Installation

You need Docker with Compose (Linux server, Synology …). Download `docker-compose.yml`, `env.example` and `update.sh` from the latest release, then:

```
mkdir helfi && cd helfi
cp env.example .env
nano .env                 # adjust at least HELFI_BASE_URL
docker compose up -d
docker compose logs helfi # shows the setup code
```

Afterwards only four things live on the server:

| | |
|---|---|
| `docker-compose.yml` | usually stays the same across updates |
| `.env` | your settings (address, port, version, PUID/PGID) |
| `update.sh` | manual update (optional) |
| `data/` | database, attachments, logo, backups, key file |

Important entries in `.env`:

| | |
|---|---|
| `HELFI_BASE_URL` | address under which Helfi.Space is reachable (for links in emails and SSO) |
| `HELFI_HTTP_PORT` | port on the server (default 8080) |
| `HELFI_VERSION` | `latest` (with the monthly security updates on patch day) or a fixed version, e.g. `1.5.0` or `1.5.0-s1` |
| `PUID` / `PGID` | owner of the files in `data/` (Synology e.g. `1026` / `100`; not 0) |
| `HELFI_TRUST_PROXY` | address of the reverse proxy (see next section) |
| `HELFI_FORCE_HTTPS` | `auto` (default: switch once HTTPS came through the proxy), `1` (always) or `0` (never) |
| `HELFI_HSTS` | `on` (default: only where Helfi.Space had to redirect itself), `off` (never) or `preload` |
| `HELFI_SECRET` | optional: own key for stored passwords – set it only during setup |
| `TZ` | time zone, e.g. `Europe/Berlin` |

## 5.2 HTTPS and reverse proxy

Helfi.Space itself speaks HTTP on port 8080. In production put a reverse proxy with HTTPS in front of it (Synology reverse proxy, Nginx Proxy Manager, Caddy, Traefik …). Then set

`HELFI_BASE_URL=https://helfi.firma.local` – cookies are marked `Secure` automatically.

```
helfi.firma.local {
    reverse_proxy 10.0.10.20:8080
}
```

> Enter the proxy's address in `HELFI_TRUST_PROXY`, e.g. `10.0.10.5` (several separated by commas: `loopback, 10.0.10.5`). Without it Helfi.Space only trusts `X-Forwarded-For` from its own host (`loopback`); values such as `true` or a number are ignored with a warning in the log. If a proxy listed by its address (not a range) forwards a request without `X-Forwarded-For`, it counts as from outside. If the proxy runs on another host, Helfi.Space then only sees the proxy: the brake against password guessing then hits everyone together, and every request counts as "from outside" to be safe.

#### Redirect http:// to https://

Many proxies – including the Synology reverse proxy – cannot redirect custom addresses to HTTPS themselves; whoever types `http://…` otherwise lands on the Synology's start page. Helfi.Space does it: as soon as the `https://` address from `HELFI_BASE_URL` has been opened once over HTTPS through the configured proxy, requests the proxy forwards over HTTP (`X-Forwarded-Proto: http`) are redirected there permanently. A proxy without any HTTPS is therefore never redirected into nothing; `HELFI_FORCE_HTTPS=1` redirects right away, `0` never. The target never comes from the request. Add a second rule for this:

1. *Control Panel → Login Portal → Advanced → Reverse Proxy → Create*.
2. Source: protocol **HTTP**, hostname as in the HTTPS rule, port **80**.
3. Destination: exactly as in the HTTPS rule (HTTP, host, port 8080). The Synology sets the `X-Forwarded-Proto` header itself.
4. **Check Let's Encrypt:** port 80 stays open on the router. Afterwards renew the certificate once under *Control Panel → Security → Certificate*. If that fails, delete the HTTP rule again – the existing certificate stays valid.

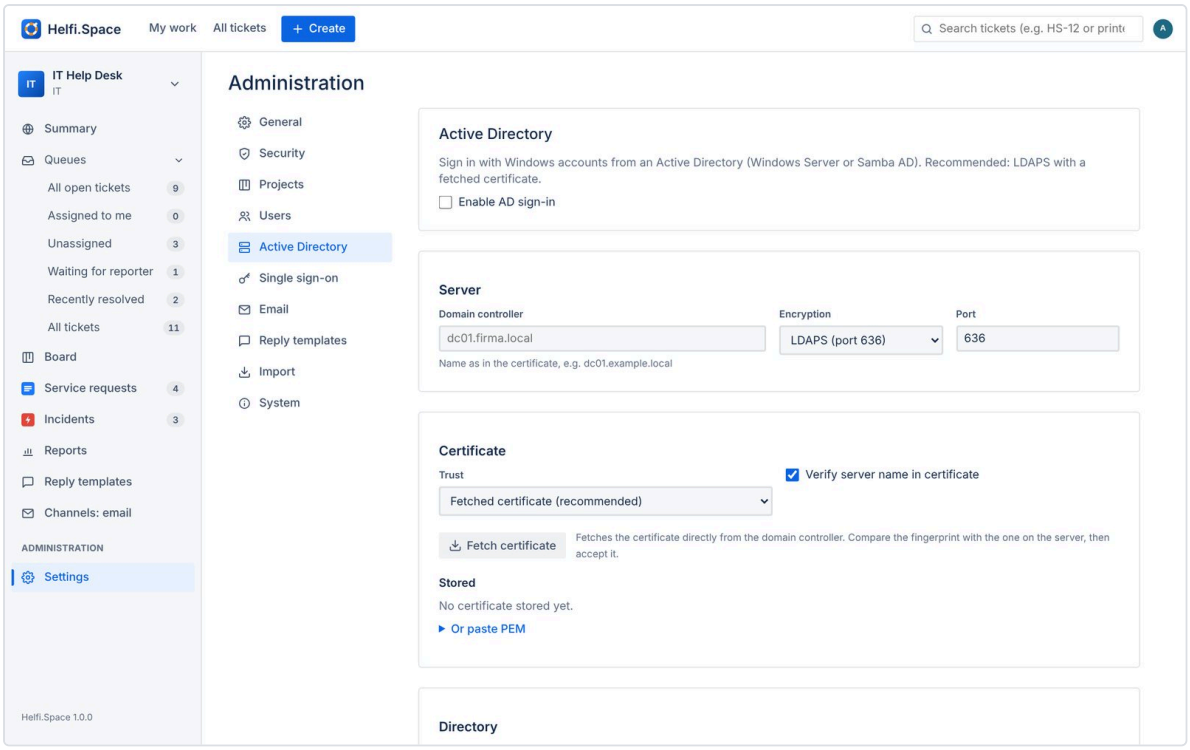
**HSTS** (one year, this address only) is sent by Helfi.Space only where it had to redirect itself – as behind the Synology. A proxy that redirects on its own (Nginx Proxy Manager with "Force SSL") keeps its own HSTS header, so there are no duplicate headers, which checkers such as the Mozilla HTTP Observatory count as an error. `HELFI_HSTS=off` switches HSTS off entirely. `HELFI_HSTS=preload` adds `preload` (only if all subdomains speak HTTPS for good).

## 5.3 Initial setup

On the first visit to `HELFI_BASE_URL` the setup wizard starts. It asks for the **setup code** from the container log (`docker compose logs helfi`) and creates the **local administrator** and the first project. It then continues straight to the AD setup.

## 5.4 Active Directory

*Administration → Active Directory*




Active Directory settings

1. Enter the **domain controller** with the name in its certificate (e.g. `dc01.firma.local`). Encryption: LDAPS (636) or StartTLS (389).
2. **Fetch certificate**, compare the SHA-256 fingerprint with the one on the server, then **Accept certificate**.
   Windows: `certlm.msc` → Personal → certificate of the DC. Samba: `openssl x509 -in /var/lib/samba/private/tls/cert.pem -noout -fingerprint -sha256`
3. Enter a **service account**, e.g. `svc-helfi@firma.local` – a normal domain account without special rights.
4. **Groups** for administrators and agents, e.g. `Helfi-Admins` and `Helfi-Agents` (name or DN, nested groups count). Optionally a group of "Allowed users". In addition every project can have a **project group** (Agents per project).
5. **Test connection**, ideally with a test sign-in – the test also checks the project groups and, with a test sign-in, names the person's projects. An empty base DN is filled in; a Microsoft certificate authority (AD CS) that is found can be accepted too, so renewed DC certificates stay valid.
6. **Enable AD sign-in** and save.

Users are created at their first sign-in or when an agent adds them as a participant from AD (**Suggest participants from AD**, on by default, requires the service account). Their role follows the groups at every sign-in unless an admin locked it under *Users* with "Role locked". Helfi.Space checks daily whether the certificate still matches and whether it expires soon, and shows problems as a notice.

## 5.5 Single sign-on

*Administration* → *Single sign-on* (OpenID Connect: Microsoft Entra ID, ADFS, Keycloak, Authentik ...)

1. Create an app at the provider and enter the redirect URI shown, e.g. `https://helfi.firma.local/auth/oidc/callback`.

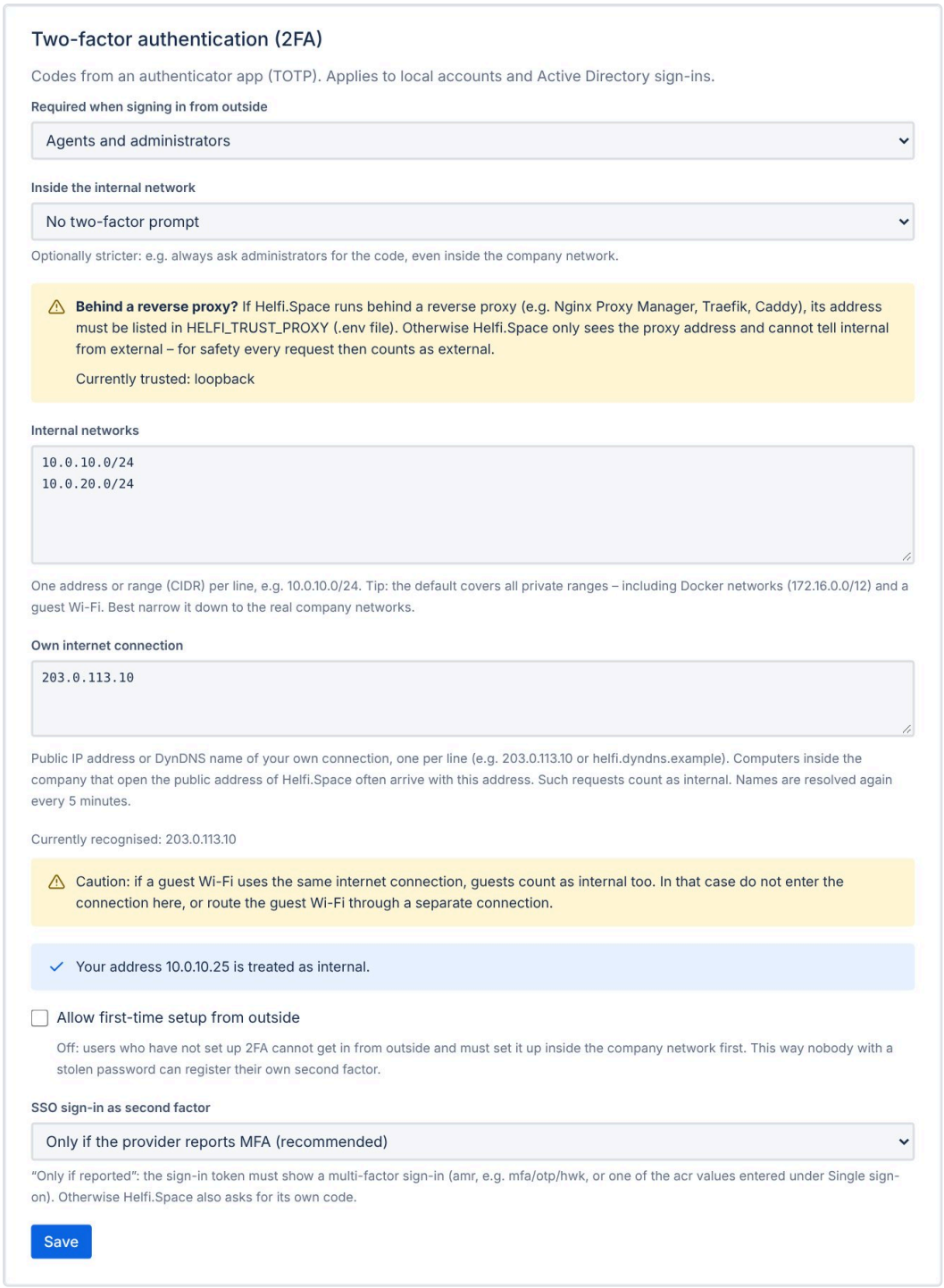
2. Enter issuer URL, client ID and client secret and click **Test**. Entra ID: `https://login.microsoftonline.com/<tenant ID>/v2.0` and add the groups claim under *Token configuration*.
3. Enter the admin and agent groups (group IDs for Entra ID).
4. Optionally “redirect straight to SSO”. The normal sign-in remains available at `/login?local=1`.

**SSO and AD together:** with “Merge with AD accounts for domain” (e.g. `firma.local`), `m.muster@firma.local` from SSO is the same account as the AD user `m.muster`. Only enter it if the provider does not allow guest accounts from other organisations. If the token contains only the email address (no `preferred_username`/UPN), accounts are only merged if the provider reports it as verified (`email_verified`). Email addresses from SSO are only taken over if they are verified or equal the user name of the allowed domain.

**Stable via the AD identifier (recommended):** under *Claim with AD identifier* enter the claim holding `objectSid` or `objectGUID` – `onprem_sid` (optional claim) for Entra ID, a custom mapping for Keycloak/Authentik. Then accounts are merged only this way, not by freely chosen names. With a service account the AD account is checked on every SSO sign-in (disabled, group, role); if the AD is unreachable, the SSO sign-in is cancelled.

## 5.6 Two-factor authentication and networks

*Administration → Security*




Two-factor requirement, internal networks and own internet connection

| Setting | Options |
|---|---|
| Required when signing in from outside | all users (default) · agents and administrators · administrators only · nobody |
| Inside the internal network | no prompt (default) · administrators only · agents and administrators · all users |

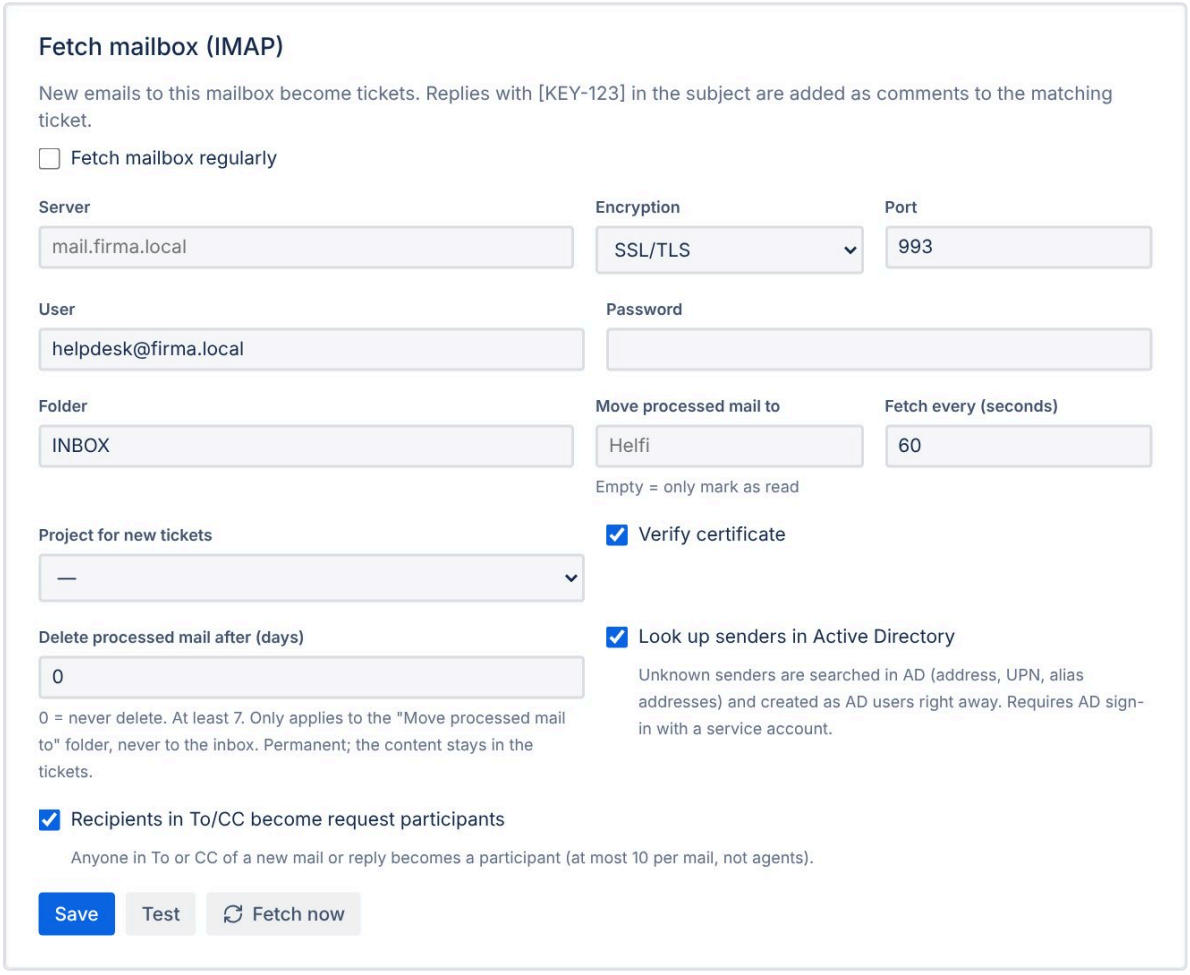
- **Internal networks:** one address or range per line. The default covers all private ranges, including Docker networks and a guest Wi-Fi – best narrow it down to the real company networks, e.g.

`10.0.10.0/24`.

- **Own internet connection:** public IP or DynDNS name (e.g. `203.0.113.10` or `helfi.dyndns.example`). Requests from there count as internal; names are resolved again every 5 minutes. Careful with a guest Wi-Fi on the same connection.
- **Reverse proxy:** without a matching `HELFI_TRUST_PROXY` every request counts as external; the page shows the current value.
- **Allow first-time setup from outside:** off by default.
- **SSO sign-in as second factor:** *Only if the provider reports MFA* (default: the token shows a multi-factor sign-in via `amr` or an entered `acr` value), *Always* or *Never*. Otherwise Helfi.Space asks for its own code after the SSO sign-in. Installations with SSO enabled keep "Always" on update.
- **Reset 2FA** for a person: *Users → Reset 2FA*.

## 5.7 Email

*Administration → Email*

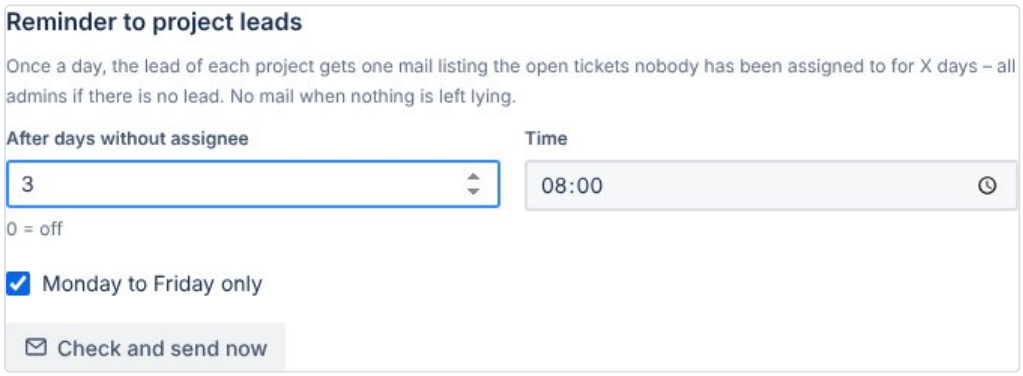



Fetching a mailbox (IMAP)

- **Fetch mailbox (IMAP):** server, user and password of the help desk mailbox. New emails become tickets, replies become comments. Processed emails are marked as read or moved to a folder – a separate folder is recommended. Skipped emails and delivery failure reports stay in the inbox, marked as read, and are not deleted automatically. Processed emails get their own flag; if the

mailbox is read in Outlook in parallel, they are still not lost (if the mail server supports custom flags, otherwise better not read it in parallel).

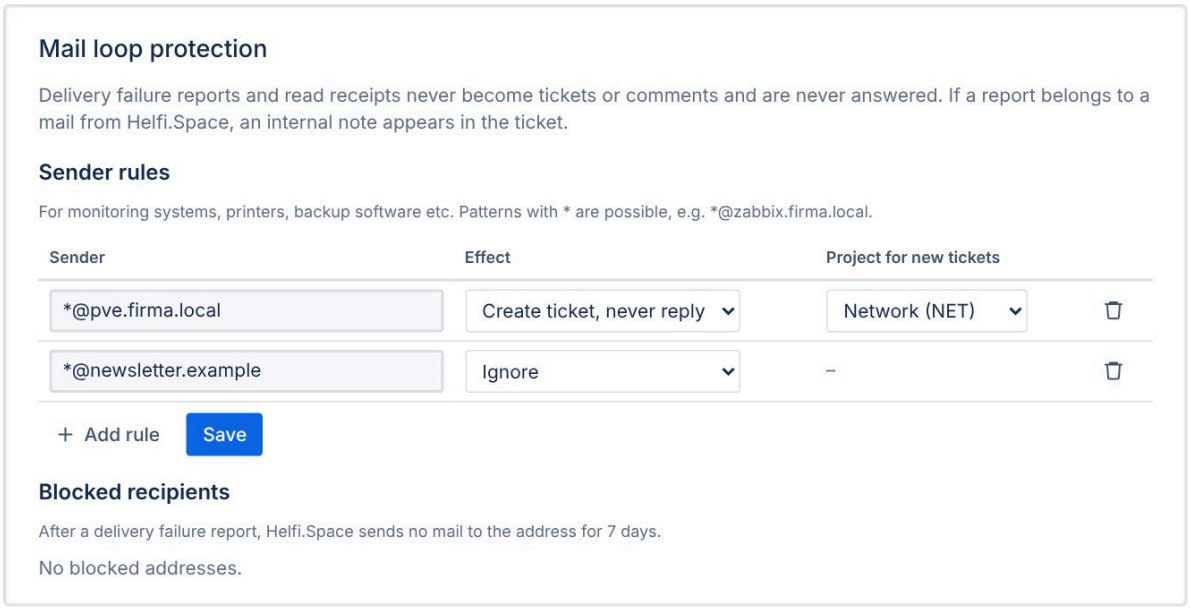
- **Replies by ticket key** in the subject are only accepted from the reporter, request participants and everyone who has already written in the ticket; anyone who merely guesses the key gets a new ticket. Replies to mails from Helfi.Space are matched via the Message-ID.
- **Sending (SMTP):** ideally use the mailbox address as sender, so replies come back into the ticket.
- **The sender is always the ticket system** – never an agent's address, no personal signatures. So the reporter knows who looks after them, reply and resolved mails end with "Kind regards", the agent's name and the system name; if the ticket is assigned to someone else, "Your contact person: …" is added. In the browser, users see the field as "Your contact".
- **Look up senders in Active Directory:** unknown senders are searched in AD and created as AD users right away. Otherwise they become *email contacts*; when they sign in later, their tickets are theirs.
- **Notifications:** each can be switched off – receipt confirmation, replies and resolved to reporters and participants, assignment to agents, mail to newly added participants and to people @-mentioned.
- **Reminder to project leads:** whoever does not look into the ticket system every day would otherwise miss new tickets. Enter a number under *After days without assignee* (0 = off); then the lead of each project gets one mail a day at the set time (optionally Monday to Friday only): all open tickets nobody has been assigned to for that many days, with a link to each ticket and to the filtered list. If a project has no lead (or the lead is not an agent of the project), the mail goes to all admins. If nothing is left lying, no mail is sent. *Check and send now* sends right away – handy for trying it out.




Reminder to project leads

- **Recipients in To/CC become participants** (can be turned off) – for new tickets and replies to mails from Helfi.Space, not when a reply is matched only by the key in the subject.
- **Retention:** "Delete processed emails after … days" (default: never). Only the folder for processed emails is cleaned, never the inbox.

### Mail loop protection and sender rules

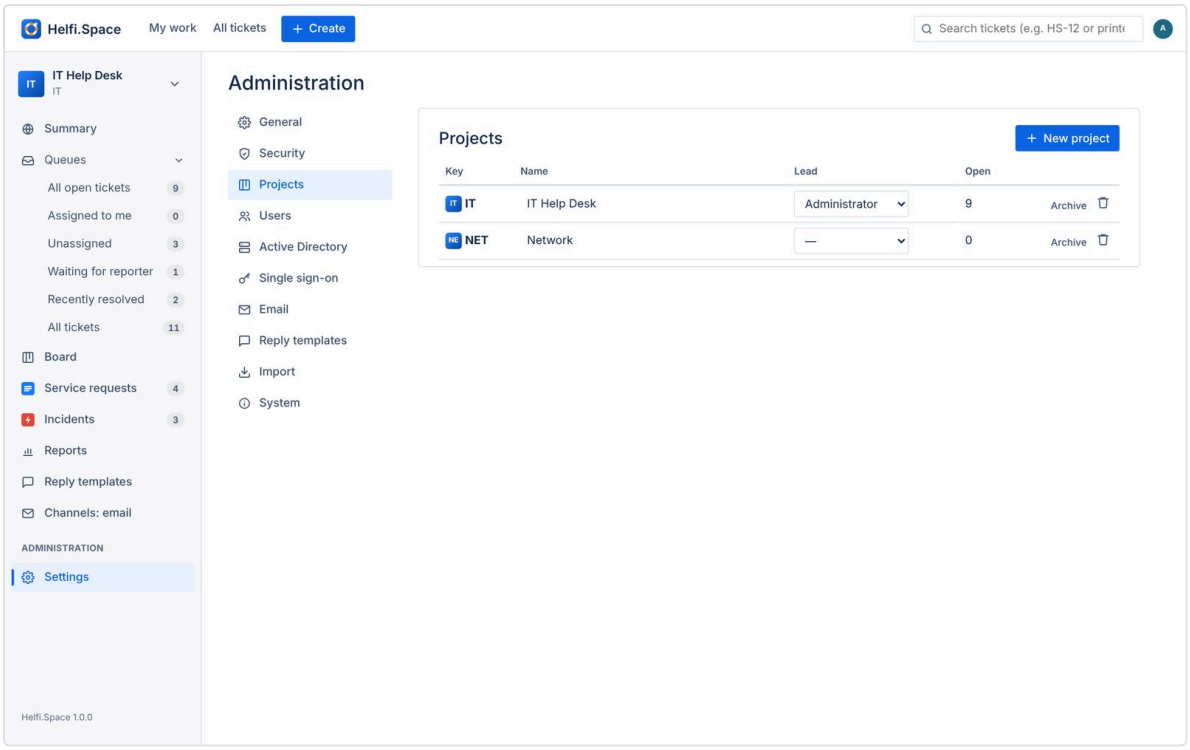



Sender rules and blocked recipients

- Delivery failure reports and read receipts never become tickets and are never answered. The affected address gets no emails for 7 days (*Blocked recipients*, can be released early there). Only addresses from real header lines of the report or from its `message/delivery-status` part are blocked – and only if the report refers to a mail Helfi.Space sent to exactly that address.
- **Sender rules** (address or pattern like `*@pve.firma.local`): “Create ticket, never reply” – optionally into a separate project – or “Ignore”.
- **Automatically generated emails** (Proxmox, NAS, monitoring; recognised by `Auto-Submitted`, `Precedence: bulk` or `noreply@`) are **only accepted with the rule “Create ticket, never reply”**. *Recently skipped emails* lists sender and reason.
- More than 20 new tickets per hour from the same sender are ignored.

## 5.8 Projects

*Administration → Projects*: create projects (key such as `IT` or `NET` and a name), set the lead, **archive** (kept with all tickets, disappears from the selection) or delete. A project with tickets can only be deleted if its tickets are moved to another project; a backup is made automatically beforehand.



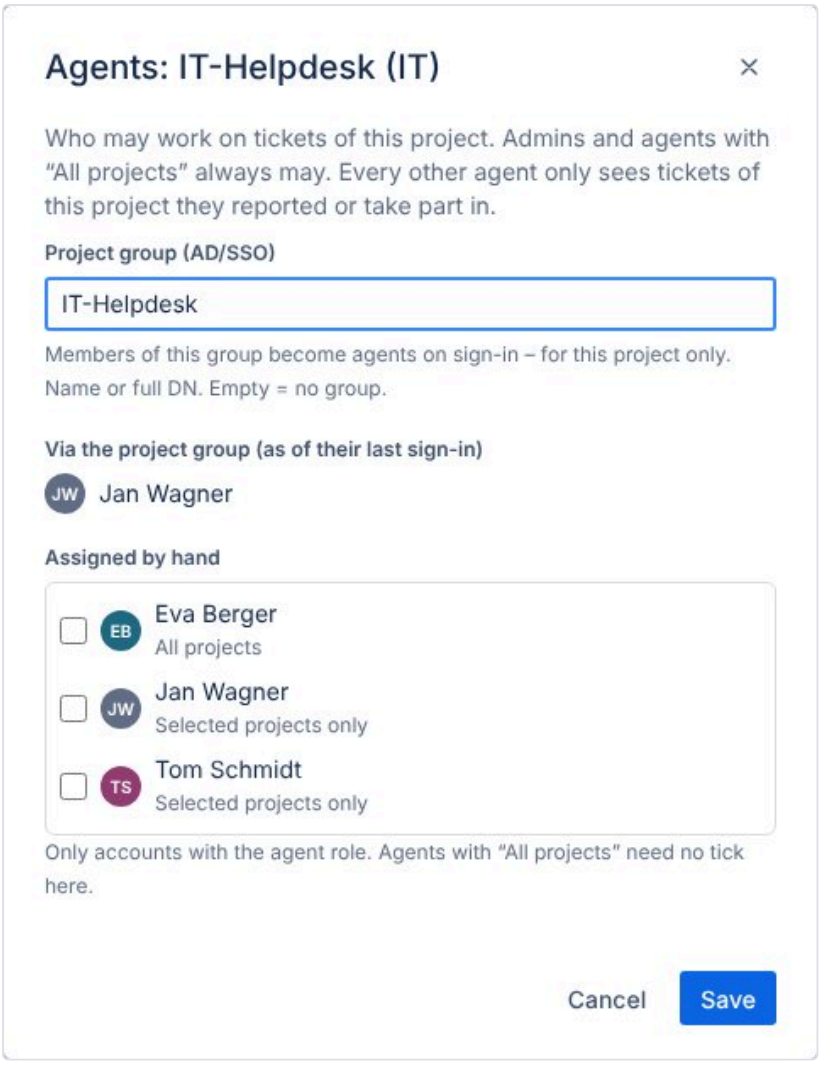

Projects

## Agents per project

By default every agent works on **all projects** – as before; nothing changes with the update. With several departments, e.g. IT and facilities, you decide who works on which tickets. An agent for **selected projects only** is a normal user for every other project: there they only see their own requests (and those they take part in) – without internal notes, history and agent tools.

In *Projects* the **Agents** button opens the window for it:

- **Project group (AD/SSO):** e.g. `IT-Helpdesk` for the IT project. Members of this group become agents of this project on sign-in; whoever is in several project groups works on several projects. Whoever is also in the general agent group stays responsible for all projects. A project group never turns anyone into a plain user and never removes admin rights. Membership is checked at every sign-in; when you change the group, the old memberships stop counting at once (until the next sign-in).
- **Assigned by hand:** tick agents – for accounts without AD or in addition. Agents with “All projects” need no tick.



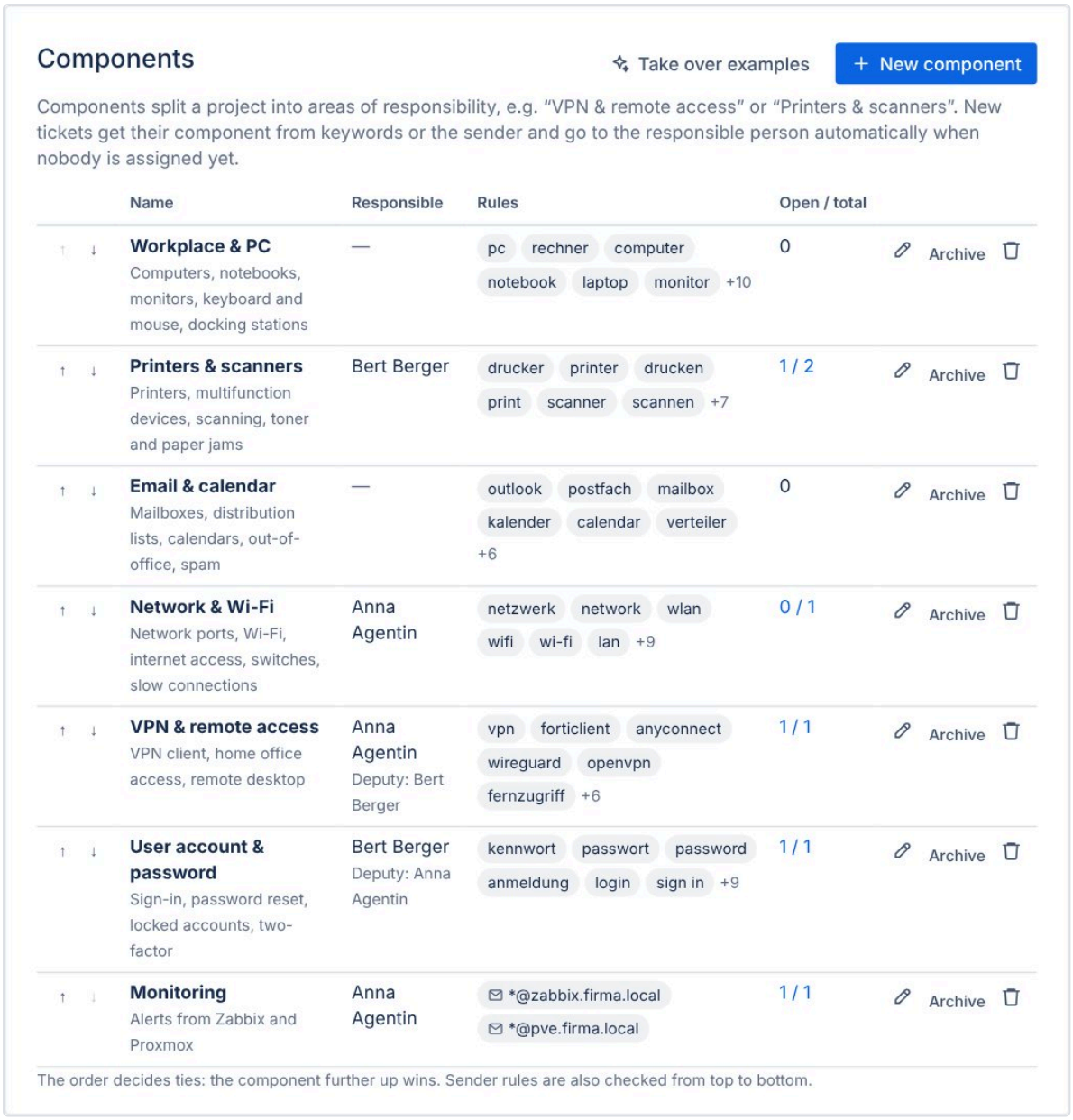

Agents of a project: project group and assignment by hand

**Restricted everywhere:** ticket lists and board, search (internal notes of other projects find nothing), queues and counts, summary, reports, SLA, problems, changes and major incidents, links, the ticket history of a device, AI functions and @-mentions in internal notes. Tickets can only be assigned to agents of the project and only be moved into one's own projects; a ticket moved into a project its assignee does not work on is unassigned afterwards. **Shared** are the knowledge base, reply templates, devices & services and the people search. Admins always work on all projects.

**Example:** enter the project group `IT-Helpdesk` for the IT project and `Facilities` for the HM project, and put the accounts into the matching AD groups; whoever works in both departments joins both. Take accounts out of the general agent group – otherwise they stay responsible for all projects.

## 5.9 Components

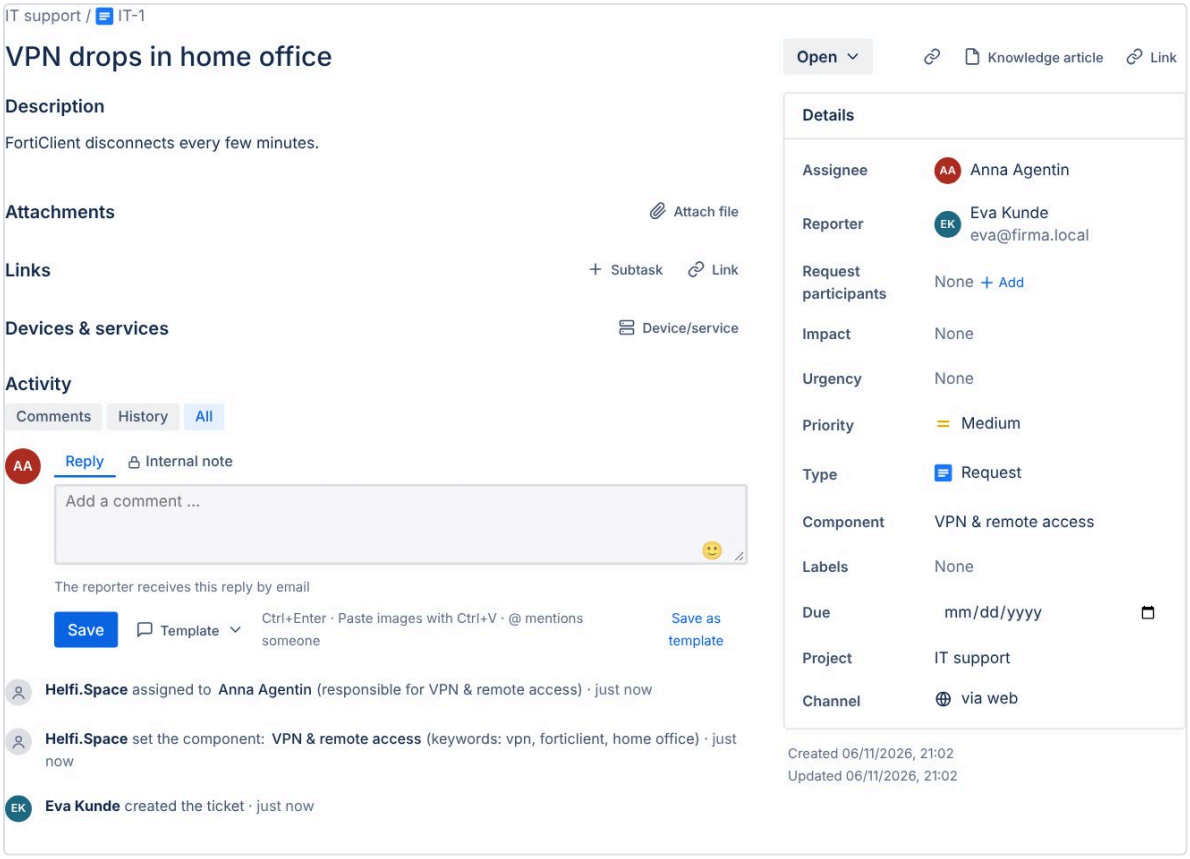
*Administration → Components* (or the *Components* button in *Projects*) splits a project into areas of responsibility, e.g. "VPN & remote access" or "Printers & scanners". Each component has a name, a short description, a **responsible person**, optionally a **deputy**, **keywords** and optionally **senders**. Only agents see components.




Components with responsible people and rules

- **How a ticket gets its component:** a matching **sender** (address or pattern such as `*@zabbix.firma.local`) counts first. Otherwise the **keywords** decide: a hit in the title counts 3, in the description 1, each keyword once. The component with the most points wins, on a tie the one further up – change the order with ↑/↓. From 4 letters a keyword also counts inside a word (“print” finds “printer”), shorter ones only as a whole word (“pc” does not find “PCIe”). If nothing fits, the field stays empty.
- **Assignment:** if nobody is assigned yet, the responsible person gets the ticket – if they are deactivated, the deputy. If an agent picks someone when creating the ticket, that choice stays.
- **The history** says why: “component VPN & remote access (keywords: vpn, forticlient)” and “assigned to Anna Agentin (responsible for VPN & remote access)”.
- **Changing:** agents pick the component when creating a ticket (“Automatic” lets the rules decide) and change it in the details; on phones under *All details*. Moving the ticket to another project clears it, because components belong to a project.

- **Archive or delete:** archived components stay visible on old tickets but are no longer assigned. Deleting empties the field on the tickets; the tickets themselves stay.
- **Import:** if the source system provides components, the first component of a ticket comes along and is created in the target project if it does not exist yet. Add the responsible person afterwards.




Ticket with component; the history shows the reason and the assignment

## Taking over examples

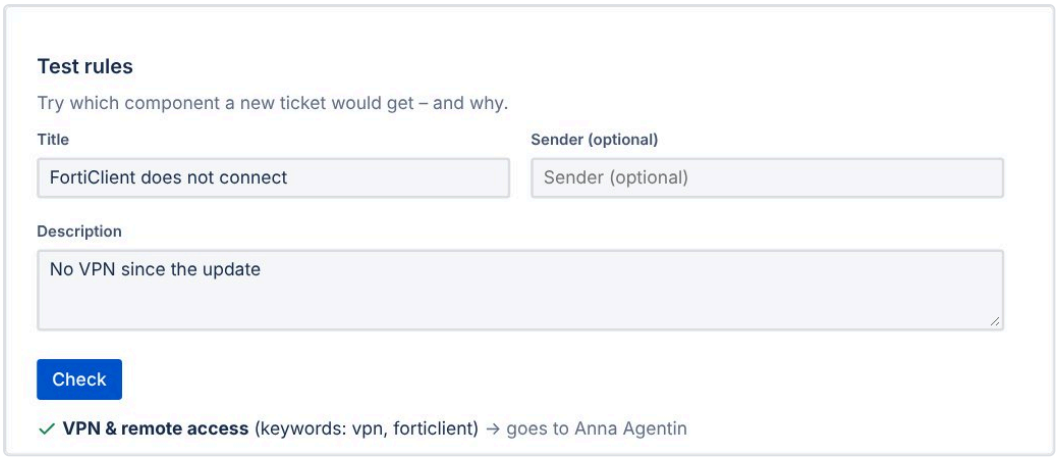
*Take over examples* offers ready-made components with description and German and English keywords: **IT support** (workplace & PC, printers & scanners, email & calendar, network & Wi-Fi, VPN & remote access, user account & password, telephony & mobile devices, software & licences, servers & storage, website & intranet), **facilities** (heating & air conditioning, electrics & lighting, locks & keys, rooms & furniture) and **administration / service** (access & permissions, procurement, people / on-/offboarding). Those that already exist are greyed out.




Examples to tick

### Testing the rules

Below the list, enter a title, a description and optionally a sender. *Check* shows which component a new ticket would get, why, and who it would go to.

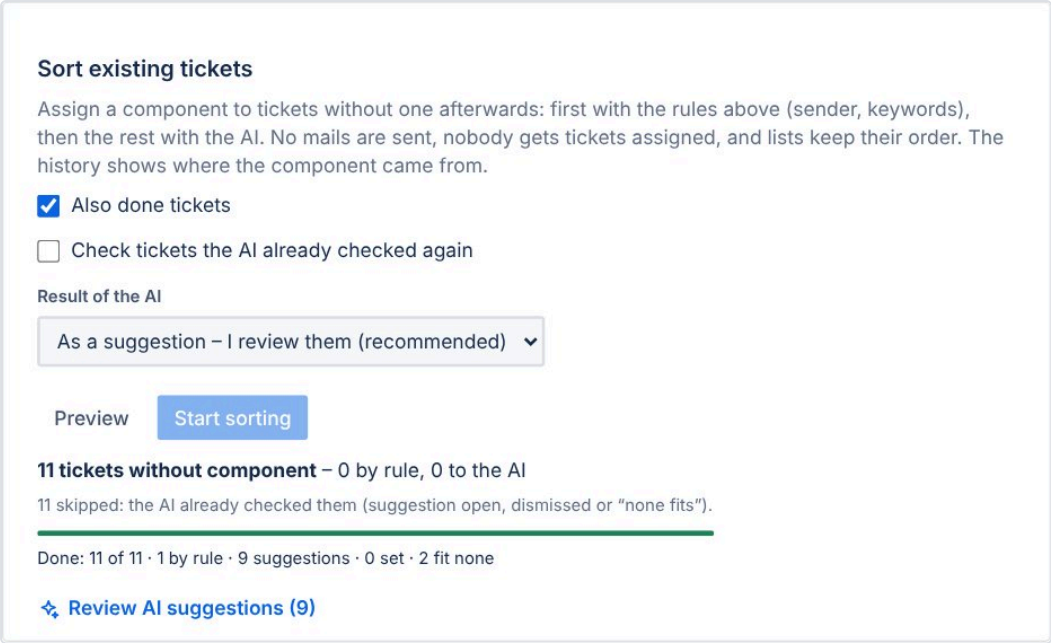



Testing the rules

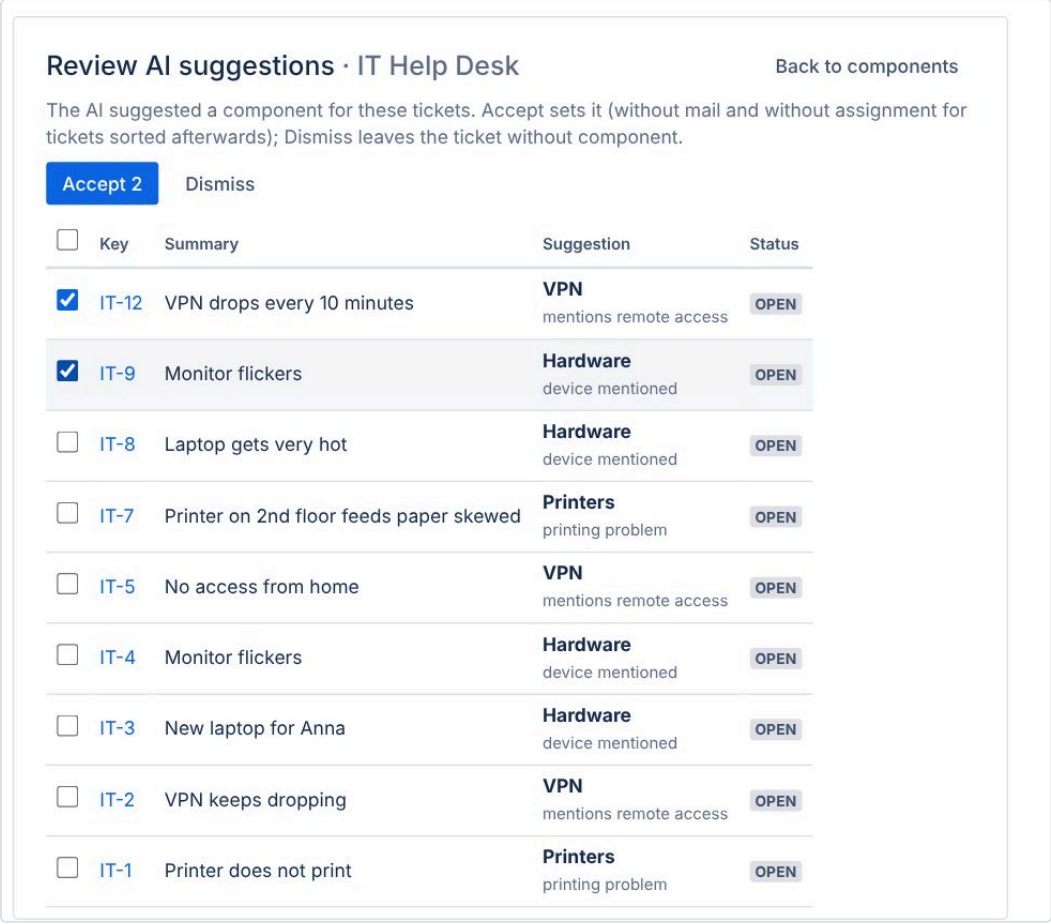
### Sorting existing tickets

After the import or when components are added, many tickets have no component yet. At the bottom of the page you assign one afterwards:

1. **Preview** shows how many tickets have no component, how many a rule matches and how many go to the AI. By default only open tickets count; done ones only with *Also done tickets* ticked. At most 1000 tickets go to the AI per run (start again for the rest); the confirmation before starting says how much text goes to the AI and points out possible costs for an AI on the internet.
2. **Start sorting** applies the rules (sender, keywords) first. If the AI function *Detect component* is on, all remaining tickets then go to the AI one after the other – in the background, with a progress bar, can be cancelled at any time. Helfi.Space asks first, as the AI receives title and description.
3. **Review AI suggestions:** by default the AI sets nothing itself. The list shows ticket, suggestion and reason; you accept or dismiss single ones or several at once. With *Set directly* the AI enters the component right away (“by AI” in the history).




Sorting existing tickets: preview and result




Reviewing AI suggestions and accepting several at once

When sorting afterwards, **no mails** are sent, nobody gets tickets **assigned**, and the "Updated" time stays – lists keep their order. Tickets the AI already checked (suggestion open, dismissed or "none fits") are skipped by a new run; *check again* asks once more, e.g. after you improved the component descriptions.

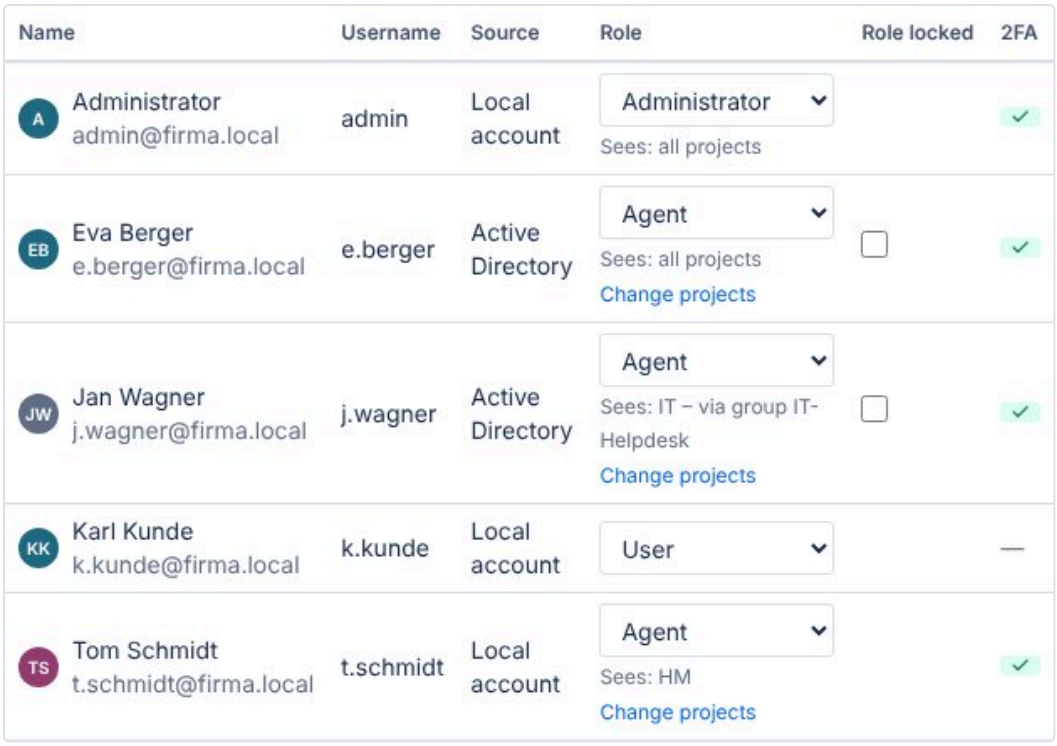
### Cutting components sensibly

- **By responsibility, not by technology:** a component is what one person (or a small team) is in charge of. Whoever looks after printers and scanners does not need two components.
- **6–12 rather than 40:** components that are too fine get picked wrongly and clutter the reports. Labels handle finer distinctions.
- **Every component needs a responsible person,** ideally with a deputy – otherwise tickets end up “Unassigned” again.
- **Keywords from real tickets:** use the words your reporters actually write (“Outlook”, “FortiClient”, “password expired”), not IT jargon. Avoid words found in almost every mail (e.g. “email” or “phone” from signatures).
- **Check with the test field** and after a few weeks look at *Reports → By component*: many tickets “Without component” mean keywords are missing.
- **Senders for systems:** monitoring, backup or printers that send mails themselves are matched reliably with a sender pattern.

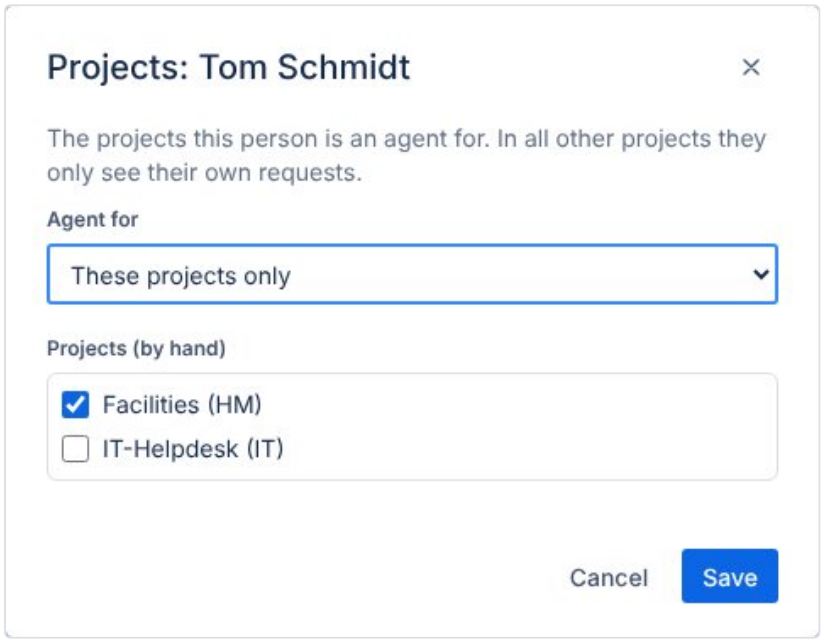
## 5.10 Users

*Administration → Users* lists all accounts with origin (local, AD, SSO, email contact), role, 2FA status and last sign-in. Here you create local users, change roles (“Role locked” prevents AD groups from overriding them), deactivate accounts and reset 2FA. Self-registered accounts are marked; if approval is switched on under *Security → Registration and forgotten passwords*, new accounts show “Awaiting approval” with an **Approve** button.

Under each agent and admin role the list shows what the person sees, e.g. “Sees: IT – via group IT-Helpdesk”. **Change projects** sets “All projects” or “These projects only” and assigns projects by hand; projects from a project group are noted there. A change here sets “Role locked”, so AD groups do not overwrite it (Agents per project).




What someone sees is shown under the role



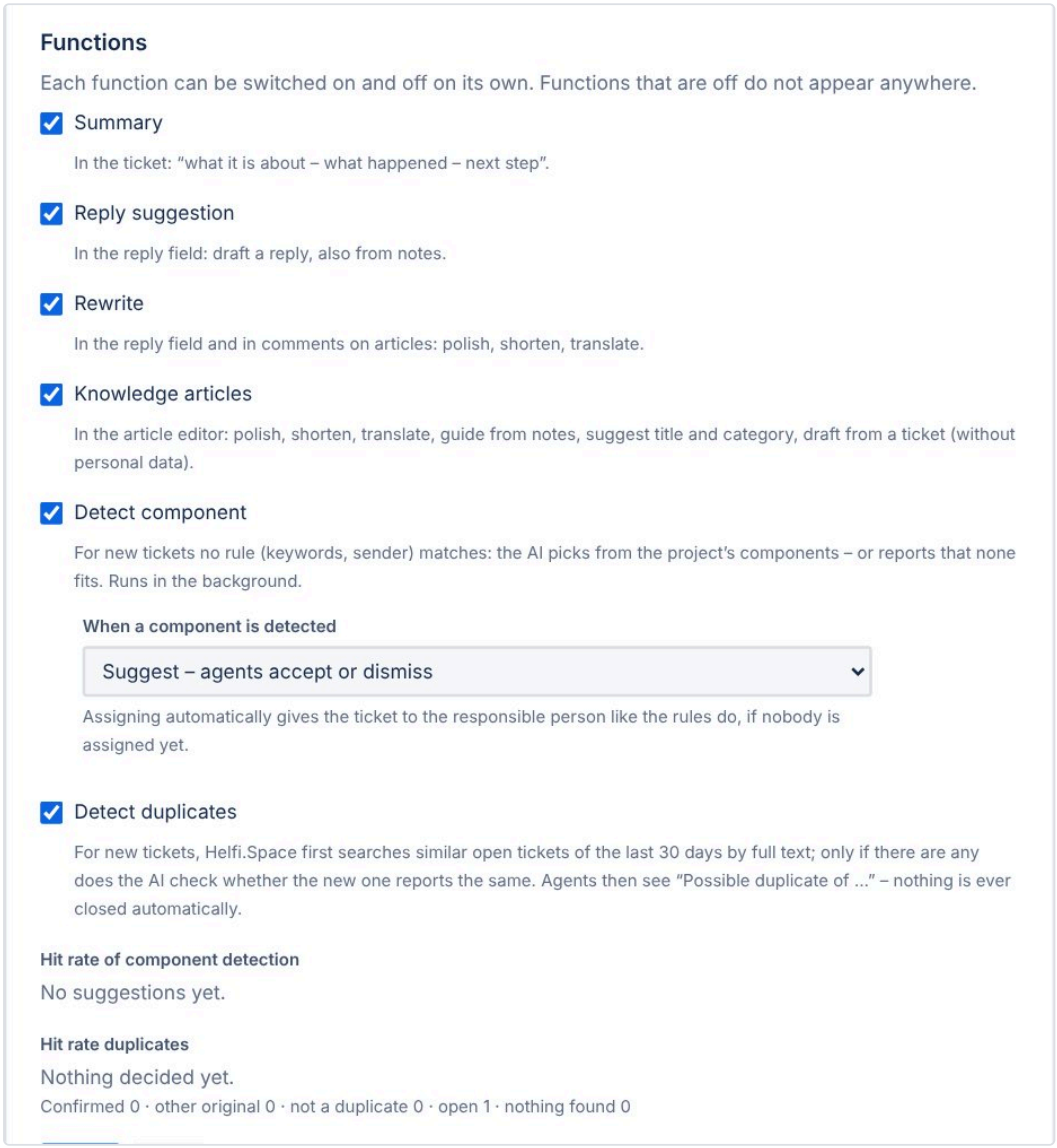

Setting the projects of an agent

**Automatic sign-out** (*Administration → Security*): period without activity for everyone (default 12 hours from `HELFI_SESSION_HOURS`), for admins (default 60 minutes) and optionally for access from outside – the shortest applicable one counts. Two minutes before, Helfi asks “Still there?”; whoever keeps working stays signed in. Optional: “Sign out when the browser is closed”.

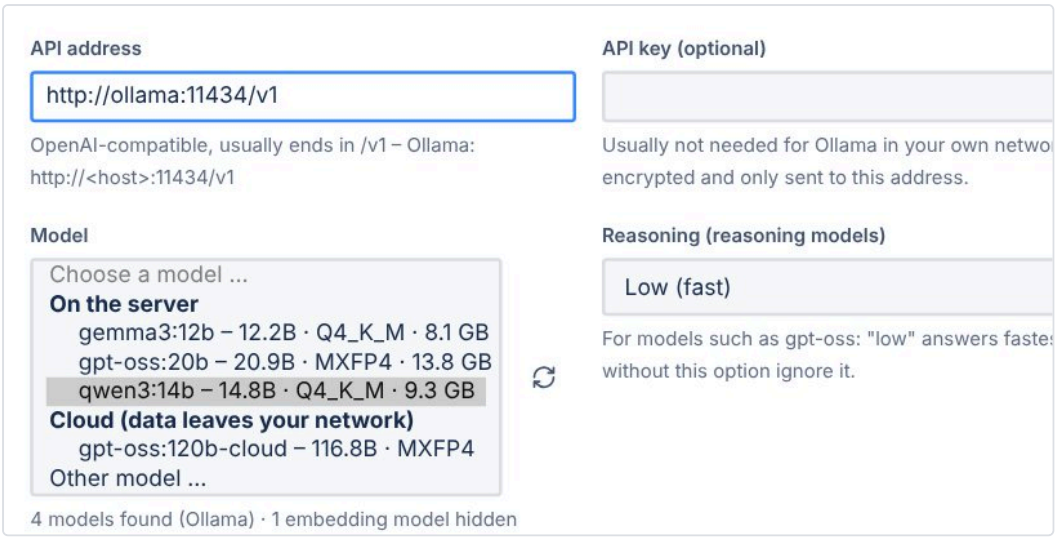
**Registration and forgotten passwords** (*Administration → Security*, registration off by default): external people create a local account with the role “User” themselves; the email address is confirmed via a link, and an existing email contact becomes the account with its requests. Optional: allowed mail domains and approval by an admin. “Forgot password” applies by default to local accounts with the role “User”, optionally to all local accounts or not at all. Both require outgoing mail.

**AI** (*Administration → AI*, off by default): address of an OpenAI-compatible API (e.g. Ollama: `http://<host>:11434/v1`) and optionally an API key (stored encrypted). The **installed models are fetched automatically** and offered in a list – for Ollama with parameter count, quantisation and size, cloud models in a group of their own. Embedding models are hidden; if a model is missing, use *Reload* or *Other model …*. Recommended for production is a self-hosted model in your own network; for cloud models or external addresses the page points out that ticket content leaves the house. Internal notes are only sent if switched on; attachments never. *Test* checks the connection.

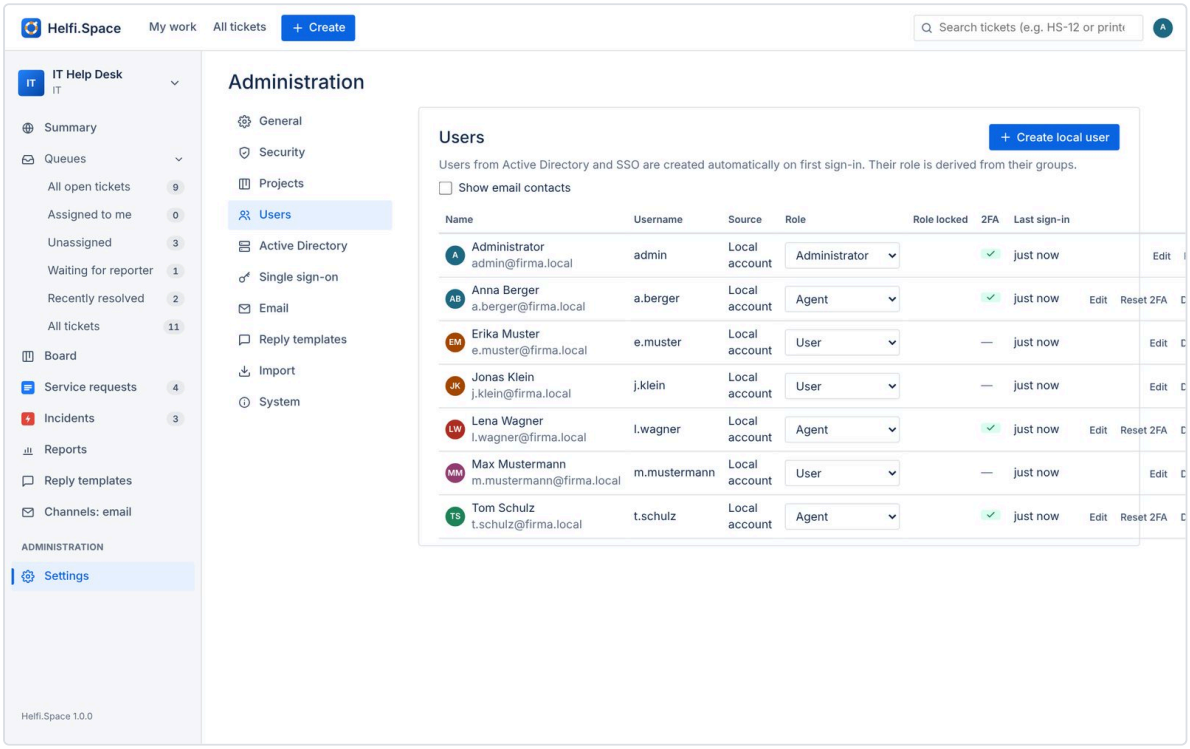
Under **Functions** you switch *Summary*, *Reply suggestion* and *Rewrite* and *Knowledge articles* (on by default) and *Detect component* and *Detect duplicates* (off by default) one by one. For detection you choose *Suggest* or *Assign automatically*. Below is the **hit rate** of components: accepted suggestions and automatic ones nobody changed, against corrected and dismissed ones; *Detect duplicates* has a hit rate of its own next to it. With a cloud model, *Detect component* sends the title and description of every new ticket without a rule match to the provider, *Detect duplicates* also the title and description (the first 1500 characters each) of up to 8 possible originals – the page points it out.




Switching AI functions one by one, with hit rate



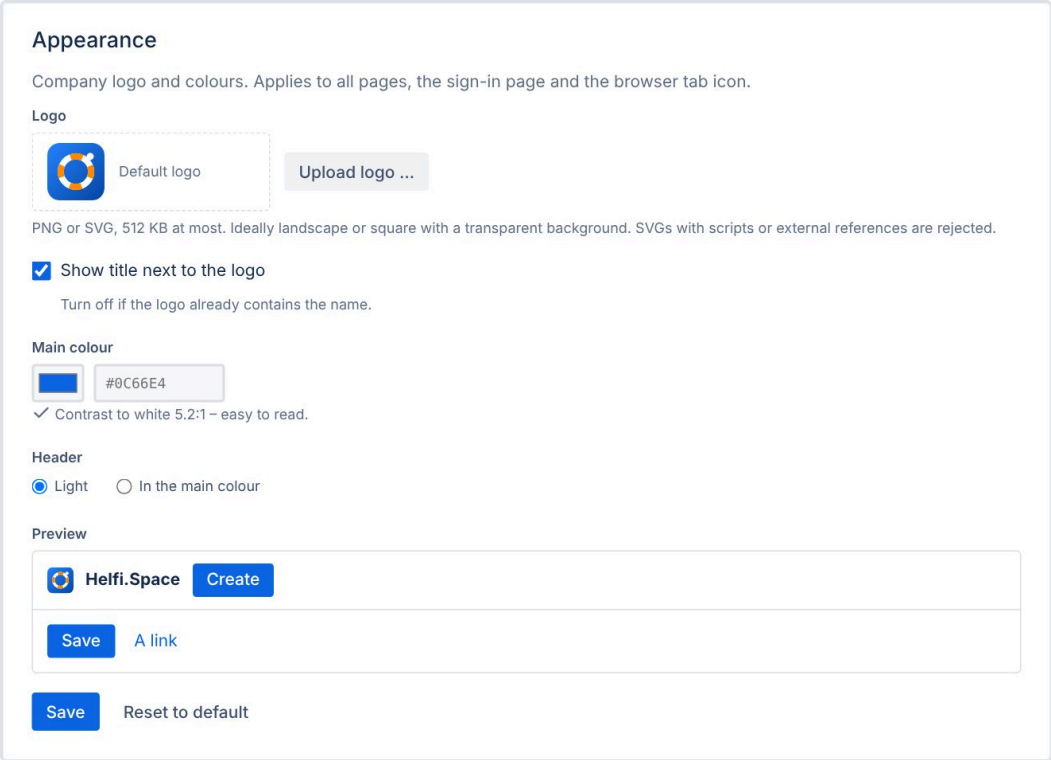

The models of the Ollama server to choose from – cloud models separate




User management

## 5.11 Appearance

*Administration → General → Appearance*

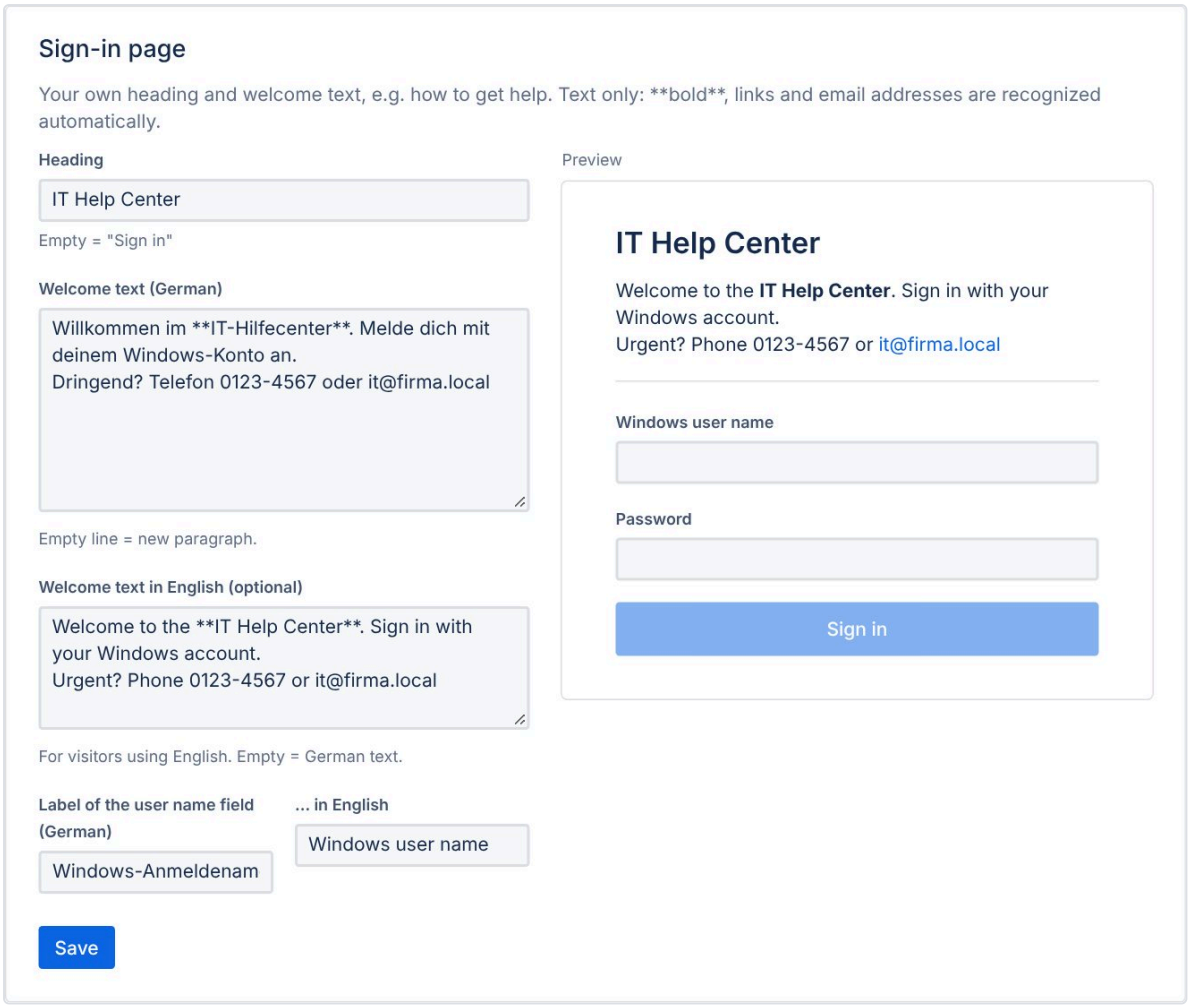



Logo, main colour and header with preview

- **Logo** as PNG or SVG (512 KB at most) for header, sign-in page and browser tab. SVGs with scripts or external references are rejected.
- **Main colour** with contrast check (warning below 4.5:1) and **header** light or in the main colour.
- *Reset to default* restores everything.

## 5.12 Sign-in page

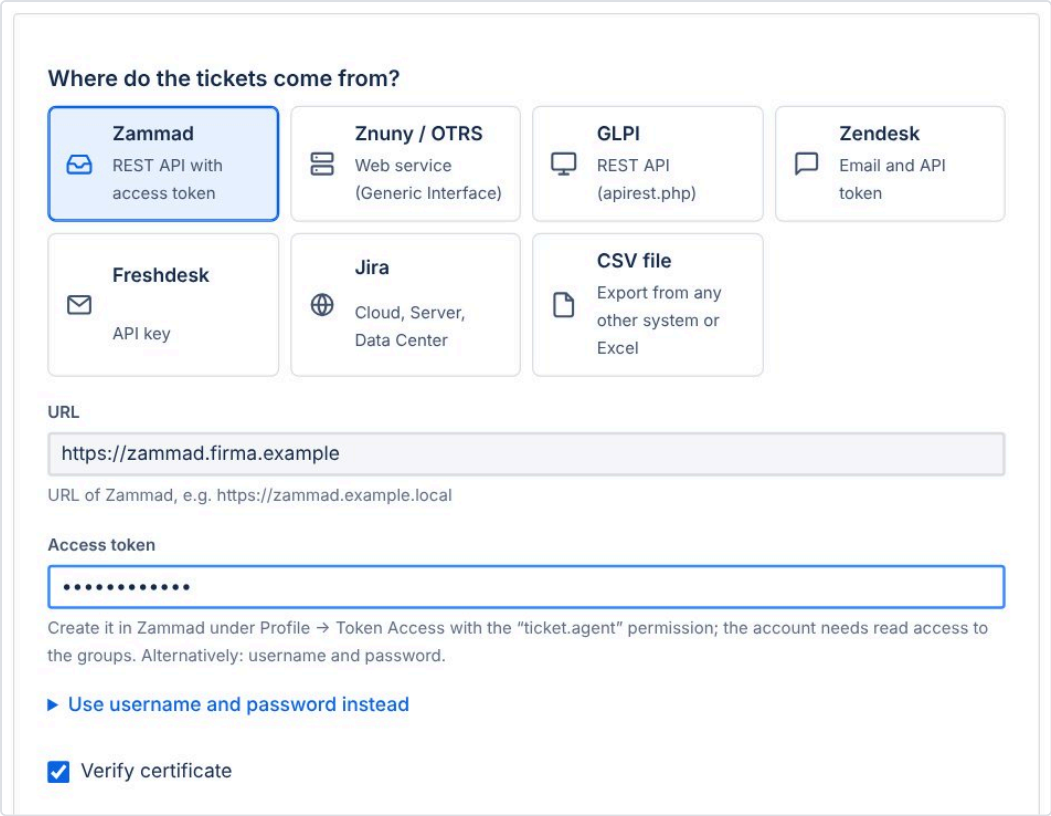
*Administration → General → Sign-in page*: your own heading, a welcome text in German and English (e.g. how to get help in an emergency) and your own label for the user name field (“Windows user name”). Text only: `**bold**`, links and email addresses become clickable, HTML is never executed.




Customising the sign-in page with preview

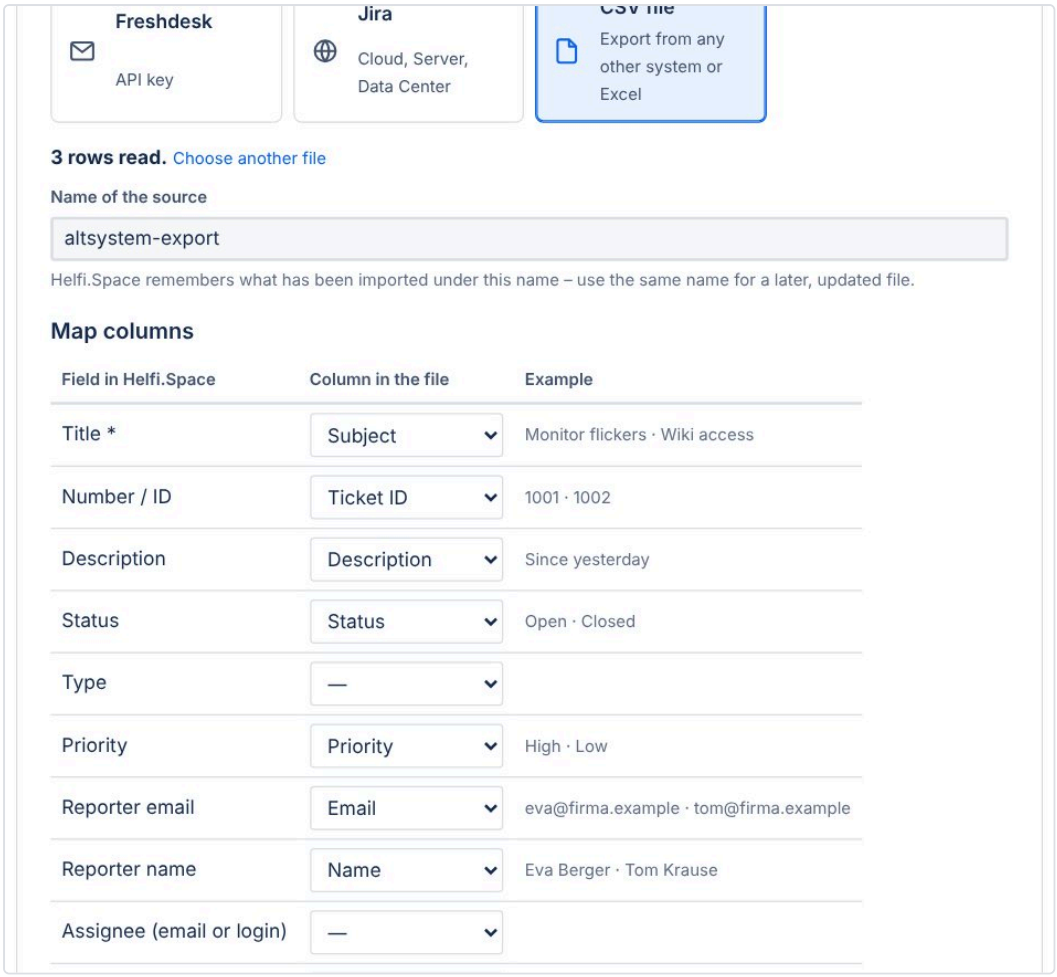
## 5.13 Import from another ticket system

*Administration → Import* takes over tickets with comments and attachments from **Zammad**, **Znuny / OTRS**, **GLPI**, **Zendesk**, **Freshdesk**, **Jira** (cloud, Server/Data Center) or a **CSV file**. Credentials are not stored.




Choosing the source – here Zammad with an access token

1. **Choose the source and connect:**
   - *Zammad:* address and access token (in Zammad under *Profile → Token Access*, permission `ticket.agent`).
   - *Znuny / OTRS:* once in Znuny, import the bundled example `GenericTicketConnectorREST` under *Admin → Web Services*; in Helfi.Space enter the full web service URL and login and password of an agent with read access.
   - *GLPI:* enable the REST API in GLPI (*Setup → General → API*), address up to `apirest.php` and the personal token (*Settings → remote access key*). Entities become projects.
   - *Zendesk:* address, email and API token; with an administrator account archived tickets come along too.
   - *Freshdesk:* address and API key from the profile (at most 30,000 tickets via the list).
   - *Jira:* address and API token (cloud) or personal access token (Server/Data Center).
   - *CSV file:* upload it and map the columns – recognised columns are prefilled, only the title is required. A *project / queue* column spreads the tickets over several projects.
2. **Choose projects** (groups, queues, entities or projects) and the target project.
3. **Map statuses and types** (suggestions are pre-filled).
4. **Import** – with progress, can be cancelled.




CSV file: mapping columns with example values

Ticket numbers are kept where possible (`SUP-123` stays `SUP-123`, Zammad `#31011` becomes `SD-31011`; very long, date-based numbers get a new one, the old one is kept in the history). Internal notes and their files stay internal. People are matched by email address, and no emails are sent. Before every import starts, Helfi.Space creates a backup. A second run only updates what changed and never creates duplicates – so you can import once as a trial and again on the cut-over day (for CSV with the same *name of the source*).

# 6 For administrators: operation ADMIN

## 6.1 Updates

### With container management (recommended)

Dockhand, Watchtower or Tugtainer detect new images of `ghcr.io/soleil0815/helfi.space.ticket:latest` and update the container – ideally at night. Besides new versions, a security update `<version>-s<n>` appears once a month on patch day (the second Tuesday): same program, only newer packages, tested and checked like every release. Helfi.Space backs up the database itself when a new version starts ( `data/backups/helfi-vor-<version>-….sqlite` ) and then migrates it.

### Manually

```
./update.sh             # latest version
./update.sh 1.5.0       # specific version (recorded in .env)
./update.sh -y          # without questions
./update.sh --cleanup   # show leftovers of earlier updates and delete them after
confirmation
./update.sh --offline helfi-1.5.0-amd64.tar.gz 1.5.0   # without internet
```

`update.sh` fetches the new version, takes over a newer `docker-compose.yml` (the old one is kept as `docker-compose.yml.alt-…` ), starts the container and waits for the health check. If the new version does not start, it restores the previous version together with its database backup; `.env` then holds the previous version again. It runs with `/bin/sh` , also on a Synology. `--cleanup` never touches `data/` , backups, `.env` or the current `docker-compose.yml` .

- **`docker compose up` fails:** error message, exit code 1, `.env` and `docker-compose.yml` are restored. If there is a previous version, `update.sh` first asks whether to start it again (automatically with `-y` ).
- **Still starting:** if the new version is still “starting” after `HELFI_UPDATE_WAIT` , `update.sh` asks whether to wait once more (automatically once with `-y` ).
- **Database not restored:** `update.sh` says so explicitly, e.g. if the backups are not readable (then `sudo ./update.sh -y` ) or there is no backup or only an incomplete one.
- **`--offline` without a version:** the version is taken from the image and recorded in `.env` , otherwise `update.sh` aborts. A relative path is taken from the current folder.
- `HELFI_UPDATE_WAIT` (time to wait, default 120 seconds) and `HELFI_IMAGE_REPO` are set on the command line, not in `.env` , e.g. `HELFI_UPDATE_WAIT=300 ./update.sh` . For your own registry, also change `image:` in `docker-compose.yml` ; it then counts as customized and is no longer replaced automatically.

> After an update the browser loads the new interface immediately (from version 1.5.0). If an older version still shows the old interface, press `Ctrl` + `Shift` + `R` once.

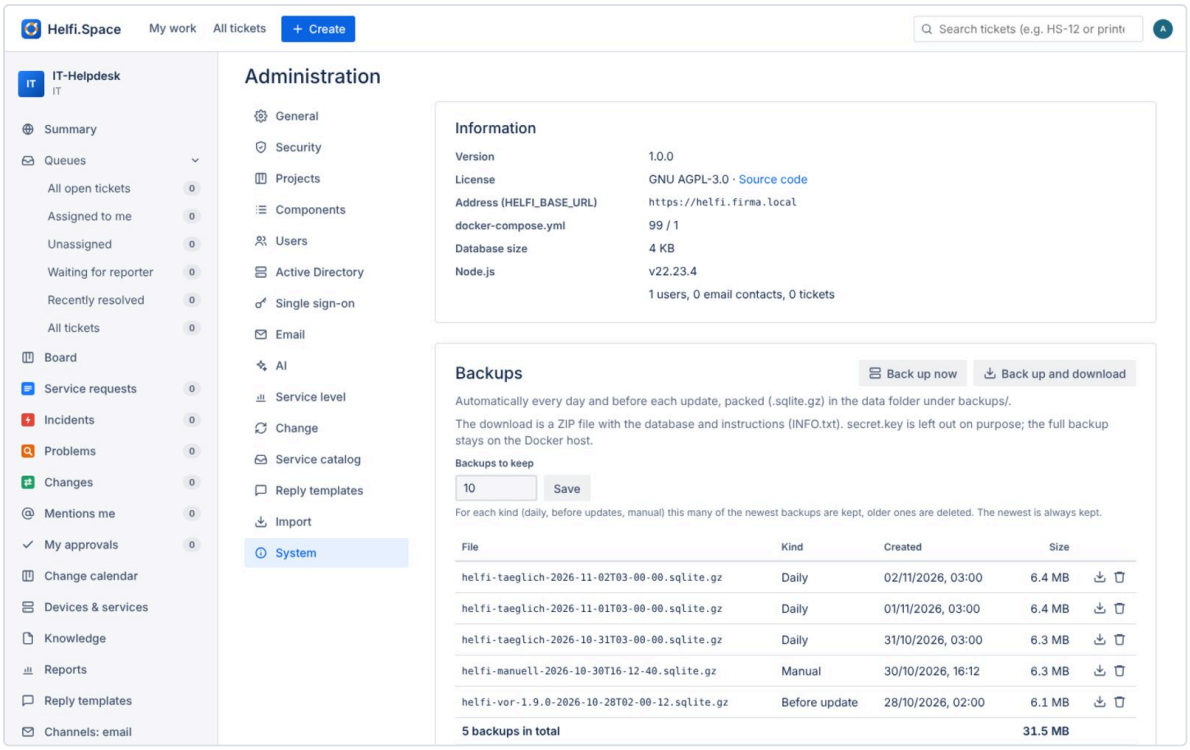
## 6.2 Pinning a version and going back

1. Enter a fixed version in `.env`, e.g. `HELFI_VERSION=1.4.0`.
2. If the newer version has already migrated the database, the older one does not start and names the matching backup in the log. Then:

```
docker compose down
cp data/helfi.sqlite data/helfi.sqlite.new
cp data/backups/helfi-vor-1.5.0-<date>.sqlite data/helfi.sqlite
rm -f data/helfi.sqlite-wal data/helfi.sqlite-shm
docker compose up -d
```

## 6.3 Backup and restore

- **Automatic:** daily, before every version change, before deleting a project and before every import starts, in `data/backups/`. A refused project deletion creates no backup. **Manual:** *Administration → System → Back up now*.
- **Backups to keep** (default 10): this many of the newest backups are kept per kind – daily, manual, before update, before project deletion and before import. Older ones are deleted when saving and from then on, at least the newest of each kind always stays. The list below shows kind, date, size and the total space used; 🗑️ deletes a single backup.
- **Download:** ⇓ next to a backup downloads it as a ZIP file, *Back up and download* creates a new one and downloads it right away. The ZIP holds the database (`helfi.sqlite`) and an `INFO.txt` with restore instructions. Sign-in sessions are removed from it so nobody can take over a signed-in session with the file – after restoring, everyone signs in again. `secret.key` and attachments are left out on purpose – the key would decrypt the stored passwords. A new installation therefore also needs `secret.key` from the Docker host. Every download is logged.
- **Packed:** backups are stored as `helfi-….sqlite.gz` in `data/backups/` and need only a fraction of the space. Each packed file is unpacked and compared with the original before it is stored. The backup before a version change stays unpacked for a day, because `update.sh` restores exactly this one if the new version does not start; afterwards it is packed too. Older unpacked backups are packed once in the background after startup. Restoring by hand: `gunzip -c data/backups/helfi-….sqlite.gz > data/helfi.sqlite` with the container stopped.
- Each backup is first written as a `.part` file and only renamed once complete. A `.part` file left behind after a crash is never used and is removed by the next clean-up after one hour.
- The backups contain only the database. **Always back up the whole `data/` folder** to another drive – including attachments, logo and `secret.key`.
- Without `secret.key` (or the `HELFI_SECRET` you set) stored passwords and 2FA keys can no longer be decrypted.




System: version, database, backups

## 6.4 Locked out? (emergency access)

Try the normal ways first:

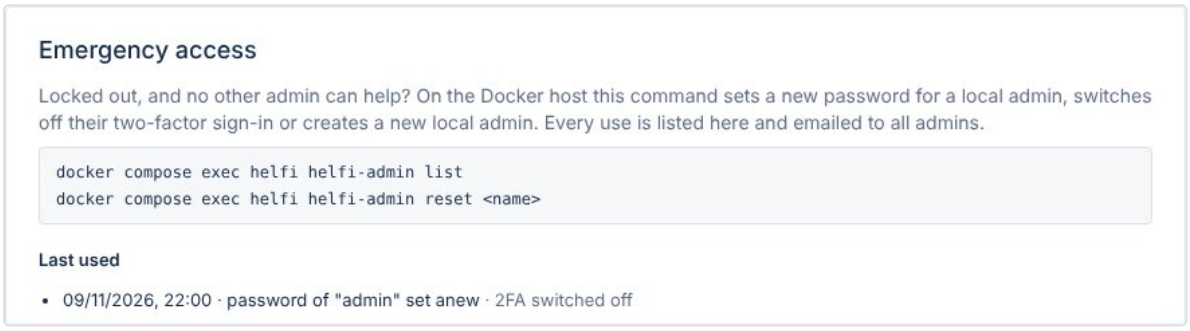
| Situation | Way out |
|---|---|
| AD unreachable | Sign in with the **local emergency admin** from the initial setup. |
| SSO redirects right away and fails | `https://…/login?local=1` opens the normal sign-in page. |
| Phone with the authenticator lost | Sign in with a **recovery code**, or another admin resets 2FA under *Users*. |
| Password forgotten | *Forgot password* on the sign-in page – for admins only if *Security → Registration and forgotten password* is set to "all local accounts" and mail goes out. |
| AD admin lost the admin group | The emergency admin or another admin gives the role back under *Users*. |
| Too many failed attempts | Wait 15 minutes or `docker compose restart helfi`. |

If none of that helps, there is the **emergency command** on the Docker host – in the folder with `docker-compose.yml`:

```
docker compose exec helfi helfi-admin list             # show local admin accounts
docker compose exec helfi helfi-admin reset admin      # new password, 2FA off, sessions
ended
docker compose exec helfi helfi-admin create notfall   # create a new local admin
```

- `reset` asks for the new password twice without echo (at least 10 characters). `--generate` creates a random password and shows it once; `--keep-2fa` keeps two-factor sign-in set up. A deactivated account becomes active again. Then sign in at `/login?local=1` and set up 2FA again.
- `create` adds a local admin (`--name "First Last"`, `--email address`), e.g. if the emergency admin was deleted.
- Without a terminal (script, scheduler): `printf '%s\n' "$PW" | docker compose exec -T helfi helfi-admin reset admin --password-stdin`.
- The command only works on **local** accounts; AD and SSO accounts are managed there.

> The command is only reachable with a shell on the Docker host – whoever has that can read the data folder anyway. It does not work over the network, and it runs as `PUID` / `PGID`, never as root. Every use is listed under *Administration → System → Emergency access* and emailed to all admins with an email address.




Administration → System: emergency access with the last use

## 6.5 Security

- Passwords of local accounts are stored with scrypt, stored credentials (service account, mailbox, SSO) are encrypted with AES-256-GCM and never sent to the browser.
- Anyone who changes server, port, encryption, service account or user, issuer or client ID for AD, IMAP, SMTP or SSO, weakens the certificate check (including “Fetched certificate” → “System certificate authorities”) or stores a different AD certificate, must enter the password or secret again in the same step, otherwise nothing is saved. This way a stored secret never goes to a different server; connection tests with a weakened check do not send it either.
- Sessions end after the time set under *Automatic sign-out* without use (without a setting `HELFI_SESSION_HOURS`, default 12 hours; administrators 60 minutes) and at the latest 30 days after sign-in. Disabled accounts get the same message at sign-in as for a wrong password.
- Two-factor authentication per RFC 6238, each code works once, recovery codes are stored only as hashes.
- Brake on wrong passwords, cookies `HttpOnly` / `SameSite`, CSRF protection, strict Content Security Policy. The interface loads nothing from the internet.
- HTML emails are cleaned and shown in an isolated frame without scripts.
- The container runs without root rights, with a read-only program part, without additional Linux capabilities and with `no-new-privileges`.
- Tickets are never deleted; every change is recorded in the history.

## 6.6 License and source code

Helfi.Space is licensed under the **GNU Affero General Public License v3.0 or later** (AGPL-3.0). You may use it free of charge – also in a company – and adapt it. Whoever **modifies it and offers it to others**, even only as a service over the network, must publish the source code of their version under the same license.

The *Source code* link is in the sidebar next to the version, on the sign-in page, on phones under *Me* and under *Administration → System*. If you run a modified version, set `HELFI_SOURCE_URL` in `.env` to the address of your source code.

Versions up to and including 1.29.2 were released under the MIT license; it continues to apply to them.

## 6.7 Questions and solutions

| Problem | Solution |
|---|---|
| Emails from Proxmox, NAS or monitoring are skipped | Add a sender rule “Create ticket, never reply”. The exact sender address is listed under *Email → Recently skipped emails*. |
| Every request counts as “from outside” | Set `HELFI_TRUST_PROXY` to the address of the reverse proxy; check the internal networks. The *Security* page shows how your address is classified. |
| Lost the phone with the authenticator | Sign in with a recovery code or use *Users → Reset 2FA*; then set it up again inside the company network. |
| AD sign-in stopped working, certificate notice | The DC certificate was renewed: *Active Directory → Fetch certificate* and accept it (or store the certificate authority). |
| An agent does not see a project's tickets | *Users* shows under the role which projects they see. If the project is missing: check the project group under *Projects → Agents* (*Test connection* with their sign-in shows their projects) or assign them by hand. After a group change in AD, sign out and in once. |
| An AD user’s role is wrong | Check the group membership in AD (*Test connection* shows which DN a group name resolves to); if the group name is ambiguous, enter the full DN. With “Role locked” the role from Helfi.Space applies. |
| Notice “docker-compose.yml is outdated” | `./update.sh` takes over the new file, or download it from the release. |
| Helfi.Space does not start after a restore | Does `secret.key` or `HELFI_SECRET` match the database? The log ( `docker compose logs helfi` ) states the cause. |
| Setup code lost | It is in the log: `docker compose logs helfi` . |